IRRATIONALITY OF $G \pm 5\pi^2/48$ AND IRRATIONALITY EXPONENTS

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ABSTRACT. Let G be Catalan's constant and put $\lambda = G/2 - 5\pi^2/96$, so that $G - 5\pi^2/48 = 2\lambda$. We prove that λ is irrational and that $|\lambda - a/b| > b^{-46.45}$ for every integer a and every sufficiently large integer b; thus $\mu(\lambda) \le 46.45$ (earlier versions of this paper gave 126, 79.35 and 57.85). The mirror constant $\lambda' = G/2 + 5\pi^2/96$, so that $G + 5\pi^2/48 = 2\lambda'$, is irrational with $\mu(\lambda') \le 52.69$ (the hybrid improvement for λ' has not been done, so this is the best bound for λ'; the two constants satisfy $\lambda + \lambda' = G$, but the two measures do not combine to a measure for G). The proof uses determinants of Löwner matrices of $\psi(z+1) - \psi(z+3/4)$ at integer nodes. In the square frame ($h \times h$ nodes, $\mu \le 52.69$ for both λ and λ'; earlier versions gave 57.85) the determinant $P_h(\lambda)$ of $T_h = \lambda I + M_0$ has common denominator with $\log \operatorname{den} P_h \le (313/105 + o(1))h^2$, a contour representation and Andréief's identity give $\log |P_h(\lambda)| \le -(3.1113\ldots - o(1))h^2$ (the energy step uses the conditionally positive definite kernel $-\frac{1}{2}\log(|z-w|^2 + \rho^2)$ and an interval-arithmetic certificate), a node-removal bound gives an exponentially wide window around λ, and an ℓ-adic Schur-complement argument with a closed-form evaluation of a family of Krylov determinants gives nonvanishing for a pair of consecutive sizes $h \in \{H, H+1\}$ with H prescribed in advance; this freedom removes the length of the interval for ℓ from the exponent, which is why the square-frame exponent is c/η. For λ the further improvement to 46.45 uses a *rectangular hybrid frame*: a common block of r nodes together with two interleaved blocks of $m = r/8$ extra row and column nodes; the new ingredients are a sharper p-adic content bound (a rank saving at primes in two intervals, on top of an asymmetric tropical certificate), a decay and window certificate for the rectangular frame, and a bordered Krylov determinant, which is also evaluated in closed form. The Lean 4 formalization proves the irrationality and $\mu(\lambda) \le 77.28$ (exponent $31682445459/409965000$, with the simpler choice of h), unconditionally; the mirror theorem is formalized with the exponents 78 for λ' and 79 for $G + 5\pi^2/48$. The determinant evaluation of §6, the square-frame bound 52.69 and the hybrid theorem 46.45 are *not* formalized; the Lean bounds are unchanged by the 52.69 improvement, since Lean uses the weaker nonvanishing route.

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1. Introduction

OpenAI [11] recently proved that Catalan's constant

$$G = \sum_{j=0}^{\infty} \frac{(-1)^j}{(2j+1)^2} = 0.91596\,55941\ldots$$

is irrational, and formalized the proof in Lean. (An earlier claimed proof [14] was found to contain a gap [16].) Their argument, and the irrationality-measure results built on it [5, 2, 7], use a determinant of large size whose denominators and decay are balanced very closely.

In the present paper we study a different, much smaller, family of determinants, attached to the number

$$\lambda \;=\; \frac{G}{2} - \frac{5\pi^2}{96} \;=\; \sum_{k\ge 1}\Big(\frac{1}{16k^2} - \frac{1}{(4k-1)^2}\Big) \;=\; -0.05605\ldots$$

(see §2). Equivalently $G - 5\pi^2/48 = 2\lambda$. We also treat the "mirror" constant $\lambda' = G/2 + 5\pi^2/96$ (§9), for which $G + 5\pi^2/48 = 2\lambda'$ and $\lambda + \lambda' = G$.

Theorem 1.1. *The number $\lambda = G/2 - 5\pi^2/96$ is irrational. Equivalently, $G - 5\pi^2/48 \notin \mathbb{Q}$.*

Theorem 1.2. *For every integer a and every sufficiently large integer b,*

$$\left|\lambda - \frac{a}{b}\right| > b^{-52.69}.$$

In particular $\mu(\lambda) \le 52.69$ (the computed value is 52.6872; versions 1–7 of this paper gave 57.85 here). This is the square-frame *bound; for λ the hybrid bound 46.45 of Theorem 1.3 is better.*

Theorem 1.3 (the hybrid bound; Part 1)**.** *For every integer a and every sufficiently large integer b,*

$$\left|\lambda - \frac{a}{b}\right| > b^{-46.45}.$$

*In particular $\mu(\lambda) \le 46.45$ (the computed value is 46.447) and $\mu(G - 5\pi^2/48) \le 46.45$. This is proved in Part 1 (§10–§17) with a rectangular frame. It is **not formalized in Lean**, and it is established only for λ (primes $\ell \equiv 3 \pmod 4$): the hybrid for λ' (primes $\ell \equiv 1 \pmod 4$) has not been done, so for λ' the square-frame bound 52.69 of Theorem 1.4 below is the best one proved here.*

A simpler choice of h (without the determinant evaluation of §6) gives the weaker bound 77.13; see Remark 8.3. The Lean formalization proves, unconditionally, the irrationality and $\mu(\lambda) \le 77.28$ (exponent 31682445459/409965000; with the simpler choice of h; the value 77.28 rather than 77.13 comes from rounding the certified constants to short rationals); the determinant evaluation of §6 and the bound 52.69 are not formalized. The improvement from 57.85 to 52.69 does not change the Lean

bound, because the Lean proof uses the weak route of Theorem 5.4, with three values of h spread by a factor $4/3$, and not the free choice of h of Theorem 6.2.

Theorem 1.4 (the mirror constant)**.** *The number $\lambda' = G/2 + 5\pi^2/96$ is irrational, and for every integer a and every sufficiently large integer b, $|\lambda' - a/b| > b^{-52.69}$; thus $\mu(\lambda') \le 52.69$ (versions 1–7 gave 57.85). Consequently $G + 5\pi^2/48 \notin \mathbb{Q}$ and $\mu(G + 5\pi^2/48) \le 52.69$.*

The proof of Theorem 1.4 (§9) is the same architecture with the frame $\psi(z+1) - \psi(z+\frac{1}{4})$; the constants (content 313/105, the decay certificate, c_{cert}) are identical, and only the ℓ-adic part and some $o(1)$ bookkeeping change. Since $\lambda + \lambda' = G$ it is natural to ask whether Theorems 1.2 and 1.4 give a measure for G; they do not (Remark 9.7). The Lean formalization does not cover Theorem 1.4.

Here the *irrationality exponent* $\mu(\xi)$ of a real number ξ is the infimum of the μ such that $|\xi - a/b| > b^{-\mu}$ for all integers a and all sufficiently large b. We stress that Theorem 1.1 concerns one specific linear form in $1, G, \pi^2$; it is not a statement about G alone, whose irrationality is OpenAI's theorem.

A search of the literature (including work of Rivoal and Zudilin and of Calegari, Dimitrov and Tang on linear forms involving G) did not turn up a previous proof of the irrationality of $G - 5\pi^2/48$ or of $G + 5\pi^2/48$.

Related work on determinants. Irrationality proofs built on determinants of large size with positivity (Hankel determinants of moment sequences) have a precedent in W. Zudilin, *A determinantal approach to irrationality* [15]: that paper proves an irrationality criterion for sequences of integer approximations that carry Hankel structure, uses it to reprove the irrationality of π and of $\zeta(3)$, and in its Section 5 ("Commentary") explains a limit of the method for integer-valued data: the integer transfinite diameter of the cube $[0,\varepsilon]^n$ tends to $\varepsilon/4$, so that the denominator–decay balance obtained from a Vandermonde-squared construction cannot be improved by that route. Its last section treats Catalan's constant and q-analogues of Apéry's constants. We do not use Zudilin's criterion; our matrices are Löwner matrices of a digamma difference rather than moment matrices, and the balance between denominator and decay is obtained from the p-adic and energy estimates below. We mention [15] as prior art for the determinantal point of view, not as a source of any step of our proofs.

1.1. **Method.** For $h \ge 1$ let M_0 be the explicit rational $h \times h$ matrix of §2 and $T_h = \lambda I + M_0$; thus $P_h(X) = \det(XI + M_0) \in \mathbb{Q}[X]$ and $P_h(\lambda) = \det T_h$.

(i) *Content* (§3). A prime-by-prime tropical (assignment) bound, with an explicit dual certificate, gives $\log \operatorname{den} P_h \le (313/105 + o(1))h^2$.

(ii) *Decay* (§4). T_h is $\frac{1}{16}$ times a Löwner matrix. Moving the Hermite contour integral to a two-ray contour produces finitely many signed atoms plus an integral against an exponentially small analytic weight, and Andréief's identity then bounds $|P_h(\lambda)| \le 16^{-h} \det \mathcal{G}_h$ with $\mathcal{G}_h$

a positive definite Gram matrix. The energy of the resulting point configuration is bounded below with the kernel $k_\rho(z,w) = -\frac{1}{2}\log(|z-w|^2+\rho^2)$, $\rho = 1/1024$, which is conditionally positive definite on finite sums (Frullani's integral and Gaussian sums), against a certificate measure σ of 300 dyadic point masses; the resulting finite inequalities are checked in interval arithmetic and give $\det \mathcal{G}_h \le \exp(-(3.1113 - o(1))h^2)$. The decay beats the content by the margin $3.1113790 - 313/105 > 0.1304$.

(iii) *Nonvanishing* (§5). For a prime $\ell \equiv 3 \pmod 4$, $\ell \ge 7$, and any rational r with $\ell \nmid \operatorname{den} r$, one of $P_{\ell-1}(r), P_{\ell-2}(r), P_{(3\ell-1)/4}(r)$ is nonzero, because the ℓ-adic Schur complement of $D(rI + M_0)D$ has a transparent Cauchy-matrix form. This gives irrationality with no information on the smallest eigenvalue of T_h.

(iv) *A determinant evaluation and the choice of h* (§6). Lemma CC says that two consecutive Gram-type matrices N_K, N_{K+1} over $\mathbb{F}_\ell$ have no common eigenvalue. It is reduced to the non-vanishing modulo ℓ of a Krylov determinant $u(K,K) = \det[w, N_K w, \dots, N_K^{K-1} w]$, and Theorem A evaluates $|u|$ as a product of powers of integers $\le 2K < \ell$ (zeros and poles in two shift parameters by Weyl's inequality and a cluster lemma, the constant by a recursion). Consequently, for any rational r with $\ell \nmid \operatorname{den} r$ and any H with $K+1 \le H \le \ell - K - 1$, one of $P_H(r), P_{H+1}(r)$ is nonzero (Theorem 6.2); the target H may therefore be fixed first and ℓ chosen afterwards among the primes $\ell \equiv 3 \pmod 4$ in $[\frac{4H+5}{3}, 4H-5]$.

(v) *Measure* (§7–8). The proof of the decay bound extends to every principal minor of T_h: removing $j = \beta h$ nodes costs at most $e^{c\beta h^2}$, with $c = 2(m' - g_{\min}) \le 6.8717994$ and $g_{\min}$ certified in interval arithmetic (Theorem 7.1). Expanding $\det(T_h + tI)$ in principal minors, this gives $|P_h(\lambda + t)| \le e^{-(3.1113 - o(1))h^2}$ for all $|t| \le e^{-ch}$. A rational p/q very close to λ lies in this window, and the arithmetic of the choice of $h \in \{H, H+1\}$, $H = \lceil kL \rceil$, and of a prime $\ell \nmid q$ in the interval above yields $\mu(\lambda) \le c/\eta < 52.69$, $\eta = 3.1113790\ldots - 313/105 = 0.1304266\ldots$ (the computed value is 52.6872; the earlier selection of ℓ, which needed an interval of length $(1+\delta)L$ and a factor $h/\ell \approx \frac{3}{4}$, gave $c\,(1/\eta + 3/4) < 57.85$; with the simpler choice of h of Theorem 5.4 one gets $c\,(4/(3\eta) + 1) < 77.13$, Remark 8.3; an earlier circle-smoothed energy step gave 79.35 with that choice, Remark 4.12).

(vi) *The hybrid frame* (Part 1). Replace the square $h \times h$ Löwner matrix by a rectangular one: rows $C \sqcup X'$, columns $C \sqcup Y'$, with a long common block C of r nodes and two interleaved blocks X', Y' of $m = r/8$ nodes. The content constant rises to $\Lambda_F \le 3.7211401$ (it is $313/105$ for $m = 0$), but the decay constant rises further to $D_F \ge 3.9062695$; the margin is $\eta_{\mathrm{hyb}} \ge 0.1851294$ instead of 0.1304266. The window constant is $c_{\mathrm{hyb}} \le 7.6540647$ and the ℓ-adic window contributes $\kappa =$

$2/3$ (the square frame now has $\kappa = 0$; the hybrid keeps $\kappa = 2/3$ because its pair of members is tied to the geometry, see Remark 17.1): $\mu(\lambda) \le c_{\mathrm{hyb}}(1/\eta_{\mathrm{hyb}} + \kappa) \le 46.45$.

1.2. **Status of the proofs.** The proofs of Theorems 1.1, 1.2 and 1.4 were derived by Claude (Anthropic) agents working with the author, and each part was re-derived and spot-checked numerically by separate Claude audit agents; no human referee has checked them. Two steps are computer-assisted (interval-arithmetic certificates, §4.4 and §7.4). The Krylov determinant evaluation (Theorem A) and the choice of h of §6 were derived and then audited by separate Claude agents (exact rational checks; see §18). The Lean formalization proves the irrationality and the weaker bound 77.13, conditional on the numerical certificate facts; Theorem A and the deduction of 52.69 are not yet formalized, and the mirror theorem, Theorem 1.4, is formalized with the same weaker exponent (78 for λ', 79 for $G + 5\pi^2/48$). See §18 for an exact account.

The observation behind the improvement from 57.85 to 52.69 in the square frame (version 8) is that the nonvanishing theorem of §6 holds for a pair $\{H, H+1\}$ with H arbitrary in a range, so the interval for ℓ costs nothing; it was made by a Claude agent and then audited by a separate Claude agent (exact checks in §18).

The hybrid theorem, Theorem 1.3, and the whole of Part 1 were derived by Claude (Anthropic) agents working with the author, in the same way, and each part was then audited by separate Claude agents (adversarial passes with independent exact computations; see §18); no human referee has checked them. One step is computer-assisted (the interval-arithmetic certificate of §13.2). **The hybrid theorem is not formalized in Lean**; the Lean results are the square-frame results with the weak nonvanishing route, 77.28 (for λ), 78 (for λ') and 79 (for $G + 5\pi^2/48$), which the 52.69 improvement does not change. A list of the statements for which our written proof is weaker than the rest is in §17.3.

1.3. **Notation.** ψ is the digamma function, $H_n = \sum_{k\le n} 1/k$, $H_n^{(2)} = \sum_{k\le n} 1/k^2$, and $\vartheta(x)$, $\vartheta(x;4,a) = \sum_{p\le x,\, p\equiv a\,(4)} \log p$ are Chebyshev functions. For $P \in \mathbb{Q}[X]$, $\operatorname{den} P$ is the least positive integer d with $dP \in \mathbb{Z}[X]$. Steps resting on numerical certificates are labelled "computer-assisted".

2. The matrices

2.1. **The number λ.** Since $\sum_{n\equiv 1(4)} n^{-2} - \sum_{n\equiv 3(4)} n^{-2} = G$ and the sum of both is $\pi^2/8$ (sums over $n \ge 1$),

$$\sum_{n\equiv 3\,(4)} n^{-2} = \frac{\pi^2}{16} - \frac{G}{2}, \qquad \sum_{k\ge 1} \frac{1}{16k^2} = \frac{\pi^2}{96},$$

so

$$\lambda := \sum_{k\ge 1}\Big(\frac{1}{16k^2} - \frac{1}{(4k-1)^2}\Big) = \frac{\pi^2}{96} - \frac{\pi^2}{16} + \frac{G}{2} = \frac{G}{2} - \frac{5\pi^2}{96}. \tag{2.1}$$

2.2. **Löwner structure.** Let

$$F(z) = \psi(z+1) - \psi(z+\tfrac{3}{4}) = \sum_{j\ge 0}\Big[\frac{1}{z+j+\frac{3}{4}} - \frac{1}{z+j+1}\Big],$$

meromorphic with simple poles at $-(j+\frac{3}{4})$ (residue $+1$) and $-(j+1)$ (residue -1), $j \ge 0$. For nodes $x_0, \dots, x_{h-1}$ the *Löwner matrix* of F is $L_F[a][b] = (F(x_b) - F(x_a))/(x_b - x_a)$ for $a \ne b$ and $L_F[a][a] = F'(x_a)$. Take the nodes to be $x_a = a$, $a = 0, \dots, h-1$.

From $F(k) - F(k-1) = \frac{1}{k} - \frac{1}{k-1/4} = -\frac{1}{k(4k-1)}$ and the corresponding identity for F',

$$L_F[a][b] = -\frac{1}{b-a}\sum_{a<k\le b}\frac{1}{k(4k-1)} \qquad (a<b), \tag{2.2}$$

$$L_F[a][a] = F'(0) + \sum_{k\le a}\Big(\frac{16}{(4k-1)^2} - \frac{1}{k^2}\Big), \tag{2.3}$$

and $F'(0) = \sum_{j\ge 0}[(j+1)^{-2} - (j+\frac{3}{4})^{-2}] = \frac{\pi^2}{6} - 16\sum_{n\equiv 3(4)} n^{-2} = 8G - \frac{5\pi^2}{6} = 16\lambda$.

Definition 2.1. Let $M_0 = M_0^{(h)}$ be the symmetric rational $h \times h$ matrix with, for $0 \le a < b < h$,

$$M_0[a][b] = -\frac{1}{4(b-a)}\sum_{k=a+1}^{b}\frac{1}{4k(4k-1)}, \qquad M_0[a][a] = -\sum_{k=1}^{a}\Big(\frac{1}{16k^2} - \frac{1}{(4k-1)^2}\Big),$$

and set $T_h = \lambda I + M_0$ and $P_h(X) = \det(XI + M_0) \in \mathbb{Q}[X]$, monic of degree h.

Proposition 2.2. *$T_h = \frac{1}{16}L_F$ at the nodes $0, \dots, h-1$. In particular $P_h(\lambda) = 16^{-h}\det L_F$, and $P_h(\lambda + t) = \det(T_h + tI)$ for $t \in \mathbb{R}$.*

Proof. Compare (2.2), (2.3) with the definition, using $F'(0) = 16\lambda$ and $\frac{1}{16k^2} - \frac{1}{(4k-1)^2} = \frac{1}{16}(\frac{1}{k^2} - \frac{16}{(4k-1)^2})$. □

Remark 2.3. The coefficients of P_h are sums of principal minors of M_0, whose entries are harmonic-type partial sums; there is no factorial closed form. Also L_F is not a Gram matrix of a positive measure, since the "Stieltjes measure" of F consists of the signed atoms $+1$ at $-(j+\frac{3}{4})$ and -1 at $-(j+1)$; this is the reason for the contour step in §4.

Lemma 2.4 (integrality)**.** *Let $Q_h = \mathrm{den}(P_h)\cdot P_h \in \mathbb{Z}[X]$. If a/b is in lowest terms and $Q_h(a/b) \ne 0$, then $|Q_h(a/b)| \ge b^{-h}$.*

Proof. Q_h has degree h and integer coefficients, so $b^h Q_h(a/b) \in \mathbb{Z}\setminus\{0\}$. □

3. The denominator bound

Theorem 3.1. $\log \operatorname{den} P_h \le (\frac{313}{105} + o(1))h^2$ *as* $h \to \infty$.

We write $\Lambda = 313/105 = 2.98095\ldots$. The proof is prime by prime. For a prime p put $W_{ab} = \max(0, -v_p(M_0[a][b]))$.

3.1. **Tropical bound.**

Lemma 3.2. *Let* τ_p *be the maximum of* $\sum_a W_{a,\sigma(a)}$ *over permutations* σ *of* $\{0,\ldots,h-1\}$. *Then* $v_p(\operatorname{den} P_h) \le \tau_p$. *If* $u_a, v_b \ge 0$ *satisfy* $W_{ab} \le u_a + v_b$ *for all* a, b, *then* $\tau_p \le \sum_a u_a + \sum_b v_b$.

Proof. The coefficient of X^{h-j} in $\det(XI + M_0)$ is a sum of $j \times j$ principal minors, each a signed sum of products $\prod_{a\in S} M_0[a][\sigma(a)]$ over permutations σ of a j-set S. Each product has $-v_p \le \sum_{a\in S} W_{a\sigma(a)}$, and since $W \ge 0$, extending σ by the identity off S shows this is at most τ_p. For the second statement, $\sum_a W_{a\sigma(a)} \le \sum_a (u_a + v_{\sigma(a)})$. □

3.2. **Valuation majorant.** Let p be odd and $\nu(k) = v_p(k(4k-1)) = \max(v_p(k), v_p(4k-1))$ (the factors k and $4k-1$ are coprime). Let $\mu(a) = \max_{1\le k\le a} \nu(k)$. By the ultrametric inequality applied to the formulas of the Definition,

$$W_{ab} \le v_p(b-a) + \max_{a<k\le b} \nu(k) \ (a<b), \qquad W_{aa} \le 2\mu(a). \tag{3.1}$$

For $q = p^e$ set

$$\operatorname{div}_q(a,b) = \mathbf{1}[q \mid b-a,\ a \ne b],$$
$$\operatorname{supp}_q(a,b) = \mathbf{1}[\exists k \in (\min(a,b), \max(a,b)] :\ q \mid k(4k-1)] \ (a \ne b), \qquad \operatorname{supp}_q(a,a) = 2\cdot\mathbf{1}[\mu(a) \ge e].$$

Because $v_p(n) = \sum_{e\ge1} \mathbf{1}[p^e \mid n]$ and $\max_k \nu(k) = \sum_e \mathbf{1}[\exists k :\ p^e \mid k(4k-1)]$, (3.1) says $W \le \sum_{e\ge1} w^{(e)}$ entrywise with $w^{(e)} = \operatorname{div}_{p^e} + \operatorname{supp}_{p^e} \in \{0,1,2\}$. Since a maximum of a sum is at most the sum of maxima, $\tau_p \le \sum_e \tau(w^{(e)})$.

3.3. **Per-level dual certificate.** Let $q = p^e$ be an odd prime power. The least $k \ge 1$ with $q \mid k(4k-1)$ is $k_q = (j_q q + 1)/4$, where $j_q = 1$ if $q \equiv 3 \pmod 4$ and $j_q = 3$ if $q \equiv 1 \pmod 4$ (indeed $4k - 1 = jq$ forces $jq \equiv 3 \pmod 4$, and the branch $q \mid k$ gives $k \ge q > k_q$). Hence $\operatorname{supp}_q(a,b) \ne 0$ forces $\max(a,b) \ge k_q$. Put $n_q = \min(h, k_q)$ and define potentials

$$u_a = v_a = \begin{cases} 1 & a \ge k_q \ (\text{“high”}), \\ \mathbf{1}[a \le h-1-q] & a < k_q \ (\text{“low”}). \end{cases}$$

Lemma 3.3. $w^{(e)}_{ab} \le u_a + v_b$ *for all* a, b. *Consequently*

$$\tau(w^{(e)}) \le \delta(q) := \min\bigl(2h,\ 2(h-n_q) + 2\min(n_q, (h-q)_+)\bigr),$$

and $\delta(q) = 0$ *if* $q \ge h$ *and* $k_q \ge h$; *in particular* $\tau_p = 0$ *for* $p > 4h$.

Proof. Note $k_q < q$. *Both low:* $|b-a| < k_q < q$ so div $= 0$, and $(a,b] \subset [1, k_q - 1]$ so supp $= 0$ (diagonal likewise). *Both high:* $w \le 2 = u_a + v_b$. *a low, b high:* supp $\le 1 = v_b$; if div $= 1$ then $b \ge a + q$, $b \le h - 1$, so $a \le h - 1 - q$ and $u_a = 1$; the case a high, b low is symmetric. The diagonal is 0 at low and at most 2 at high indices. Now apply Lemma 3.2: $\sum u + \sum v = 2\#\{\text{high}\} + 2\#\{a < n_q : a \le h - 1 - q\}$, and the trivial potentials $u = v \equiv 1$ give $2h$. □

By Lemma 3.2, $v_p(\operatorname{den} P_h) \le \sum_{e \ge 1} \delta(p^e)$ for odd p.

3.4. **Summation. Prime 2.** Every entry of M_0 has $-v_2 \le 4 + 2\log_2 h$ (from the factors $1/(4(b-a))$, $1/(4k(4k-1))$, $1/(16k^2)$), so the prime 2 contributes $O(h \log h)$ to $\log \operatorname{den} P_h$.

Levels $e \ge 2$. Here $\delta \le 2h$, so these contribute at most $2h\,(\sum_{p^e \le 4h, e \ge 2} \log p) = 2h(\psi_{\mathrm{Ch}}(4h) - \vartheta(4h)) = O(h^{3/2})$ (Chebyshev: $\psi_{\mathrm{Ch}}(x) - \vartheta(x) = O(\sqrt{x})$).

Level $e = 1$. For $q = p$ a direct computation from Lemma 3.3 (using $|k_q - j_q q/4| \le 1/4$ and that δ is 2-Lipschitz in n_q) gives $\delta(p) \le h\, g_j(p/h) + \frac{1}{2}$, $j = j_p$, where

$$g_j(\theta) = \begin{cases} 2 & 0 \le \theta \le \frac{4}{4+j}, \\ 4 - \frac{(4+j)\theta}{2} & \frac{4}{4+j} \le \theta \le 1, \\ 2 - \frac{j\theta}{2} & 1 \le \theta \le \frac{4}{j}, \\ 0 & \theta \ge \frac{4}{j}. \end{cases}$$

Indeed for $\theta \le 4/(4+j)$ one has $n_q \le h - q$ and $\delta = 2h$; for $4/(4+j) \le \theta \le 1$, $\delta = 2(h - n_q) + 2(h - q)$; for $\theta \ge 1$, $\delta = 2(h - n_q)$. The function g_j is continuous, non-increasing and piecewise linear, with $\int_0^\infty g_j = \frac{4}{j} + \frac{j}{4+j}$, that is

$$\int g_1 = \frac{21}{5} \ (p \equiv 3 \bmod 4), \qquad \int g_3 = \frac{37}{21} \ (p \equiv 1 \bmod 4).$$

By partial summation, since $-g_j' \ge 0$ is supported on $\theta \ge 4/7$,

$$\frac{1}{h} \sum_{p \equiv a\,(4)} \log p\; g_j(p/h) = \frac{1}{h} \int (-g_j'(\theta))\, \vartheta(\theta h; 4, a)\, d\theta \le \frac{1 + \varepsilon_h}{2} \int g_j,$$

where $\varepsilon_h = \sup_{4h/7 \le x \le 4h} (2\vartheta(x;4,a)/x - 1) \to 0$ by the prime number theorem in arithmetic progressions mod 4 (de la Vallée Poussin; see e.g. [3, Ch. 20] or [10, Cor. 11.19]); only an upper bound for $\vartheta(x;4,a)$ is needed because the weights are non-negative. Collecting terms,

$$\frac{\log \operatorname{den} P_h}{h^2} \le (1 + \varepsilon_h)\Big(\frac{1}{2} \cdot \frac{21}{5} + \frac{1}{2} \cdot \frac{37}{21}\Big) + O\Big(\frac{1}{\sqrt{h}}\Big) = (1 + \varepsilon_h)\frac{313}{105} + o(1).$$

This proves Theorem 3.1. □

Remark 3.4 (computer-assisted supplement; not used). Evaluating the per-h bound $\sum_q \delta(q) \log p$ (plus a bound for $p = 2$) exactly for every $h \le 2 \cdot 10^5$ gives $\log \operatorname{den} P_h \le 2.98635\, h^2$, with maximum at $h = 1844$ (script `scan.py`). This is not needed for the theorems. The dual certificate is also essentially

sharp: for odd primes and $h = 64, \dots, 256$ the model value equals the exact tropical optimum except at $p = 3$.

4. The decay bound

4.1. **A contour representation of L_F.** Hermite's formula gives, for any contour C enclosing the nodes and no pole of F,

$$L_F = \frac{1}{2\pi i} \oint_C F(z)\, v(z) v(z)^\top dz, \qquad v_a(z) = \frac{1}{z-a}$$

(the residue at $z = a \neq b$ gives $(F(a) - F(b))/(a-b)$, the double pole gives $F'(a)$). By the reflection formulas $\psi(1+z) = \psi(-z) - \pi\cot\pi z$ and $\psi(\frac{3}{4}+z) = \psi(\frac{1}{4}-z) + \pi\cot(\frac{\pi}{4} - \pi z)$,

(4.1)
$$F = \tilde{F} + E, \qquad \tilde{F}(z) = \psi(-z) - \psi(\tfrac{1}{4} - z), \quad E(z) = -\pi\big[\cot\pi z + \cot(\tfrac{\pi}{4} - \pi z)\big] = \frac{-\pi\sin(\pi/4)}{\sin\pi z\, \sin(\pi/4 - \pi z)}.$$

Here $\tilde{F}$ is analytic in $\operatorname{Re} z < 0$ and $O(1/|z|)$ there, while E is 1-periodic, has poles only on the real axis, and $|E(x+iy)| \le Ce^{-2\pi|y|}$ for $|y| \ge 1$.

Fix $r_0 = 7/50$, $N_0 = \lfloor r_0 h \rfloor$, $x^* = -N_0 - \frac{3}{8}$ (which lies in the gap $(-N_0 - \frac{3}{4}, -N_0)$ between poles of F). Let Γ_h be the two rays $x^* + u(-1 \pm 3i)$, $u \ge 0$ (the *rational direction* $(-1, \pm 3)$, at angle $\pi - \arctan 3 = 0.6024\pi$ from the positive axis), traversed downward, and let $\mathcal{A} = \{j + \frac{3}{4},\ j+1 :\ 0 \le j < N_0\}$, $|\mathcal{A}| = 2N_0$. For $\mu \in \mathcal{A}$ let $c_\mu = +1$ for $\mu = j + \frac{3}{4}$ and -1 for $\mu = j+1$, and $u_\mu(a) = 1/(a+\mu)$. The direction enters only through three facts: the left wedge bounded by Γ_h has half-angle $\arctan 3 < \pi/2$, so $\operatorname{Re} z \le x^* < 0$ on it; E and $x^* \equiv 5/8 \pmod 1$ are periodic; and on the part of Γ_h with $|\operatorname{Im} z| < 1$ (that is, $u < \frac{1}{3}$) one has $\operatorname{Re} z \in (x^* - \frac{1}{3}, x^*] = (-N_0 - 0.7083\ldots, -N_0 - \frac{3}{8}]$, strictly inside the gap between the poles $-N_0 - \frac{3}{4}$ and $-N_0$.

Proposition 4.1 (contour identity)**.** *For every h,*

$$L_F = -\sum_{\mu \in \mathcal{A}} c_\mu\, u_\mu u_\mu^\top + \frac{1}{2\pi i} \int_{\Gamma_h} E(z)\, v(z) v(z)^\top dz.$$

Proof. Move the contour in Hermite's formula from around the nodes to Γ_h, closing on the right where the integrand is $O(|z|^{-3})$; this picks up the residues of F at the $2N_0$ poles $-\mu$, $\mu \in \mathcal{A}$, which lie to the right of x^*, each contributing $\pm c_\mu u_\mu u_\mu^\top$ according to orientation. Then split $F = \tilde{F} + E$; the $\tilde{F}$ part integrates to 0 over Γ_h, by closing in the left wedge where $\tilde{F} v v^\top$ is analytic and $O(|z|^{-3})$. □

(The identity, with the stated orientation and signs, was checked entrywise to 10^{-17} with 25-digit arithmetic at $N_0 = 2$ and $N_0 = 7$, for the direction $(-1, 3)$ and for the earlier angle $3\pi/5$.)

4.2. **Absolute-value Andréief bound.** Let $|m| = \sum_{\mu\in\mathcal{A}} \delta_{-\mu} + \frac{1}{2\pi}|E(z)||dz|$ on Γ_h, and

$$\mathcal{G}_h = \sum_{\mu\in\mathcal{A}} u_\mu u_\mu^\top + \frac{1}{2\pi}\int_{\Gamma_h} |E|\, v\, v^*|dz|,$$

a Hermitian positive definite matrix (the v_a are linearly independent functions).

Proposition 4.2. *$|\det L_F| \le \det \mathcal{G}_h$; hence $|P_h(\lambda)| \le 16^{-h}\det \mathcal{G}_h$. More generally, for real t, $|P_h(\lambda+t)| \le \det(\mathcal{G}_h/16 + |t|I)$. Moreover*

$$\det \mathcal{G}_h = \frac{1}{h!}\int \prod_{i<j}|z_i - z_j|^2 \prod_{a<b}(b-a)^2 \prod_{i,a}|z_i - a|^{-2}\prod_i d|m|(z_i).$$

Proof. By Proposition 4.1, $L_F[a][b] = \int v_a v_b\, dm$ for a complex measure m with total variation $|m|$. Andréief's identity [4] gives $\det L_F = \frac{1}{h!}\int \det[v_a(z_i)]^2 \prod dm(z_i)$, so $|\det L_F| \le \frac{1}{h!}\int |\det[v_a(z_i)]|^2 \prod d|m|(z_i) = \det \int v_a\overline{v_b}\, d|m| = \det \mathcal{G}_h$, and the displayed formula is Cauchy's determinant $\det[1/(z_i - a)] = \pm\prod_{i<j}(z_i - z_j)\prod_{a<b}(b-a)/\prod(z_i - a)$. For the last assertion, $T_h + tI$ is the Gram integral $\int \Phi_a\Phi_b\, dM$ on the disjoint union of $\Gamma_h \cup (-\mathcal{A})$ and the node set, with $\Phi_a = v_a/4$ on the first part and $\Phi_a = \mathbf{1}_{\{a\}}$ on the second, and $M = m \sqcup t\cdot\#$; the same argument bounds $|\det|$ by $\det(\mathcal{G}_h/16 + |t|I)$. □

4.3. **The kernel k_ρ.** Fix $\rho = 1/1024$ and put

$$k_\rho(z,w) := -\tfrac{1}{2}\log(|z-w|^2 + \rho^2) \qquad (z,w\in\mathbb{C}), \qquad k_\rho(z,z) = -\log\rho.$$

The energy step below uses k_ρ in place of the logarithmic kernel. Its virtue is that the conditional positive definiteness it needs is a statement about *finite* sums and is proved by three elementary integrals (no Fourier transform and no measure on $\mathbb{C}$), and that no regularity constant for the potential of the certificate measure is ever needed.

Lemma 4.3 (comparison)**.** *For $z \ne w$, $k_\rho(z,w) \le -\log|z-w|$.*

Proof. $|z-w|^2 < |z-w|^2 + \rho^2$ and log is increasing. □

Lemma 4.4 (Frullani form of the logarithm)**.** *For $a > 0$, $\log a = \displaystyle\int_0^\infty \frac{e^{-t} - e^{-at}}{t}\, dt$, and the integrand is integrable on $(0,\infty)$.*

Proof. For $t > 0$, $(e^{-t} - e^{-at})/t = \int_1^a e^{-tu}\, du$ (oriented integral); the integrand is bounded by $|a-1|$ near 0 and by $2e^{-\min(1,a)t}$ on $[1,\infty)$. If $a \ge 1$ the function $(t,u) \mapsto e^{-tu}$ on $(0,\infty)\times[1,a]$ is non-negative and $\int_1^a(\int_0^\infty e^{-tu}\, dt)\, du = \int_1^a du/u = \log a < \infty$, so Tonelli lets us exchange the integrals. If $0 < a < 1$, interchange with reversed orientation (the integrand is then $-\int_a^1 e^{-tu}du$). □

Lemma 4.5 (finite Gaussian sums)**.** *For $t \geq 0$, finitely many points $p_\alpha = (x_\alpha, y_\alpha) \in \mathbb{R}^2$ and reals w_α, $\sum_{\alpha,\beta} w_\alpha w_\beta e^{-t|p_\alpha - p_\beta|^2} \geq 0$.*

Proof. Put $v_\alpha = w_\alpha e^{-t|p_\alpha|^2}$; the sum is $\sum v_\alpha v_\beta e^{2t\langle p_\alpha, p_\beta\rangle} = \sum_{n\geq 0} \frac{(2t)^n}{n!} \sum_{\alpha,\beta} v_\alpha v_\beta \langle p_\alpha, p_\beta\rangle^n$. By the binomial theorem $\langle p, p'\rangle^n = \sum_{k=0}^n \binom{n}{k} (x^k y^{n-k})(x'^k y'^{n-k})$, hence

$$\sum_{\alpha,\beta} v_\alpha v_\beta \langle p_\alpha, p_\beta\rangle^n = \sum_{k=0}^{n} \binom{n}{k} \Big(\sum_\alpha v_\alpha x_\alpha^k y_\alpha^{n-k}\Big)^2 \geq 0,$$

a finite identity. A finite sum of convergent series is the series of the finite sums, and a series of non-negative terms is non-negative. □

Lemma 4.6 (conditional positive definiteness)**.** *Let $p_1, \dots, p_N \in \mathbb{C}$ (repetitions allowed) and $q_1, \dots, q_N \in \mathbb{R}$ with $\sum q_\alpha = 0$. Then $\sum_{\alpha,\beta} q_\alpha q_\beta\, k_\rho(p_\alpha, p_\beta) \geq 0$.*

Proof. Put $a_{\alpha\beta} = |p_\alpha - p_\beta|^2 + \rho^2 > 0$. By Lemma 4.4 and linearity, $\sum q_\alpha q_\beta(-\log a_{\alpha\beta}) = \int_0^\infty \sum_{\alpha,\beta} q_\alpha q_\beta \frac{e^{-t a_{\alpha\beta}} - e^{-t}}{t}\, dt$. The e^{-t} part is $(\sum q_\alpha)^2 e^{-t}/t = 0$ pointwise. What remains is $\frac{e^{-t\rho^2}}{t} \sum q_\alpha q_\beta e^{-t|p_\alpha - p_\beta|^2} \geq 0$ by Lemma 4.5. Multiply by $\frac{1}{2}$. □

For a finite set $X \subset \mathbb{C}$ with $|X| = h$ and a certificate measure $\sigma = \sum_{k \leq n_\sigma} m_k \delta_{c_k}$ ($m_k > 0$, $\sum m_k = 1$) put

$$U_\rho^\sigma(z) = \sum_k m_k\, k_\rho(z, c_k), \qquad I_\rho(\sigma) = \sum_{k,l} m_k m_l\, k_\rho(c_k, c_l).$$

Corollary 4.7. $\displaystyle \frac{1}{h^2} \sum_{x \neq y \in X} k_\rho(x, y) \;\geq\; \frac{2}{h} \sum_{x \in X} U_\rho^\sigma(x) - I_\rho(\sigma) + \frac{\log \rho}{h}.$

Proof. Apply Lemma 4.6 to the list $(X, c_1, \dots, c_{n_\sigma})$ with weights $(\frac{1}{h}, \dots, \frac{1}{h}, -m_1, \dots, -m_{n_\sigma})$, which sum to 0. Expanding, $\frac{1}{h^2} \sum_{x,y\in X} k_\rho(x,y) - \frac{2}{h} \sum_x U_\rho^\sigma(x) + I_\rho(\sigma) \geq 0$, and the diagonal terms $x = y$ contribute $\frac{1}{h^2} \cdot h \cdot (-\log \rho)$ to the left, which we move across. □

4.4. **The energy functional and the certificate.** Scale $z = hs$ and put τ = Lebesgue measure on $[0,1]$ and $U^\tau(s) = -\int_0^1 \log|s - t|\, dt$, so that $I(\tau) = 3/2$. For $\operatorname{Re} z < 0$ and $y = \operatorname{Im} z \geq 0$ one has the closed form

$$U^\tau(x + iy) = 1 - x \log|z| - (1 - x) \log|1 - z| - y \arctan \frac{y}{x^2 + y^2 - x},$$

symmetric in $y \mapsto -y$; on the real line $U^\tau(t) = 1 - t\log|t| - (1-t)\log|1-t|$. Write Γ for the rays $\{-r_0 + u(-1 \pm 3i) : u \geq 0\}$ and $\kappa = 10^{-3}$. For a fixed *certificate measure* σ define

$$g(s) = U_\rho^\sigma(s) - U^\tau(s) + (1 - \kappa)\pi |\operatorname{Im} s|.$$

The only requirement on σ is that its quantities are computed rigorously; its relation to an optimizer is irrelevant.

The measure σ. $\sigma = \sum_{k=1}^{n_\sigma} m_k \delta_{c_k}$ is a sum of $n_\sigma = 300$ point masses (84 real points and 108 conjugate pairs, so σ and g are conjugation-symmetric), with dyadic data: coordinates of c_k of the form $X/2^{24}$, $Y/2^{24}$ with integers X, Y of at most 23 bits, and masses $n_k/2^{32}$ with $\sum m_k = 1$ *exactly*. The data are in the file `sigma300.json`. Everything below is a finite computation with these rationals; in particular there are no circles, no Lipschitz constant and no continuum energy $I(\sigma)$.

What is certified. Put $\omega = 2^{-30}$ (a box half-width absorbing the horizontal shift of Γ_h/h) and $u_{\max} = \frac{3}{2}$.

(R) A cover of $u \in [0, u_{\max}]$ by intervals $[u_a, u_b]$ such that $g \geq m'$ on each box $[-r_0 - u_b - \omega, -r_0 - u_a + \omega] \times [3u_a, 3u_b]$, together with the tail bound $T(u_{\max}) \geq m'$: for $u \geq u_{\max}$, $|z| \leq r_0 + \omega + u\sqrt{10}$, $U_\rho^\sigma(z) \geq -\frac{1}{2}\log((|z| + R_\sigma)^2 + \rho^2)$ with $R_\sigma = \max|c_k|$, $U^\tau(z) \leq -\log(|z| - 1)$, and the field term is $\geq (1 - \kappa)3\pi u$; the resulting lower bound $T(u)$ is increasing.

(S) A partition of $[-r_0, 0]$ into cells $C_\ell = [a_\ell, b_\ell]$ with $g \geq \gamma_\ell$ on C_ℓ. Put $S_{\text{cert}} := 2\sum_\ell (b_\ell - a_\ell)\min(\gamma_\ell - m', 0)$.

(L) A cover of $[0, 1]$ (in fact of $[-\frac{1}{64}, 1]$) by cells on which $U_\rho^\sigma - U^\tau \geq g_{\min}$.

Then put

$$D_{\text{cert}} := 2(m' + S_{\text{cert}}) - I_\rho(\sigma) + \tfrac{3}{2}, \qquad c := 2(m' - g_{\min}).$$

On a box or cell the lower bound for U_ρ^σ is $-\frac{1}{2}\sum_k m_k \log(d_{x,k}^2 + d_{y,k}^2 + \rho^2)$ with $d_{x,k}, d_{y,k}$ the largest horizontal and vertical distances from c_k to the box, which makes every logarithm argument as large as possible on the box (the two coordinates decouple, so the maximum is attained); U^τ is bounded above on ray boxes by ball arithmetic on the closed form, on segment cells by its value at the right end (it is increasing on $(-\infty, 0]$), and on cells of $[0, 1]$ by monotonicity on each side of $\frac{1}{2}$ (the supremum is $U^\tau(b)$ if $b \leq \frac{1}{2}$, $U^\tau(a)$ if $a \geq \frac{1}{2}$, and $1 + \log 2$ otherwise).

Certified numbers (computer-assisted). The script `kcert.py` (Arb [1], 96-bit precision; ray boxes bisected adaptively) was run on `sigma300.json` with the following results; S_{cert} is assembled from exact rationals, and all other entries are lower or upper bounds as indicated.

quantity	certified value	cells
$I_\rho(\sigma)$ (exact finite sum)	≤ 2.1107199269	$\binom{n_\sigma+1}{2}$ terms
m': inf g over the ray boxes, $u \in [0, \frac{3}{2}]$	≥ 1.9718000062	26,977 boxes
tail $T(\frac{3}{2})$	≥ 13.735 (analytic)	–
S_{cert}	≥ -0.1107505186	14,000 cells
$g_{\min}$ on $[-\frac{1}{64}, 1]$	≥ -1.4640996549	4,423 cells
D_{cert}	≥ 3.1113790	
$c = 2(m' - g_{\min})$	≤ 6.8717994	
$\eta = D_{\mathrm{cert}} - \Lambda$	≥ 0.1304266	
c/η (Theorem 8.1; free choice of h)	$\le$ 52.69 (computed: 52.6872)	
$c\,(1/\eta{+}3/4)$ (versions 1–7; fixed $h/\ell \approx \frac{3}{4}$, with Lemma 5.5)	$\le$ 57.85 (computed: 57.841)	
$c\,(1/\eta + 3/2)$ (versions 1–7, without Lemma 5.5)	$<$ 63 (computed: 62.995)	
$c\,(4/(3\eta){+}1)$ (simpler h, Remark 8.3)	≤ 77.1217 (unrounded: 77.1212)	

(The rational data of the certificate were also re-derived in exact arithmetic by `mkdata.py`, with the masses rounded down to 2^{-32}; this gave $D \ge 3.11137905$ and $c \le 6.87179932$.) The certificate is valid for $h \ge 1/\omega = 2^{30}$. It is a computation about the fixed function g and does not depend on h.

4.5. **The decay theorem.**

Lemma 4.8 (pointwise energy inequality A_ρ)**.** *Let $w_\kappa = |E|^{1-\kappa}$ on Γ_h and $w_\kappa = 1$ at the atoms. For every configuration $z_1, \dots, z_h \in \Gamma_h \cup (-\mathcal{A})$ of pairwise distinct points (otherwise the Cauchy determinant vanishes), writing $s_i = z_i/h$,*

$$-\log\Big(\prod_{i<j}|z_i - z_j|^2 \prod_{a<b}(b-a)^2 \prod_{i,a}|z_i - a|^{-2} \prod w_\kappa(z_i)\Big) \ge h^2\Big[\tfrac{2}{h}\textstyle\sum_i g(s_i) - I_\rho(\sigma) + \tfrac{3}{2} - \eta_1(h)\Big],$$

with $\eta_1(h) = \frac{|\log\rho|}{h} + \frac{2\log(1+8h/3)}{h} + \varepsilon_2(h) + \frac{\log^+ C_E}{h} \to 0$, where $C_E = \sup_{\Gamma_h}|E|e^{2\pi|\operatorname{Im} z|} < \infty$ is independent of h and $\varepsilon_2(h) = O((\log h)/h)$.

Proof (this is the case $S = \{0, \dots, h-1\}$ of the proof of Theorem 7.1). The exponent is $2h\log h + h^2 H$, and the term $2h\log h \ge 0$ is dropped. We write $H = H_1 + H_2 + H_3 + H_4$. ($H_1$) The h points s_i are distinct, so by Lemma 4.3 and Corollary 4.7, $\frac{1}{h^2}\sum_{i\ne j} - \log|s_i - s_j| \ge \frac{1}{h^2}\sum_{i\ne j} k_\rho(s_i, s_j) \ge \frac{2}{h}\sum_i U^\sigma_\rho(s_i) - I_\rho(\sigma) - \frac{|\log\rho|}{h}$. ($H_2$) Since $\operatorname{Re} z_i \le -3/8$, $t \mapsto \log|s-t|$ is increasing on $[0,1]$ and the left Riemann sum for U^τ loses at most $\log(1+1/|s|)/h \le \log(1+8h/3)/h$; this gives $\frac{2}{h}\sum_i(-U^\tau(s_i)) - \frac{2\log(1+8h/3)}{h}$. ($H_3$) With $t_a = a/h$,

$\frac{1}{h^2}\sum_{a\neq b} -\log|t_a-t_b| = \frac{1}{h^2}[h(h-1)\log h - 2\sum_{n<h}\log n!] \geq \frac{3}{2} - \varepsilon_2(h)$, using $\log n! \leq (n+\frac{1}{2})\log n - n + 1$. (H_4) $-\log|E|^{1-\kappa} \geq (1-\kappa)(2\pi|\operatorname{Im} z| - \log^+ C_E)$, with C_E independent of h because E is 1-periodic and $x^* \equiv 5/8 \pmod 1$. Since $|\operatorname{Im} z| = h|\operatorname{Im} s|$, the sum of the field terms is $h^2 \cdot \frac{2}{h}\sum_i (1-\kappa)\pi|\operatorname{Im} s_i| - h\log^+ C_E$. Adding (H_1)–(H_4) gives the claim. □

Lemma 4.9 (finite-cell bound B_ρ)**.** *Let $h \geq 2^{30}$ and let $s_1, \dots, s_k$ be the scaled points of any admissible configuration of $k \leq h$ points of $\Gamma_h \cup (-\mathcal{A})$ (atoms pairwise distinct). Then*

$$\frac{1}{h}\sum_{i\leq k} g(s_i) \;\geq\; \frac{k}{h}\, m' + S_{\mathrm{cert}} - \frac{2}{h}\sum_{\ell} (m' - \gamma_\ell)^+.$$

Proof. Write $\frac{1}{h}\sum_i g(s_i) = \frac{k}{h}m' + \frac{1}{h}\sum_i (g(s_i) - m')$. *Contour points.* Write $x^*/h = -r_0 + \delta_h$, $\delta_h = (r_0 h - N_0 - \frac{3}{8})/h \in [-\frac{3}{8h}, \frac{5}{8h})$, so $|\delta_h| < 1/h \leq \omega$ and $s_i = -r_0 + \delta_h + u(-1 \pm 3i)$. For $u \leq u_{\max}$ the point s_i (or its conjugate; g is conjugation-symmetric) lies in a box of (R), and for $u > u_{\max}$ the tail bound applies; in either case $g(s_i) \geq m'$. *Atoms.* Atoms lie in $-(\mathbb{Z}_{\geq 0} + \frac{3}{4})/h \cup -\mathbb{Z}_{\geq 1}/h \subset [-r_0, 0]$ and are distinct. Assign each to one cell containing it (an atom on a shared endpoint is assigned once). The two progressions have spacing $1/h$, so a closed cell C_ℓ receives $N_\ell \leq 2h(b_\ell - a_\ell) + 2$ atoms. Since $g \geq \gamma_\ell$ on C_ℓ and $\min(\gamma_\ell - m', 0) \leq 0$, $\frac{1}{h}\sum_{\text{atoms}}(g - m') \geq \sum_\ell \frac{N_\ell}{h}\min(\gamma_\ell - m', 0) \geq S_{\mathrm{cert}} - \frac{2}{h}\sum_\ell (m' - \gamma_\ell)^+$. Here $N_\ell/h \leq 2(b_\ell - a_\ell) + 2/h$ and $2\sum_\ell (b_\ell - a_\ell)\min(\gamma_\ell - m', 0) = S_{\mathrm{cert}}$. □

Theorem 4.10 (decay; computer-assisted)**.** $\det \mathcal{G}_h \leq \exp(-(D^+ - o(1))h^2)$ *with $D^+ = D_{\mathrm{cert}} \geq 3.1113790$. Hence $|P_h(\lambda)| \leq \exp(-(D^+ - o(1))h^2)$.*

Proof. Split $d|m| = w_\kappa\, d\Omega_\kappa$ with $\Omega_\kappa = \sum_{\mathcal{A}} \delta + \frac{1}{2\pi}|E|^\kappa |dz|$, of total mass $\leq 2h + c_\kappa$, $c_\kappa = \frac{1}{2\pi}\int_{\Gamma_0} |E|^\kappa |dz| < \infty$ independent of h by periodicity. By Lemmas 4.8 and 4.9 (with $k = h$), the integrand of Proposition 4.2 against $\prod w_\kappa$ is at most $\exp(-h^2[D_{\mathrm{cert}} - \eta_1 - 2\eta_2])$ pointwise, where $\eta_2(h) = \frac{2}{h}\sum_\ell (m' - \gamma_\ell)^+$ (the cell number is fixed, so this is $O(1/h)$). Therefore

$$\det \mathcal{G}_h \leq \frac{1}{h!}(2h + c_\kappa)^h \exp(-h^2[D_{\mathrm{cert}} - \eta_1 - 2\eta_2]),$$

and $\log(2h + c_\kappa)/h \to 0$. Finally $|P_h(\lambda)| \leq 16^{-h} \det \mathcal{G}_h$ by Proposition 4.2. □

Remark 4.11. The error terms η_1, η_2 are $O((\log h)/h)$ with moderate explicit constants; no explicit h_0 is claimed (the certificate itself requires $h \geq 2^{30}$). The exponent $D^+ \geq 3.1113790$ is certified; numerics suggest that the true limit is $\frac{9}{2}\log 2 = 3.11916\ldots$ (not used). The margin for the deductions below is

$$\eta := D^+ - \Lambda \;\geq\; 3.1113790 - 2.9809524 = 0.1304266,$$

and the argument of §8 uses the decay theorem only through this margin.

Remark 4.12 (cross-check by circle smoothing). An earlier version of the energy step replaced each point by a uniform measure on a circle of radius $1/h$ and used a certificate measure σ supported on 448 disjoint circles together with the inequality $I(\nu-\sigma) \geq 0$ for signed measures of mass zero [13, Lemma I.1.8] and a Lipschitz bound for U^σ ($L_\sigma \leq 1522.04$). That route, with an independent interval-arithmetic certificate, gives $D^+ \geq 3.1072419$, $m' \geq 1.9681060$, $S \geq -0.1101022$, $g_{\min} \geq -1.4643531$, $c \leq 6.8650$ and $\mu(\lambda) \leq 79.35$ (the value 79.3439). We keep it only as an independent cross-check of the decay exponent: the two certificates use different kernels, different measures σ and a different ray angle ($3\pi/5$), and agree on $D^+ \approx 3.11$. The k_ρ route gives the larger D^+ because its certificate measure was re-optimised.

5. ℓ-adic nonvanishing

Let $\ell \equiv 3 \pmod 4$ be a prime with $\ell \geq 7$, $K = (\ell+1)/4$ (an integer ≥ 2), and let $K+1 \leq h \leq \ell-1$. Let $\mathbb{Z}_{(\ell)}$ be the localization of $\mathbb{Z}$ at ℓ. Call the indices $a < K$ *low* and $a \geq K$ *high*.

Lemma 5.1 (ℓ-singular structure)**.** *Among the quantities k, $4k-1$, $b-a$, 4 occurring in the entries of M_0 ($0 \leq a < b < h$, $1 \leq k \leq h-1$) only $4k-1$ can be divisible by ℓ, and this occurs only for $k = K$ (with $4K-1 = \ell$). Consequently every high diagonal entry of M_0 contains the term $+\ell^{-2}$, every low–high entry contains $-1/(16K\ell(b-a))$, and all other entries lie in $\mathbb{Z}_{(\ell)}$.*

Proof. $k, b-a < \ell$. If $\ell \mid 4k-1$ with $4k-1 \leq 4h-5 < 4\ell$ then $4k-1 \in \{\ell, 2\ell, 3\ell\}$; 2ℓ is even and $3\ell \equiv 1 \pmod 4$, so $4k-1 = \ell$, i.e. $k = K$. The sum $\sum_{a<k\leq b}$ contains $k = K$ iff $a < K \leq b$ (low–high); the diagonal sum $\sum_{k\leq a}$ contains it iff $a \geq K$. □

Lemma 5.2. *Let $D = \operatorname{diag}(I_K, \ell I_{h-K})$. Then $D(XI+M_0)D \in M_h(\mathbb{Z}_{(\ell)}[X])$ and, modulo ℓ,*

$$D(XI+M_0)D \equiv \begin{pmatrix} XI_K + A & \hat{B} \\ \hat{B}^\top & I \end{pmatrix}, \qquad A = M_0|_{\text{low}} \bmod \ell, \quad \hat{B}_{ab} \equiv -\frac{1}{4(b-a)} \ (a < K \leq b),$$

and hence $\ell^{2(h-K)} P_h(X) \equiv \det(XI_K + A - \hat{B}\hat{B}^\top) \pmod \ell$. The right-hand side is monic of degree K; in particular $v_\ell(\operatorname{den} P_h) = 2(h-K)$.

Proof. By Lemma 5.1: the high diagonal entries times ℓ^2 are $\equiv 1$ (the term ℓ^{-2}), the low–high entries times ℓ are $\equiv -1/(16K(b-a)) \equiv -1/(4(b-a))$ since $16K = 4\ell+4$, the high–high off-diagonal entries times ℓ^2 and the term $X\ell^2$ are $\equiv 0$, and the low block is unchanged. Take the Schur complement of the identity block; $\det D = \ell^{h-K}$. □

Lemma 5.3 (key identity)**.** *Let $s = \ell - h \in [1, \ell-K-1]$, $m = \ell - K$, and $C_s = [1/(a+j)]_{0\leq a<K,\ 1\leq j\leq s}$ (entries in $\mathbb{F}_\ell$; note $a+j \leq K-1+s < \ell$). Put $c = -H_m^{(2)} \in \mathbb{F}_\ell$ and $Y = 16X + c$. Then, in $\mathbb{F}_\ell[X]$,*

$$16^K \ell^{2(h-K)} P_h(X) \equiv \det(Y\, I_K + C_s C_s^\top).$$

Proof. Modulo ℓ, $4K = \ell + 1 \equiv 1$, so $\frac{4}{4k-1} = \frac{1}{k-1/4} \equiv \frac{1}{k-K} \equiv \frac{1}{k+m}$ and thus $-\frac{1}{k(4k-1)} \equiv \frac{1}{k} - \frac{1}{k+m}$ and $\frac{16}{(4k-1)^2} \equiv \frac{1}{(k+m)^2}$. For $a < b < K$ (so $b + m < \ell$) summing over $a < k \le b$ and using $\frac{1}{(b-a)}\big(\frac{1}{a+j} - \frac{1}{b+j}\big) = \frac{1}{(a+j)(b+j)}$,

$$16A_{ab} \equiv \frac{1}{b-a}\sum_{k=a+1}^{b}\Big(\frac{1}{k}-\frac{1}{k+m}\Big) = \sum_{j=1}^{m}\frac{1}{(a+j)(b+j)}, \qquad 16A_{aa} \equiv \sum_{k\le a}\Big(\frac{1}{(k+m)^2}-\frac{1}{k^2}\Big) = \sum_{j=1}^{m}\frac{1}{(a+j)^2}$$

Thus $16A \equiv cI + C_m C_m^\top$. Next, $16(\hat{B}\hat{B}^\top)_{aa'} \equiv \sum_{b=K}^{h-1}\frac{1}{(b-a)(b-a')}$. Substituting $j = \ell - b$ (which runs over $s+1,\dots,m$ as b runs over $h-1,\dots,K$) we have $a + j \equiv -(b-a)$, so this equals $\sum_{j=s+1}^{m}\frac{1}{(a+j)(a'+j)}$. Subtracting, $16(A - \hat{B}\hat{B}^\top) \equiv cI + C_s C_s^\top$, and Lemma 5.2 gives the claim, since $\det(XI + (cI + C_sC_s^\top)/16) = 16^{-K}\det(YI + C_sC_s^\top)$. □

(This was checked by exact characteristic-polynomial reduction for $h = 12, 20$ and every $\ell \equiv 3 \pmod 4$ in $(h, 4h]$.)

Write $\Xi_s = C_s^\top C_s$, i.e. $(\Xi_s)_{ij} = \sum_{a=0}^{K-1}\frac{1}{(a+i)(a+j)}$. By Sylvester's determinant identity, for $s \le K$,

$$\det(YI_K + C_sC_s^\top) = Y^{K-s}\det(YI_s + \Xi_s). \tag{5.1}$$

Theorem 5.4 (ℓ-adic nonvanishing)**.** *Let $\ell \equiv 3 \pmod 4$ be prime, $\ell \ge 7$, $K = (\ell+1)/4$, and let $r \in \mathbb{Q}$ with $\ell \nmid \operatorname{den} r$. Then at least one of*

$$h_1 = \ell - 1, \qquad h_2 = \ell - 2, \qquad h_3 = \ell - K = \tfrac{3\ell-1}{4}$$

satisfies $v_\ell(P_h(r)) = -2(h-K)$; in particular $P_h(r) \neq 0$.

Proof. All three h_i satisfy $K+1 \le h_i \le \ell-1$ (for h_3: $\frac{3\ell-1}{4} \ge \frac{\ell+5}{4} \iff \ell \ge 3$), and have $s = 1, 2, K$ respectively, all $\le K$ since $K \ge 2$. By Lemmas 5.2 and 5.3, $\ell^{2(h-K)}P_h(r)$ is an ℓ-adic unit iff the polynomial (5.1) does not vanish at $Y = y := 16r + c \in \mathbb{F}_\ell$. Suppose it vanishes at y for all three values of s.

- $s = K$: at $Y = 0$ the value of (5.1) is $\det(\Xi_K) = \det(C_K)^2$, where $C_K = [1/(a+j)]$ is a Cauchy matrix with $\det C_K = \prod_{a<a'}(a'-a)\prod_{j<j'}(j'-j)/\prod_{a,j}(a+j)$, a ratio of products of integers in $[1, 2K-1]$, and $2K-1 < \ell$. So $\det C_K \neq 0$ in $\mathbb{F}_\ell$, the polynomial does not vanish at 0, and hence $y \neq 0$.
- $s = 1$: the value is $y^{K-1}(y + \Xi_{11})$ with $\Xi_{11} = \sum_{a<K}(a+1)^{-2} = H_K^{(2)}$. As $y \neq 0$, $y = -\Xi_{11} \neq 0$.
- $s = 2$: the value is $y^{K-2}\det\begin{pmatrix} 0 & \Xi_{12} \\ \Xi_{12} & \Xi_{22}-\Xi_{11}\end{pmatrix} = -y^{K-2}\Xi_{12}^2$, and $\Xi_{12} = \sum_{a<K}\big(\frac{1}{a+1} - \frac{1}{a+2}\big) = \frac{K}{K+1} \neq 0$ in $\mathbb{F}_\ell$ since $1 \le K < K+1 < \ell$. This is nonzero, a contradiction.

(If $\ell = 7$ then $h_2 = h_3$, $K = 2$, and the argument is unchanged.) Hence for some i the reduction is nonzero, so $v_\ell(P_{h_i}(r)) = -2(h_i - K)$. □

(Exact characteristic polynomials for $\ell = 7, 11, 19, 23, 31, 43$ confirm that the three reductions have gcd = 1 and the stated valuations.)

Lemma 5.5 (primes dividing the denominator)**.** *Let $\ell \equiv 3 \pmod 4$ be prime, $\ell \geq 7$, $K+1 \leq h \leq \ell-1$, and $r = p/q$ with $e = v_\ell(q) \geq 1$; put $u = \ell^e r \in \mathbb{Z}_{(\ell)}^\times$. If $e = 1$, or $e \geq 3$, or $e = 2$ and $u \not\equiv -1 \pmod \ell$, then $P_h(r) \neq 0$ for every admissible h.*

Proof. Let $N = D(rI+M_0)D$. By Lemmas 5.1, 5.2, its blocks are $rI+M_{0,LL}$, $\ell M_{0,LH} \equiv \hat{B}$, and $\ell^2 rI + J$ with $J \equiv I$, all entries other than the r's being ℓ-integral. Multiply the low rows by ℓ^e (and, if $e \geq 3$, the high rows by ℓ^{e-2}). Modulo ℓ the matrix becomes $\begin{pmatrix} uI & 0 \\ \hat{B}^\top & I \end{pmatrix}$ for $e = 1$ (determinant u^K), $\begin{pmatrix} uI & 0 \\ \hat{B}^\top & (u+1)I \end{pmatrix}$ for $e = 2$ (determinant $u^K(u+1)^{h-K}$), uI_h for $e \geq 3$ (determinant u^h). These are units in the stated cases. □

(Checked on 1044 exact cases; the exception $e = 2$, $u \equiv -1$ does occur.)

5.1. **Proof of Theorem 1.1.** Suppose $\lambda = a/b$ in lowest terms. Choose a prime $\ell \equiv 3 \pmod 4$, $\ell \geq 7$, $\ell \nmid b$, as large as we please (there are infinitely many primes $\equiv 3 \pmod 4$). Theorem 5.4 gives $h \geq (3\ell-1)/4$ with $P_h(\lambda) \neq 0$. By Lemma 2.4, Theorem 3.1 and Theorem 4.10,

$$1 \leq b^h \operatorname{den}(P_h)\,|P_h(\lambda)| \leq \exp\bigl(h \log b - (D^+ - \Lambda - o(1))h^2\bigr),$$

which is false for large h because $D^+ - \Lambda > 0.130$. □

6. A determinant evaluation and the choice of h

Theorem 5.4 uses three values of h whose ratio $h_{\max}/h_{\min}$ is $4/3$ (the factor $4/3$ in Remark 8.3). In this section we show that already two consecutive values

$$h \in \{H,\ H+1\}, \qquad K+1 \leq H \leq \ell-K-1 \qquad (\text{that is, } s = \ell-h \in \{s_0, s_0+1\},\ s_0 = \ell-H-1 \in [K, \ell-K$$

suffice, for *any* such H; the case $s_0 = K$, $\{H, H+1\} = \{\ell-K-1, \ell-K\}$, was the only one considered in versions 1–7. The ratio of the two values is $1 + O(1/H)$, and since H may be prescribed in advance and ℓ chosen afterwards among the primes $\equiv 3 \pmod 4$ in $[\frac{4H+5}{3}, 4H-5]$ (§8), the factor h/ℓ no longer matters. The proof rests on a statement about two consecutive Gram-type matrices over $\mathbb{F}_\ell$ (Lemma CC below), which in turn reduces to the non-vanishing modulo ℓ of an explicit Krylov determinant; the latter is evaluated in closed form (Theorem A). By (5.1) the polynomial $\det(YI_K + C_sC_s^\top)$ has the factor Y^{K-s} for $s < K$, so the value $y = 0$ cannot be excluded for any $h > \ell-K$; this is why the smallest usable s is K (that is, the largest usable h is $\ell-K$).

Throughout this section $\ell \equiv 3 \pmod 4$ is a prime, $\ell \geq 7$, $K = (\ell+1)/4$, and for $s \geq 0$ with $K+s < \ell$ we use, over $\mathbb{F}_\ell$,

$$C_s = \Bigl[\frac{1}{a+j}\Bigr]_{0\leq a<K,\ 1\leq j\leq s}, \qquad N_s = C_sC_s^\top \in M_K(\mathbb{F}_\ell), \qquad w_j = \Bigl(\frac{1}{a+j}\Bigr)_{0\leq a<K},$$

so that $N_{s+1} = N_s + w_{s+1}w_{s+1}^\top$. All denominators are ℓ-units because $a+j \le K-1+s+1 < \ell$.

6.1. **Statement.**

Lemma 6.1 (Lemma CC)**.** *Let $s \ge K$ with $K+s<\ell$. Then N_s and N_{s+1} have no common eigenvalue in $\overline{\mathbb{F}}_\ell$ (the eigenvalue 0 included). Equivalently, the polynomials $\det(YI_K+N_s)$ and $\det(YI_K+N_{s+1})$ have no common root in $\overline{\mathbb{F}}_\ell$. This holds for every s with $K \le s \le \ell-K-1$; in particular for $s=K$ (note $2K<\ell$).*

The proof (§6.6) is the combination of the Krylov reduction of §6.2 with the evaluation of Theorem A below.

Theorem 6.2 (Choice of h)**.** *Let $\ell \equiv 3 \pmod 4$ be prime, $\ell \ge 7$, $K=(\ell+1)/4$, let $r \in \mathbb{Q}$ with $\ell \nmid \operatorname{den} r$, and let H be an integer with*

$$K+1 \le H \le \ell-K-1 \qquad \Big(\text{equivalently } \tfrac{4H+5}{3} \le \ell \le 4H-5\Big).$$

Then at least one of $h \in \{H, H+1\}$ satisfies $v_\ell(P_h(r)) = -2(h-K)$; in particular $P_h(r) \neq 0$. For $H=\ell-K-1$ this is the pair $h_1=\ell-K=\frac{3\ell-1}{4}$, $h_2=\ell-K-1=\frac{3\ell-5}{4}$ of versions 1–7.

6.2. **The Krylov reduction.**

Lemma 6.3. *Let F be a field of characteristic $\neq 2$, $N \in M_K(F)$ symmetric, $w \in F^K$. If $z \in F$ is a common root of $\det(z-N)$ and $\det(z-N-ww^\top)$, then there is $x \neq 0$ with $Nx=zx$ and $w^\top x=0$.*

Proof. The matrix determinant lemma gives $\det(z-N-ww^\top) = \det(z-N) - w^\top \operatorname{adj}(z-N)\,w$, so $w^\top \operatorname{adj}(z-N)\,w = 0$. Since $\det(z-N)=0$, $\operatorname{rank}(z-N) \le K-1$. If $\operatorname{rank}(z-N) \le K-2$, then $\ker(z-N)$ has dimension ≥ 2 and so contains some $x \neq 0$ with $w^\top x = 0$. If $\operatorname{rank}(z-N) = K-1$, then $\operatorname{adj}(z-N)$ is symmetric of rank one with columns in $\ker(z-N) = \langle x \rangle$, so $\operatorname{adj}(z-N) = \gamma\, xx^\top$ with $\gamma \neq 0$, and $\gamma(w^\top x)^2 = 0$ forces $w^\top x = 0$. □

(This covers isotropic eigenvectors, $x^\top x = 0$, which genuinely occur modulo ℓ.)

Lemma 6.4. *Let $N \in M_K(F)$ be symmetric and $w \in F^K$. If some $x \neq 0$ satisfies $Nx=zx$ and $w^\top x=0$, then $\det[w, Nw, \dots, N^{K-1}w] = 0$.*

Proof. $x^\top N^k w = (N^k x)^\top w = z^k x^\top w = 0$ for every k, so x lies in the left kernel of the Krylov matrix. □

Corollary 6.5. *Let $s \ge 1$, $K+s<\ell$, and $u(K,s) := \det[w_{s+1}, N_s w_{s+1}, \dots, N_s^{K-1} w_{s+1}] \in \mathbb{F}_\ell$. If $u(K,s) \neq 0$, then N_s and N_{s+1} have no common eigenvalue in $\overline{\mathbb{F}}_\ell$.*

Proof. Apply Lemmas 6.3 and 6.4 over $F = \overline{\mathbb{F}}_\ell$ with $N = N_s$ and $w = w_{s+1}$. □

6.3. **Theorem A.** For real $L, \rho \notin \{0, -1, -2, \dots\}$ and $n \geq 1$ let $c_y = (1/(a+y))_{a=0}^{n-1} \in \mathbb{R}^n$ and

$$N(L,\rho) := \sum_{k\geq 0} \left[c_{L+k} c_{L+k}^{\top} - c_{\rho+k} c_{\rho+k}^{\top} \right] \in M_n(\mathbb{R}),$$

which converges absolutely (each term is $O(k^{-3})$). We write $\theta = L - 1$, $s = \rho - L$ (so $\rho = \theta + s + 1$), and

$$u_n(\theta, s) := \det\left[w, Nw, \dots, N^{n-1} w \right], \qquad N = N(\theta{+}1, \theta{+}s{+}1),\ w = c_{\theta+s+1}.$$

Lemma 6.6. *(a) $N(L,\rho)$ is real symmetric, meromorphic in (L,ρ), with poles only where $L + k + a = 0$ or $\rho + k + a = 0$ for some $0 \leq a < n$, $k \geq 0$; near any point only finitely many terms are singular. (b) If $s = \rho - L \in \mathbb{Z}_{\geq 0}$, then $N(L,\rho) = \sum_{y=L}^{\rho-1} c_y c_y^{\top}$ (a sum of s terms); if $L - \rho = c \in \mathbb{Z}_{\geq 0}$, then $N(L,\rho) = -\sum_{y=\rho}^{L-1} c_y c_y^{\top}$. (c) $N(L,\rho+1) = N(L,\rho) + c_\rho c_\rho^{\top}$ and $N(L,\rho) = c_L c_L^{\top} + N(L+1,\rho)$. (d) Off-diagonal entries are rational functions of (L,ρ), and $N(L,\rho) = N^{\mathrm{rat}}(L,\rho) + (\psi'(L) - \psi'(\rho))\, I$ with N^{rat} rational. (e) u_n is a rational function of (θ, s). (f) For integer $s \geq 0$ and $\theta = 0$: $N(1, s{+}1) = N_s$ (the matrix of §6.2, over $\mathbb{Q}$) and $w = c_{s+1} = w_{s+1}$; thus $u_n(0,s) = u(n,s)$ with $K = n$.*

Proof. (a)–(c) are immediate from the definition (telescoping). For (d), $\sum_{k\geq 0} \frac{1}{(a+y+k)(b+y+k)} = \frac{1}{b-a} \sum_{j=0}^{b-a-1} \frac{1}{a+y+j}$ for $a < b$, and $\sum_{k\geq 0} (a+y+k)^{-2} = \psi'(y) - \sum_{j<a} (y+j)^{-2}$, i.e. $\psi'(y)$ minus a rational function. For (e), the Krylov determinant is invariant under $N \mapsto N + \tau I$ (the change of basis from $N^k w$ to $(N+\tau)^k w$ is unitriangular), so $\psi'(L) - \psi'(\rho)$ can be dropped, and u_n is a polynomial in the entries of N^{rat} and w. (f) is (b). □

Theorem 6.7 (Theorem A)**.** *Put $T(m) = m(m+1)/2$ and*

$$A_n(s) = \prod_{c=1}^{n-1} \left[(s+c)(s+1-c) \right]^{T(n-c)}, \qquad B_n(\theta) = \prod_{i=1}^{n-1} (\theta + i)^{2i(n-i)},$$

$$C_n(t) = \prod_{c=1}^{n} (t+c)^{\beta_c}, \quad \beta_c = 2c(n+1-c) - n, \qquad \kappa_n^{+} = \prod_{m=1}^{n-1} m^{2(n-m)(n-2m)}.$$

Then there is a constant $\kappa_n \in \mathbb{R}$ with $|\kappa_n| = \kappa_n^{+}$ such that, as rational functions of (θ, s),

$$u_n(\theta, s) = \kappa_n \, \frac{A_n(s)}{B_n(\theta)\, C_n(\theta + s)}.$$

Remark 6.8. *Only $|u_n|$ is needed.* Numerically $\kappa_n = (-1)^{\lfloor n/2 \rfloor} \kappa_n^{+}$; the sign is not proved here and is not used. The formula was also verified exactly (rational arithmetic) for every $n \leq 12$, at random rational (θ, s) including non-integer and negative s, and with the Cauchy Gram matrices N_s themselves (§18). For $n = 3$, $\theta = 0$ it gives $u(3,s) = -s^3(s-1)/(64(s+2)^4(s+3)^3)$.

6.4. **Tools.**

Lemma 6.9 (resultant identity)**.** *Let N be real symmetric $n\times n$ with eigenvalues $\lambda_1,\dots,\lambda_n$, let $w\in\mathbb{R}^n$, and let $N'=N+ww^\top$ have eigenvalues $\mu_1,\dots,\mu_n$. Then*

$$\det[w,Nw,\dots,N^{n-1}w]^2=\pm\prod_{i,k}(\lambda_i-\mu_k).$$

Proof. Write $N=U\,\mathrm{diag}(\lambda)U^\top$ with U orthogonal and $\tilde w=U^\top w$. Then $\det[w,Nw,\dots]=\det U\cdot\prod_k\tilde w_k\cdot\prod_{i<j}(\lambda_j-\lambda_i)$. On the other hand $\chi_{N'}(z)=\chi_N(z)(1-\sum_k\tilde w_k^2/(z-\lambda_k))$, so $\chi_{N'}(\lambda_k)=-\tilde w_k^2\prod_{j\neq k}(\lambda_k-\lambda_j)$, and $\prod_{i,k}(\lambda_i-\mu_k)=\pm\prod_k\chi_{N'}(\lambda_k)=\pm\prod_k\tilde w_k^2\prod_{i<j}(\lambda_i-\lambda_j)^2$. (This is the classical identity for the resultant of the characteristic polynomials of N and N'.) □

We call this identity (R). It exhibits $|u_n|^2$ as a product over pairs (i,k) of the differences $|\lambda_i-\mu_k|$.

Lemma 6.10 (cluster lemma)**.** *Let $M(\varepsilon)=\varepsilon^{-2}\sigma J+\varepsilon^{-1}Y+Z(\varepsilon)$ be real symmetric $n\times n$ matrices, $\varepsilon\to0$, where $\sigma=\pm1$, $J=I_r\oplus0_{n-r}$, Y is real symmetric with both diagonal blocks (the $r\times r$ and the $(n-r)\times(n-r)$ block) equal to zero, and $Z(\varepsilon)$ is bounded. Then $M(\varepsilon)$ has r eigenvalues $\sigma\varepsilon^{-2}+O(1)$ and $n-r$ eigenvalues $O(1)$.*

Proof. Write $\varepsilon^2M=\sigma J+\varepsilon Y+O(\varepsilon^2)$. Let $S=\begin{pmatrix}0&X\\-X^\top&0\end{pmatrix}$ with $X=-\sigma Y_{12}$ (Y_{12} the off-diagonal block of Y), and $Q=\exp(\varepsilon S)$, which is orthogonal. Since $[J,S]=\begin{pmatrix}0&X\\X^\top&0\end{pmatrix}$,

$$Q^\top(\varepsilon^2M)Q=\sigma J+\varepsilon\big(Y+\sigma[J,S]\big)+O(\varepsilon^2)=\sigma J+O(\varepsilon^2),$$

because the diagonal blocks of Y vanish and the off-diagonal block of $Y+\sigma[J,S]$ is $Y_{12}+\sigma X=0$. Weyl's inequality $|\lambda_i(X_1+X_2)-\lambda_i(X_1)|\le\|X_2\|$ shows that ε^2M has r eigenvalues $\sigma+O(\varepsilon^2)$ and $n-r$ eigenvalues $O(\varepsilon^2)$. □

6.5. **Proof of Theorem A.** Throughout, $\theta\in\mathbb{R}\setminus\mathbb{Z}$ is fixed (generic) unless said otherwise, so that the points $s=-\theta-c$ (poles) and the integers $s=j$ (zeros) below are distinct. We write λ_i, μ_k for the eigenvalues of N and $N'=N+ww^\top$ and use (R) throughout.

Step 1: zeros in s. *At $s=j$, $0\le j\le n-2$.* By Lemma 6.6(b), N is a sum of j rank-one terms, so 0 is an eigenvalue of N of multiplicity $\ge n-j$ and of N' of multiplicity $\ge n-j-1$. Near $s=j$ the matrix N is analytic in s (for $\theta\notin\mathbb{Z}$ no term is singular), so by Weyl's inequality these eigenvalues are $O(|s-j|)$. This gives at least $(n-j)(n-j-1)$ factors of size $O(|s-j|)$ in (R), all other factors being bounded; hence u_n vanishes at $s=j$ to order $\ge T(n-1-j)$. *At $s=-c$, $1\le c\le n-1$.* By Lemma 6.6(b), $N=-\sum_{y=\rho}^{L-1}c_yc_y^\top$ has rank $\le c$, and $w=c_\rho$ is one of these terms, so N' has rank $\le c-1$. The same argument gives $(n-c)(n-c+1)$ small

factors and order $\geq T(n-c)$. Since $A_n(s)$ vanishes exactly at these points to exactly these orders ($s+1-c=0$ at $s=j=c-1$, with exponent $T(n-c)=T(n-1-j)$), u_n/A_n has no pole in s at the zeros of A_n.

Step 2: poles in s. For fixed θ the entries of N and w are singular only at $\rho = 1-c$, i.e. $s=-\theta-c$, $c\geq 1$. Near $\rho = 1-c+\varepsilon$, the singular terms are $c_{\rho+k}c_{\rho+k}^\top$ with $k=c-1-a$ for a in the set $\{0,\dots,\min(c,n)-1\}$ (call its size $c^*=\min(c,n)$); entry a of $c_{\rho+k}$ has a simple pole, so each singular term contributes $-\varepsilon^{-2}e_ae_a^\top$ plus an ε^{-1} term plus a bounded part. Thus

$$N = -\varepsilon^{-2}J_{c^*} + \varepsilon^{-1}Y + O(1).$$

The ε^{-1}-coefficient Y has vanishing diagonal blocks: for $p\neq q$ in the singular set, the contributions $1/(q-p)$ (from the term singular at p) and $1/(p-q)$ (from the term singular at q) cancel; the diagonal entries have no ε^{-1} term since $(\varepsilon+m)^{-2}$ is regular for $m\neq 0$; and entries with both indices outside the singular set are regular. *Case $c>n$.* Then $J_{c^*}=I$, w is regular, and N, N' are $-\varepsilon^{-2}I+\varepsilon^{-1}Y+O(1)$ with $Y=0$ (only the first block exists); all eigenvalues of both are $-\varepsilon^{-2}+O(1)$, all differences are $O(1)$, so u_n is bounded: no pole. *Case $c\leq n$.* Here $w=c_\rho$ has the pole in coordinate $a=c-1$, and $ww^\top$ is exactly the term $k=0$ of the ρ-series, so $N'=N+ww^\top$ has the same form with J_{c-1} (and the same vanishing of the diagonal blocks of its ε^{-1} coefficient). By Lemma 6.10 (with $\sigma=-1$), N has c "big" eigenvalues $-\varepsilon^{-2}+O(1)$ and $n-c$ "small" ones $O(1)$, and N' has $c-1$ big and $n-c+1$ small ones. In (R), big–big and small–small differences are $O(1)$, and each of the

$$c(n-c+1)+(c-1)(n-c) = 2c(n+1-c)-n = \beta_c$$

big–small differences has size ε^{-2}. Hence $u_n = O(\varepsilon^{-\beta_c})$.

Step 3: poles in θ at $\theta=-i$, $1\leq i\leq n-1$, s generic. Put $L=1-i+\varepsilon$. The singular terms are $c_{L+k}c_{L+k}^\top$, $k=i-1-a$, $a\in\{0,\dots,i-1\}$, and exactly as in Step 2, but with sign $+$, $N=+\varepsilon^{-2}J_i+\varepsilon^{-1}Y+O(1)$ with Y having vanishing diagonal blocks. Now $w=c_\rho$ is regular, so N' has the same singular part. Both have i big and $n-i$ small eigenvalues, giving $2i(n-i)$ big–small differences, so $u_n=O(\varepsilon^{-2i(n-i)})$.

Step 4: identification in s. As $\rho\to\infty$ with L fixed, $N=R_\infty(L)+O(1/\rho)$ with $R_\infty(L)=\sum_{k\geq 0}c_{L+k}c_{L+k}^\top$, analytic in $1/\rho$, and $\rho w=\mathbf{1}+O(1/\rho)$ with $\mathbf{1}$ the all-ones vector. Hence $\rho^n u_n\to\det[\mathbf{1},R_\infty\mathbf{1},\dots,R_\infty^{n-1}\mathbf{1}]$, finite, and $u_n=O(s^{-n})$. Consider $F(s):=u_nC_n(\theta+s)/A_n(s)$ as a rational function of s. By Steps 1–2, F has no poles (the poles of u_n are at $s=-\theta-c$, $c\leq n$, of order $\leq\beta_c$, where $A_n\neq 0$; the zeros of A_n are integers, where u_n vanishes to at least the order of A_n and $C_n(\theta+s)\neq 0,\infty$), so F is a polynomial in s. Both A_n and C_n are monic and

$$\deg C_n - \deg A_n = \left[\tfrac{n(n+1)(n+2)}{3}-n^2\right]-\tfrac{(n-1)n(n+1)}{3} = n,$$

so $F(s) = O(s^{-n}s^n) = O(1)$, and F is a constant $K_n(\theta)$:

$$u_n(\theta, s) = K_n(\theta)\,\frac{A_n(s)}{C_n(\theta+s)}. \tag{6.1}$$

Moreover $K_n(\theta) = \lim_{s\to\infty} s^n u_n = \det[\mathbf{1}, R_\infty\mathbf{1}, \dots]$ with $L = \theta + 1$.

Step 5: identification in θ. By Lemma 6.6(d), K_n is a rational function of θ (the scalar $\psi'(L)$ in R_∞ drops out of the Krylov determinant), and its entries $\sum_{k\ge0} c_{L+k}c_{L+k}^\top - \psi'(L)I$ have poles only where $L + j = 0$ for some $0 \le j \le n-2$, that is $\theta = -i$, $1 \le i \le n-1$. Comparing (6.1) with Step 3 (at $\theta = -i$ and generic s, $A_n(s)/C_n(\theta+s)$ is finite and nonzero), the pole of K_n at $-i$ has order $\le 2i(n-i)$. Hence K_nB_n is a polynomial in θ. Now let $L \to \infty$ with ρ fixed and generic. Then $N(L,\rho) \to -\sum_{k\ge0} c_{\rho+k}c_{\rho+k}^\top$ (the L-series is $O(1/L)$) and $w = c_\rho$ is fixed, so u_n stays bounded. In (6.1), $s = \rho - L$ and $C_n(\theta+s) = C_n(\rho-1)$ is a fixed nonzero number, while $A_n(\rho - L)$ is of exact order $L^{\deg A_n}$. Hence $K_n(\theta) = O(\theta^{-\deg A_n})$. Since $\deg B_n = \sum_i 2i(n-i) = \frac{(n-1)n(n+1)}{3} = \deg A_n$, the polynomial K_nB_n is bounded, hence a constant κ_n. This proves the shape of Theorem A:

$$u_n = \kappa_n \frac{A_n(s)}{B_n(\theta)\,C_n(\theta+s)}. \tag{6.2}$$

Step 6: the constant, by recursion in n.

Lemma 6.11 (recursion)**.** *Fix non-integer s. As $\theta = -1+\varepsilon \to -1$,*

$$\varepsilon^{2(n-1)}\,|u_n(-1+\varepsilon, s)| \;\longrightarrow\; \frac{1}{|s|}\,|u_{n-1}(1,\, s-1)| \qquad (n \ge 2),$$

where on the right u_{n-1} is the $(n-1)$-dimensional function evaluated at $\theta' = 1$, $s' = s-1$.

Proof. Here $L = \varepsilon$ and $\rho = s + \varepsilon$. We have $c_L = \varepsilon^{-1}p$ with $p = e_0 + O(\varepsilon)$, and by Lemma 6.6(c) $N = \varepsilon^{-2}pp^\top + A(\varepsilon)$ with $A(\varepsilon) = N(1+\varepsilon, \rho)$ analytic at $\varepsilon = 0$. Put $N' = N + ww^\top = \varepsilon^{-2}pp^\top + A'(\varepsilon)$, $A' = A + ww^\top$; $w = c_\rho$ has first coordinate $w_0 = 1/\rho \to 1/s$. *Big eigenvalues.* The largest eigenvalues are $\lambda_{\text{big}} = \varepsilon^{-2}|p|^2 + \hat p^\top A\hat p + O(\varepsilon^2)$ and $\mu_{\text{big}} = \varepsilon^{-2}|p|^2 + \hat p^\top A'\hat p + O(\varepsilon^2)$ with $\hat p = p/|p| \to e_0$; so $\mu_{\text{big}} - \lambda_{\text{big}} \to w_0^2 = 1/s^2$. *Small eigenvalues.* $\varepsilon^2\chi_N(z) = \varepsilon^2\det(z-A) - p^\top \operatorname{adj}(z-A)p \to -\det(z - A(0)[1{:}])$, where $A(0)[1{:}]$ is the compression of $A(0) = N(1,s)$ to the coordinates $1,\dots,n-1$; likewise for N' with $A'(0)[1{:}] = A(0)[1{:}] + \tilde w\tilde w^\top$, $\tilde w = (w_a)_{a\ge1}$. Hence the other $n-1$ eigenvalues of N (resp. N') converge to those of $A(0)[1{:}]$ (resp. $A'(0)[1{:}]$). Re-indexing $a' = a-1$, the entries of $A(0)[1{:}]$ are $\sum_k\big[\frac{1}{(a'+2+k)(b'+2+k)} - \frac{1}{(a'+s+1+k)(b'+s+1+k)}\big] = N(2, s+1)_{a'b'}$ and $\tilde w = c_{s+1}$, that is, the $(n-1)$-dimensional problem with $L' = 2$, $\rho' = s+1$, i.e. $\theta' = 1$, $s' = s-1$. Now apply (R). The big–big difference tends to $1/s^2$; each of the $2(n-1)$ big–small differences is $\pm\varepsilon^{-2}(1+o(1))$; the small–small differences tend to

the corresponding product for the $(n-1)$-dimensional problem, which is $\pm u_{n-1}(1, s-1)^2$ by (R). Thus $|u_n|^2 \sim s^{-2}\varepsilon^{-4(n-1)} u_{n-1}(1, s-1)^2$. □

Lemma 6.12 (exponent bookkeeping)**.** *$C_n(s-1)\,A_{n-1}(s-1) = s\,C_{n-1}(s)\,A_n(s)$ as polynomials in s.*

Proof. Compare the exponent of $(s+d)$ on both sides ($T(m) = 0$ for $m \le 0$): for $1 \le d \le n-3$: $2(d+1)(n-d)-n+T(n-2-d) = 2d(n-d)-n+1+T(n-d)$; $d = n-2$: both sides equal $3n-4$; $d = n-1$: both equal n; $d = 0$: $n + T(n-2) = 1 + T(n-1)$; $d < 0$ (the factors $s+1-c$): both equal $T(n-1+d)$. □

(The identity of Lemma 6.12 was also checked by exact computation for $2 \le n \le 8$.)

Proof of Theorem A, conclusion. Evaluate (6.2) at $\theta = -1+\varepsilon$. Since $B_n(-1+\varepsilon) = \varepsilon^{2(n-1)} \prod_{i=2}^{n-1} (i-1+\varepsilon)^{2i(n-i)}$,

$$\varepsilon^{2(n-1)}|u_n| \to \frac{|\kappa_n|\,|A_n(s)|}{\prod_{i=2}^{n-1}(i-1)^{2i(n-i)}\,|C_n(s-1)|}.$$

By Lemma 6.11 and (6.2) for $n-1$ at $(\theta', s') = (1, s-1)$, the limit is also $\dfrac{|\kappa_{n-1}|\,|A_{n-1}(s-1)|}{|s|\,B_{n-1}(1)\,|C_{n-1}(s)|}$, with $B_{n-1}(1) = \prod_{i=1}^{n-2}(i+1)^{2i(n-1-i)}$. By Lemma 6.12, $A_{n-1}(s-1)/(s\,C_{n-1}(s)) = A_n(s)/C_n(s-1)$, so all s-dependence cancels and

$$|\kappa_n| = |\kappa_{n-1}| \cdot \frac{\prod_{i=2}^{n-1}(i-1)^{2i(n-i)}}{\prod_{i=1}^{n-2}(i+1)^{2i(n-1-i)}} = |\kappa_{n-1}| \prod_{m=1}^{n-1} m^{2(2n-1-3m)},$$

where the exponent of m is $2[(m+1)(n-m-1) - (m-1)(n-m)] = 2(2n-1-3m)$. Since $u_1 = 1/(\theta+s+1)$, $\kappa_1 = 1$, and summing the exponents of m over $N = m+1, \dots, n$ gives $2\sum_{N=m+1}^{n}(2N-1-3m) = 2(n-m)(n-2m)$. Hence $|\kappa_n| = \prod_{m=1}^{n-1} m^{2(n-m)(n-2m)} = \kappa_n^+$. □ □

6.6. **Proofs of Lemma CC and Theorem 6.2.**

Proof of Lemma CC. Let $n = K$ and $s \ge K$, $K + s < \ell$. By Lemma 6.6(f) and Theorem A at $\theta = 0$,

$$u(K, s) = \kappa_K \frac{A_K(s)}{B_K(0)\,C_K(s)} \quad \text{in } \mathbb{Q},$$

with κ_K of absolute value $\prod_{m<K} m^{2(K-m)(K-2m)}$ (exponents may be negative). The bases occurring are: the integers $m \in [1, K-1]$ (from κ_K and $B_K(0) = \prod i^{2i(K-i)}$); $s+c$ with $1 \le c \le K$, so $s+c \in [s+1, s+K]$ (from A_K and C_K); and $s+1-c \in [s-K+2, s] \subset [2, s]$ (from A_K). All are integers in $[1, K+s]$, and $K+s < \ell$, so every base is an ℓ-unit and $u(K,s)$ is a nonzero element of $\mathbb{Z}_{(\ell)}$ that is a unit. The entries of N_s and w_{s+1} are ℓ-integral, so the reduction modulo ℓ of $u(K,s)$ is the Krylov determinant over $\mathbb{F}_\ell$ of Corollary 6.5, which is therefore nonzero. Corollary 6.5 gives Lemma CC. □ □

Note that only $|u|$ enters: we use only that $u(K,s)$ is a nonzero rational number all of whose prime factors are $< \ell$.

Proof of Theorem 6.2. Put $s = \ell - H - 1$, so that $h = H+1$ has $\ell - h = s$ and $h = H$ has $\ell - h = s+1$. The hypothesis on H gives $K \le s \le \ell - K - 2$. Both values satisfy $K+1 \le h \le \ell - 1$ and $1 \le \ell - h \le \ell - K - 1$ ($H \ge K+1$ and $H+1 \le \ell - K \le \ell - 1$), so Lemmas 5.2 and 5.3 apply: $\ell^{2(h-K)} P_h(r)$ is an ℓ-adic unit iff $D_{\ell-h}(Y) := \det(Y I_K + N_{\ell-h})$ does not vanish at $Y = y := 16r + c \in \mathbb{F}_\ell$. Suppose $D_s(y) = D_{s+1}(y) = 0$. Then $-y \in \mathbb{F}_\ell$ (possibly 0) is a common eigenvalue of N_s and N_{s+1}, contradicting Lemma CC, which applies because $s \ge K$ and $K + s \le \ell - 2 < \ell$. (In versions 1–7, where $s = K$, one first showed $y \neq 0$ from $D_K(0) = \det(C_K)^2 \neq 0$. That step is not needed, and does not extend to $s > K$, where $D_s(0) = \det(C_s C_s^{\top})$ is a Gram minor and not a product; but Lemma 6.3 and Corollary 6.5 hold for every z, including $z = 0$.) Hence for some $h \in \{H, H+1\}$ the reduction is nonzero and $v_\ell(P_h(r)) = -2(h-K)$. □ □

Remark 6.13 (numerical checks). Independently of Theorem A, the nonvanishing of $u(K,K)$ modulo ℓ (and hence Lemma CC for $s = K$) was verified by exact computation of the Krylov determinant for all $\ell \equiv 3 \pmod 4$ with $7 \le \ell < 4500$ (307 primes); for 39 primes $\ell < 400$ the characteristic polynomials of N_K and N_{K+1} over $\mathbb{F}_\ell$ were checked to be coprime, and Lemma 5.3 was checked for $\ell = 7, 11, 19, 23$ and $h = \ell - K, \ell - K - 1$ and all $X \in \mathbb{F}_\ell$. These are consistency checks, not used in the proof.

For the general-s statement used in version 8 the following were checked by an independent audit (not used in the proof): $u(K,s) \neq 0$ in $\mathbb{F}_\ell$ for every $s \in [K, \ell - K - 1]$ and every prime $\ell \equiv 3 \pmod 4$, $7 \le \ell < 160$ (640 cases), with the closed form of Theorem A at $\theta = 0$ agreeing with the Krylov determinant modulo ℓ in each; $\gcd(\det(Y + N_s), \det(Y + N_{s+1})) = 1$ over $\mathbb{F}_\ell$ for all 622 pairs $s \in [K, \ell - K - 2]$ in that range; and, directly from the definition of M_0 and without Lemmas 5.2, 5.3, CC or Theorem A, for every $\ell \equiv 3 \pmod 4$, $7 \le \ell \le 151$ and every H with $K + 1 \le H \le \ell - K - 1$: $\gcd(\ell^{2(H-K)} P_H,\ \ell^{2(H+1-K)} P_{H+1}) = 1$ modulo ℓ with $v_\ell(\operatorname{den} P_h) = 2(h-K)$.

7. Node removal and the root-free window

7.1. **Why a small value is not enough.** Let $Q_h = \operatorname{den}(P_h) P_h$ and $|\lambda - p/q| = |t|$. The mean-value route, $|Q_h(p/q) - Q_h(\lambda)| \le |t| \max |Q_h'|$, with $\max |Q_h'|$ of size $e^{\Lambda h^2 + O(h)}$, only yields a bound of the shape $\exp(-C \log^2 q)$, which is not a finite irrationality measure. What is needed is smallness of $|P_h|$ on a whole window $|x - \lambda| \le e^{-ch}$ of *exponential radius*, together with $Q_h(p/q) \neq 0$. The second requirement is supplied by Theorem 5.4 (valid at every rational, including p/q). For the first, note that

$$P_h(\lambda + t) = \det(T_h + tI) = \sum_{S \subseteq \{0,\dots,h-1\}} t^{\,h - |S|} \det T_h[S], \tag{7.1}$$

by multilinearity in the columns, where $T_h[S]$ is the principal submatrix on S. So the window requires a bound for *every* principal minor, not only for the full determinant, and uniform in the number $j = h - |S|$ of removed nodes. The proof of Theorem 4.10 extends to principal minors; the only new feature is a cost for each removed node.

7.2. **Statement.** Retain the notation of §4.4: the certificate measure σ with $n_\sigma = 300$ point masses, $\rho = 1/1024$, $g = U^\sigma_\rho - U^\tau + (1-\kappa)\pi|\operatorname{Im} s|$, and the certified data $m' \geq 1.9718000062$, $S_{\mathrm{cert}} \geq -0.1107505186$, $g_{\min} \geq -1.4640996549$ and $D_{\mathrm{cert}} \geq 3.1113790$. On the real line the field term vanishes, so $g = U^\sigma_\rho - U^\tau$ there, and $g_{\min}$ bounds it below on $[0,1]$. Recall $c = 2(m' - g_{\min}) \leq c_{\mathrm{cert}} := 6.8717994$.

Theorem 7.1 (node removal N_ρ; computer-assisted)**.** *There are $\eta(h) = O((\log h)/h)$ and a constant c_κ, both independent of S, such that for every $h \geq 2^{30}$ and every $S \subseteq \{0,\dots,h-1\}$, with $k = |S|$, $j = h-k$ and $\beta = j/h$,*

$$|\det T_h[S]| \;\leq\; 16^{-k}\,(2h + c_\kappa)^k\,\exp\bigl(-h^2[D_{\mathrm{cert}} - \beta c] + h^2\eta(h)\bigr).$$

Explicitly, $\eta(h) = \frac{|\log\rho|}{h} + \frac{2\log(1+8h/3)}{h} + \varepsilon_2(h) + \frac{\log^+ C_E}{h} + \frac{4}{h}\sum_\ell (m' - \gamma_\ell)^+$.

For $S = \{0,\dots,h-1\}$ ($\beta = 0$) this is Theorem 4.10. The proof is that of Lemmas 4.8 and 4.9 line by line; we record what changes. (The coefficient of the last term of η is $4/h$ because H_S carries the factor $2/h$ in front of $\sum g$, so the error $\frac{2}{h}\sum_\ell (m' - \gamma_\ell)^+$ of Lemma 4.9 is doubled.)

7.3. **Proof of Theorem 7.1. Inputs.** By Proposition 4.1 the identity $L_F[a][b] = \int v_a v_b\, dm$ is entrywise, so it holds for principal submatrices: $L_F[S] = \int \phi_S \phi_S^\top dm$ with $\phi_S = (v_a)_{a\in S}$, and $T_h[S] = L_F[S]/16$. As m has finite total variation and $|v_a| \leq 8/3$ on its support, Andréief's identity gives $\det L_F[S] = \frac{1}{k!}\int \det[v_a(z_i)]^2_{i\leq k, a\in S}\prod dm(z_i)$ (for $k = 0$ both sides are 1), and therefore

$$|\det L_F[S]| \leq \frac{1}{k!}\int |C_S(z)|^2 \prod_{i\leq k} d|m|(z_i),$$

$$|C_S|^2 = \frac{\prod_{i\neq i'}|z_i - z_{i'}|\prod_{a\neq b\in S}|a-b|}{\prod_{i,\,a\in S}|z_i - a|^2},$$

by Cauchy's determinant. Split $d|m| = w_\kappa\, d\Omega_\kappa$ as in the proof of Theorem 4.10; then $\Omega_\kappa(\Sigma_h) \leq 2h + c_\kappa$ and c_κ is independent of h. Every point of the support of m has $\operatorname{Re} z \leq -\frac{3}{8}$ while the nodes lie in $[0, h-1]$, so free points and nodes never coincide. It therefore suffices to prove the pointwise bound

$$|C_S(z)|^2 \prod_i w_\kappa(z_i) \leq \exp(-h^2[D_{\mathrm{cert}} - \beta c - \eta(h)]), \tag{7.2}$$

since integrating it against $\prod \Omega_\kappa$ and using $1/k! \leq 1$ gives the theorem.

Scaling. Put $s_i = z_i/h$, $t_a = a/h$. Counting powers of h: $-k(k-1)$ from the free pairs, $-k(k-1)$ from the node pairs inside S and $+2k^2$ from the

cross terms, in total $2k\log h \geq 0$ in the exponent of the reciprocal, which is dropped. Hence $-\log(|C_S|^2\prod w_\kappa) \geq h^2 H_S$ with

$$h^2 H_S = \sum_{i\neq i'} -\log|s_i - s_{i'}| + \sum_{a\neq b\in S} -\log|t_a - t_b| + 2\sum_{i,\,b\in S} \log|s_i - t_b| + \sum_i -\log w_\kappa(z_i).$$

Completing the node set. Let S^c be the j removed nodes, and let all pairs be of distinct elements. Two exact identities, from adding and removing the pairs meeting S^c (for the second: the ordered pairs meeting S^c are those with first entry in S^c, plus those with second entry in S^c, minus those with both; the summand is symmetric), are

$$2\sum_{i,\,b\in S}\log|s_i - t_b| = 2\sum_{i,\,\text{all } b}\log|s_i - t_b| + 2\sum_{i,\,a\in S^c}(-\log|s_i - t_a|),$$

$$\sum_{b\neq b'\in S} = \sum_{\text{all } b\neq b'} - 2\sum_{a\in S^c}\sum_{b\neq a} + \sum_{a\neq a'\in S^c} \qquad \text{(applied to } -\log|t_b - t_{b'}|).$$

Let X be the set of the h distinct points $s_1,\dots,s_k$ and t_a ($a\in S^c$); it has $k+j=h$ elements. The free–free, free–removed and removed–removed terms assemble to $\sum_{x\neq y\in X} -\log|x-y|$, and

$$h^2 H_S = \underbrace{\sum_{x\neq y\in X} -\log|x-y|}_{[B]} + \underbrace{2\sum_{i,\,\text{all } b}\log|s_i - t_b|}_{[R1]} + \underbrace{\sum_{\text{all } b\neq b'} -\log|t_b - t_{b'}|}_{[R2]} + \underbrace{2\sum_{a\in S^c}\sum_{b\neq a}\log|t_a - t_b|}_{[R3]} + \underbrace{\sum_i -\log w_\kappa(z_i)}_{[R4]}.$$

For $S=\{0,\dots,h-1\}$ the term $[R3]$ is absent and this is the decomposition in the proof of Lemma 4.8.

The term $[B]$ (replaces circle smoothing). The pairs of removed nodes occur in $[B]$ and once more, with the opposite sign, in the second identity above; so *the interaction of the removed nodes among themselves is counted exactly once and no energy of the removed set is left over.* Apply Lemma 4.3 to every ordered pair of distinct points of X, free or removed, and then Corollary 4.7 with this X:

$$[B] = \sum_{x\neq y\in X} -\log|x-y| \;\geq\; \sum_{x\neq y\in X} k_\rho(x,y) \;\geq\; h^2\Big[\tfrac{2}{h}\textstyle\sum_{x\in X} U^\sigma_\rho(x) - I_\rho(\sigma)\Big] + h\log\rho,$$

that is, $\frac{[B]}{h^2} \geq \frac{2}{h}\sum_{i\leq k} U^\sigma_\rho(s_i) + \frac{2}{h}\sum_{a\in S^c} U^\sigma_\rho(t_a) - I_\rho(\sigma) - \frac{|\log\rho|}{h}$. There is no Lipschitz term, no smoothing and no continuum energy: only the finite-sum Lemma 4.6.

The terms $[R1]$, $[R2]$, $[R4]$. As in Lemma 4.8: $[R1]/h^2 \geq -\frac{2}{h}\sum_i U^\tau(s_i) - \frac{2k}{h^2}\log(1+\frac{8h}{3})$ (monotonicity of $u \mapsto \log|s-u|$ on $[0,1]$ for $\operatorname{Re} s \leq 0$, and $|s_i| \geq \frac{3}{8h}$), and $[R4]/h^2 \geq \frac{2}{h}\sum_i(1-\kappa)\pi|\operatorname{Im} s_i| - \frac{k}{h^2}\log^+ C_E$. The term $[R2] = h(h-1)\log h - 2\sum_{n<h}\log n!$ involves all h nodes and is independent of S. Using $\log n! \leq (n+\frac{1}{2})\log n - n + 1$ for $n \geq 1$ gives $[R2]/h^2 \geq \frac{3}{2} - \varepsilon_2(h)$ with $\varepsilon_2(h) = O((\log h)/h)$. Nodes carry no weight, so no field term arises at the removed nodes.

The term $[R3]$: the removed-node term. For a node a with $t = a/h$,

$$\sum_{b\neq a}\log|t_a - t_b| = \log(a!\,(h-1-a)!) - (h-1)\log h.$$

By $n! \geq (n/e)^n$, valid for all $n \geq 0$, applied to $a!$ and to $(h-1-a)! = (h-a)!/(h-a)$, we get $\log(a!(h-1-a)!) \geq h[t\log t + (1-t)\log(1-t)] + h\log h - h - \log h$, that is, $\sum_{b\neq a}\log|t_a - t_b| \geq -h\,U^\tau(t_a)$ *with no error term.* Hence $[R3]/h^2 \geq -\frac{2}{h}\sum_{a\in S^c} U^\tau(t_a)$.

Collecting. On the real line $U^\sigma_\rho - U^\tau = g$, so

$$H_S \;\geq\; \frac{2}{h}\sum_{i\leq k} g(s_i) + \frac{2}{h}\sum_{a\in S^c} g(t_a) - I_\rho(\sigma) + \frac{3}{2} - \eta_1(h), \tag{7.3}$$

with $\eta_1(h) = \frac{|\log\rho|}{h} + \frac{2\log(1+8h/3)}{h} + \varepsilon_2(h) + \frac{\log^+ C_E}{h}$, which uses only $k \leq h$ and so is uniform in S.

Removed nodes. Since $t_a \in [0,1)$, (L) gives $\frac{2}{h}\sum_{a\in S^c} g(t_a) \geq 2\beta\, g_{\min}$. As $g_{\min} < 0$, a removed node costs.

Free points (Lemma 4.9). The free points have mass $k/h = 1-\beta$, and Lemma 4.9 gives

$$\frac{1}{h}\sum_{i\leq k} g(s_i) \;\geq\; (1-\beta)m' + S_{\text{cert}} - \frac{2}{h}\sum_{\ell}(m' - \gamma_\ell)^+.$$

Inserting both into (7.3),

$$\begin{aligned} H_S &\geq 2(1-\beta)m' + 2S_{\text{cert}} + 2\beta g_{\min} - I_\rho(\sigma) + \tfrac{3}{2} - \eta_1 - \tfrac{4}{h}\textstyle\sum_\ell(m'-\gamma_\ell)^+ \\ &= D_{\text{cert}} - \beta\cdot 2(m' - g_{\min}) - \eta(h), \end{aligned}$$

where (since $D_{\text{cert}} = 2(m' + S_{\text{cert}}) - I_\rho(\sigma) + \frac{3}{2}$, with S_{cert} the sum of Lemma 4.9) the expression in the statement is exactly $D_{\text{cert}} - \beta c$, and $\eta = \eta_1 + \frac{4}{h}\sum_\ell(m'-\gamma_\ell)^+$ is independent of S. This is (7.2). □

7.4. **The certificate for $g_{\min}$.** The number $g_{\min}$ is part of the certificate of §4.4, produced by the same script `kcert.py` on the same `sigma300.json`. The interval $[-\frac{1}{64}, 1] \supset [0,1]$ is divided into 4423 exact cells and each cell becomes one Arb enclosure [1] (full-cell enclosures, so no Lipschitz bound is needed). On each cell the lower bound for U^σ_ρ is the one described in §4.4,

and $U^\tau(t) = 1-t\log|t|-(1-t)\log|1-t|$ has $(U^\tau)' = \log|1-t|-\log|t|$, which is positive for $t < \frac{1}{2}$ and negative for $t > \frac{1}{2}$ (also across 0), so its supremum on a cell $[a,b]$ is $U^\tau(b)$ if $b \le \frac{1}{2}$, $U^\tau(a)$ if $a \ge \frac{1}{2}$, and $U^\tau(\frac{1}{2}) = 1+\log 2$ otherwise. The certified infimum is

$$g_{\min} \;\ge\; -1.4640996549,$$

with the minimiser near $t \approx 0.7499$ in floating point. Hence $c = 2(m' - g_{\min}) \le 2(1.9718000062 + 1.4640996549) = 6.8717993\ldots \le c_{\text{cert}} := 6.8717994$.

7.5. **The window corollary.**

Corollary 7.2. *For $h \ge 2^{30}$ and every real t with $|t| \le e^{-ch}$,*

$$|P_h(\lambda+t)| \;\le\; \exp\bigl(-D_{\text{cert}}h^2 + h^2\eta(h) + h\log(2(2h+c_\kappa))\bigr) = e^{-(D_{\text{cert}}-o(1))h^2},$$

where the $o(1)$ does not depend on t. In particular this holds for $|t| \le e^{-c_{\text{cert}}h}$.

Proof. By (7.1), grouping the subsets S by $j = h - |S|$ (there are $\binom{h}{j}$ of them) and applying Theorem 7.1, with $\beta = j/h$ so that $e^{h^2\beta c} = e^{jch}$,

$$|P_h(\lambda+t)| \le (2h+c_\kappa)^h e^{h^2\eta} e^{-D_{\text{cert}}h^2} \sum_{j=0}^{h} \binom{h}{j} 16^{-(h-j)} \bigl(|t|e^{ch}\bigr)^j.$$

(Here $(2h+c_\kappa)^k \le (2h+c_\kappa)^h$ for $2h+c_\kappa \ge 1$.) If $|t|e^{ch} \le 1$ every summand is at most $\binom{h}{j}$, and the sum is at most 2^h. □

Remark 7.3 (earlier versions). An earlier version of this paper used a Gram-eigenvalue bound, giving $\mu(\lambda) \le 126$. The node-removal bound replaced it, first with the circle-smoothing energy step (Remark 4.12, $\mu(\lambda) \le 79.35$) and now with the k_ρ energy step.

8. Proof of the main theorem

Theorem 8.1. *Let $\eta = D_{\text{cert}} - \Lambda \ge 0.1304266$ and $c_{\text{cert}} = 6.8717994$. Then for every $\nu > c_{\text{cert}}/\eta$ there are only finitely many rationals p/q with $|\lambda - p/q| < q^{-\nu}$. In particular, since $c_{\text{cert}}/\eta \le 52.6872$ (it is 52.68713 for the unrounded values) and $52.6872 < 52.69$, Theorem 1.2 holds.*

Proof. Let $\nu > c_{\text{cert}}/\eta$ and suppose $|\lambda - p/q| =: |t| < q^{-\nu}$ with q large; put $L = \log q$. Choose $\delta > 0$ so small that still $\nu > c_{\text{cert}}(1+\delta)/\eta$, and set

$$k = \frac{1+\delta}{\eta}, \qquad H = \lceil kL \rceil.$$

Choice of ℓ. Call a prime $\ell \equiv 3 \pmod 4$ *usable* if $\frac{4H+5}{3} \le \ell \le 4H-5$; for such ℓ one has $\ell \ge 7$ and $K+1 \le H \le \ell - K - 1$ with $K = (\ell+1)/4$, which is the hypothesis of Theorem 6.2. By the prime number theorem

in arithmetic progressions mod 4 (de la Vallée Poussin; [3, Ch. 20], [10, Cor. 11.19]), $\vartheta(x;4,3) = x/2 + o(x)$, so the product of the usable primes is

$$\exp\Big(\tfrac{1}{2}(4H - 5 - \tfrac{4H+5}{3})(1+o(1))\Big) = \exp\Big(\tfrac{4}{3}H(1+o(1))\Big) \geq q^{\frac{4}{3}k(1+o(1))} > q$$

for large q, because $k > 1/\eta > 7$. The primes dividing q have product at most q. Hence there is a usable prime ℓ with $\ell \nmid q$. (This is where the length of the interval for ℓ enters, but only through its *product*, which is $q^{4k/3}$ and not just barely larger than q; no condition on h/ℓ is needed, so the interval costs nothing in the exponent. Lemma 5.5 is not needed at all. If one prefers to allow $\ell \mid q$, Lemma 5.5 still gives $P_H(p/q) \neq 0$ when $v_\ell(q) \neq 2$, since it holds for every admissible h.)

Choice of h. Since $\ell \nmid q$, Theorem 6.2 (applied with this H and $r = p/q$) gives $h \in \{H, H+1\}$ with $P_h(p/q) \neq 0$. Then $kL \leq h \leq kL + 2$, $h \geq 2^{30}$ for large q, and $Q_h(p/q) \neq 0$, whence by Lemma 2.4

$$|Q_h(p/q)| \geq q^{-h} = e^{-hL}.$$

Window. Since $\nu > c_{\mathrm{cert}}\, k$ and $h \leq kL + 2$, we have $c_{\mathrm{cert}}\, h < \nu L$ for large q. Hence $|t| < e^{-\nu L} \leq e^{-c_{\mathrm{cert}} h} \leq e^{-ch}$, and Corollary 7.2 and Theorem 3.1 give

$$|Q_h(p/q)| \leq \mathrm{den}(P_h)\, |P_h(\lambda + t)| \leq \exp\big((\Lambda - D_{\mathrm{cert}} + o(1))h^2\big) = \exp\big(-(\eta - o(1))h^2\big).$$

Contradiction. Combining, $(\eta - o(1))h \leq L$. But $h \geq kL$ gives $\eta h \geq (1+\delta)L$, a contradiction for large q. Note that the content, the decay, the node removal, the window and the integrality depend on h only (not on ℓ); the ℓ-adic input is used only through $P_h(p/q) \neq 0$. □

Remark 8.2 (sensitivity; history of the ℓ-selection). The exponent is c/η with $\eta = D_{\mathrm{cert}} - \Lambda$ and $c = 2(m' - g_{\min})$, i.e. 52.687. In versions 1–7 the choice of h was tied to ℓ ($h \in \{\frac{3\ell-1}{4}, \frac{3\ell-5}{4}\}$, $h/\ell \approx \frac{3}{4}$), so ℓ had to range over an interval $(X, Y]$ of length $(1+\delta)L$ in order to contain a prime $\ell \equiv 3 \pmod 4$ with $v_\ell(q) \neq 2$ (using Lemma 5.5); the length of that interval, multiplied by $h/\ell \approx \frac{3}{4}$, produced the extra term $\frac{3}{4}c$ and the bound $c(\frac{1}{\eta} + \frac{3}{4}) = 57.841$. Without Lemma 5.5 the interval had to have length $(2+\delta)L$ and the bound was $c(\frac{1}{\eta} + \frac{3}{2}) = 62.995$. The general-$s$ form of Theorem 6.2 removes this term: one fixes $h \approx kL$ first and then looks for ℓ in $[\frac{4H+5}{3}, 4H - 5]$, an interval whose prime product is about $q^{4k/3} \gg q$. The remaining term c/η is what remained in the old formula when the interval for ℓ was taken to have length 0, and it cannot be lowered by this method: it is dictated by the certified constants c_{cert} and η alone.

Remark 8.3 (a simpler choice of h gives 77.13). If in the proof of Theorem 8.1 one uses Theorem 5.4 instead of Theorem 6.2, that is $h \in \{\ell - 1, \ell - 2, \frac{3\ell-1}{4}\}$ (and $h = \ell - 1$ when $\ell \mid q$), the value of h is again tied to ℓ and only $\frac{3\ell-1}{4} \leq h \leq \ell - 1$ is available. One then chooses, as in versions 1–7, a prime $\ell \equiv 3 \pmod 4$ with $v_\ell(q) \neq 2$ in an interval $(X, Y]$ with $X = \frac{4}{3}kL$, $Y = X + (1+\delta)L$, $k = (1+\delta)/\eta$ (its prime product exceeds $q^{1/2}$ and the primes

with $v_\ell(q) = 2$ have product at most $q^{1/2}$), and one gets $kL - \frac{1}{4} \le h \le Y$, and the window condition $c_{\mathrm{cert}}\, Y < \nu L$ requires $\nu > c_{\mathrm{cert}}\big(\frac{4(1+\delta)}{3\eta} + 1 + \delta\big)$. The rest of the proof is unchanged and gives: for every $\nu > c_{\mathrm{cert}}\,(4/(3\eta)+1)$ there are only finitely many p/q with $|\lambda - p/q| < q^{-\nu}$. Since $4/(3\eta) + 1 \le 11.2229$ ($11.22286\ldots$ unrounded) and $c_{\mathrm{cert}} \cdot 11.2229 \le 77.1217 < 77.13$ (unrounded 77.1212), this gives $\mu(\lambda) \le 77.13$. The factor $\frac{4}{3}$ comes from the spread of the three values of h, and without Lemma 5.5 the bound is $c(4/(3\eta) + 2) < 84$ (83.99). The Lean formalization currently covers this version (§18), and it does not use Theorem A.

9. The mirror constant

Put

$$\lambda' \;:=\; \sum_{k\ge 1}\Big(\frac{1}{(4k-3)^2} - \frac{1}{16k^2}\Big) \;=\; \Big(\frac{\pi^2}{16} + \frac{G}{2}\Big) - \frac{\pi^2}{96} \;=\; \frac{G}{2} + \frac{5\pi^2}{96} \;=\; 0.972\ldots,$$

using $\sum_{n\equiv 1(4)} n^{-2} = \frac{\pi^2}{16} + \frac{G}{2}$ (the companion of the sums in §2); thus $G + 5\pi^2/48 = 2\lambda'$ and $\lambda + \lambda' = G$. The proof of Theorem 1.4 is the proof of Theorems 1.1 and 1.2 for the *mirror frame*

$$\bar F(z) = \psi(z+1) - \psi(z+\tfrac14) = \sum_{j\ge 0}\Big[\frac{1}{z+j+\frac14} - \frac{1}{z+j+1}\Big],$$

with poles $+1$ at $-(j+\frac14)$ and -1 at $-(j+1)$. The shift $\frac34 \to \frac14$ exchanges the roles of the residue classes 1 and 3 modulo 4. We list *every* place where the argument changes; everything not listed is used verbatim. The constants $\Lambda = 313/105$, $D_{\mathrm{cert}} \ge 3.1113790$, $c_{\mathrm{cert}} = 6.8717994$, $\eta \ge 0.1304266$ are the same, so the numerical value of the exponent is the same, $c_{\mathrm{cert}}/\eta \le 52.6872 < 52.69$ (and 77.1213 for the simpler choice of h).

9.1. **The matrices.** From $\bar F(k) - \bar F(k-1) = \frac{1}{k} - \frac{4}{4k-3} = -\frac{3}{k(4k-3)}$ and the analogous identity for $\bar F'$ one gets, exactly as for (2.2), (2.3),

$$\bar L[a][b] = -\frac{3}{b-a}\sum_{a<k\le b}\frac{1}{k(4k-3)}\ (a<b), \qquad \bar L[a][a] = \bar F'(0) + \sum_{k\le a}\Big(\frac{16}{(4k-3)^2} - \frac{1}{k^2}\Big),$$

and $\bar F'(0) = \psi'(1) - \psi'(\frac14) = \frac{\pi^2}{6} - (\pi^2 + 8G) = -16\lambda'$. Let $N_0 = N_0^{(h)}$ be the symmetric rational matrix with

$$N_0[a][b] = \frac{3}{16(b-a)}\sum_{a<k\le b}\frac{1}{k(4k-3)}\ (a<b), \qquad N_0[a][a] = \frac{1}{16}\sum_{k\le a}\Big(\frac{1}{k^2} - \frac{16}{(4k-3)^2}\Big),$$

and put $\bar T_h = \frac{1}{16}\bar L = -(\lambda' I + N_0)$, $\bar P_h(X) = \det(XI + N_0) \in \mathbb{Q}[X]$ (monic of degree h). Then $\bar P_h(\lambda' + t) = (-1)^h \det(\bar T_h - tI)$, and Lemma 2.4 holds verbatim for $\bar Q_h = \mathrm{den}(\bar P_h)\bar P_h$.

9.2. **Content.** For $p \geq 5$ and $q = p^e$ the least $k \geq 1$ with $q \mid k(4k-3)$ is $k_q = (j_q q + 3)/4$, where now $j_q = 1$ if $q \equiv 1 \pmod 4$ and $j_q = 3$ if $q \equiv 3 \pmod 4$ (indeed $4k - 3 = jq$ forces $jq \equiv 1 \pmod 4$; k and $4k-3$ are coprime for $p \geq 5$, and $k_q < q$). So the classes are *swapped* relative to the construction before Lemma 3.3; the dual certificate (the potentials u, v and Lemma 3.3) is verbatim. Since $|k_q - j_q q/4| \leq \frac{3}{4}$, the per-level bound is $\delta(q) \leq h\, g_{j_q}(q/h) + \frac{3}{2}$, and the error costs $O(h)$ in total. The two integrals $\int g_1 = 21/5$ and $\int g_3 = 37/21$ are now attached to the classes 1 and 3 mod 4 respectively; by the prime number theorem in progressions the two classes have equal weight, so $\Lambda = \frac{1}{2}(\frac{21}{5} + \frac{37}{21}) = \frac{313}{105}$ again. The primes $2, 3$ contribute $O(h \log h)$ as before. Hence $\log \operatorname{den} \bar{P}_h \leq (\frac{313}{105} + o(1))h^2$.

9.3. **Nonvanishing.** Let $\ell \equiv 1 \pmod 4$ be prime, $\ell \geq 13$, $K = (\ell+3)/4$ (so $4K - 3 = \ell$), and $K + 1 \leq h \leq \ell - 1$; low and high indices are as in §5.

Lemma 9.1 (ℓ-singular structure, mirror)**.** *Among k, $4k-3$, $b-a$, 16 and the numerator 3 occurring in the entries of N_0 only $4k-3$ can be divisible by ℓ, and this occurs only for $k = K$. Hence every high diagonal entry of N_0 contains the term $-\ell^{-2}$, every low–high entry contains $3/(16K\ell(b-a))$, and all other entries lie in $\mathbb{Z}_{(\ell)}$.*

Proof. If $\ell \mid 4k-3$ with $4k-3 \leq 4h-7 < 4\ell$, then $4k-3 \in \{\ell, 2\ell, 3\ell\}$; 2ℓ is even and $3\ell \equiv 3 \not\equiv 1 \pmod 4$. The numerator 3 is a unit since $\ell \neq 3$. The rest is as in Lemma 5.1. □

Lemma 9.2. *With $D = \operatorname{diag}(I_K, \ell I_{h-K})$, modulo ℓ*

$$D(XI + N_0)D \equiv \begin{pmatrix} XI_K + A & \hat{B} \\ \hat{B}^\top & -I \end{pmatrix}, \qquad \hat{B}_{ab} \equiv \frac{3}{16K(b-a)} \equiv \frac{1}{4(b-a)},$$

because $16K = 4\ell + 12 \equiv 12$. The Schur complement of the block $-I$ gives $\ell^{2(h-K)}\bar{P}_h(X) \equiv (-1)^{h-K} \det(XI_K + A + \hat{B}\hat{B}^\top)$, which is monic of degree K up to sign; so $v_\ell(\operatorname{den} \bar{P}_h) = 2(h-K)$. (The sign in front of $\hat{B}\hat{B}^\top$ is the opposite *of that in Lemma 5.2, because the high block is $-I$ instead of $+I$.)*

Lemma 9.3 (key identity, mirror)**.** *Let $s = \ell - h$, $m = \ell - K$, let C_s be as in Lemma 5.3 (note $a + j \leq K - 1 + s < \ell$), and put $Y = -(16X + H_m^{(2)})$. Then in $\mathbb{F}_\ell[X]$*

$$16^K \ell^{2(h-K)} \bar{P}_h(X) \equiv (-1)^h \det(YI_K + C_s C_s^\top).$$

Proof. Modulo ℓ, $4K \equiv 3$, so $\frac{4}{4k-3} = \frac{1}{k-3/4} \equiv \frac{1}{k-K} \equiv \frac{1}{k+m}$, hence $\frac{3}{k(4k-3)} \equiv \frac{1}{k+m} - \frac{1}{k}$ and $\frac{16}{(4k-3)^2} \equiv \frac{1}{(k+m)^2}$. For $a < b < K$ (so $b + m < \ell$)

$$16A_{ab} \equiv -\frac{1}{b-a} \sum_{k=a+1}^{b} \Big(\frac{1}{k} - \frac{1}{k+m}\Big) = -\sum_{j=1}^{m} \frac{1}{(a+j)(b+j)}, \qquad 16A_{aa} \equiv \sum_{k \leq a} \Big(\frac{1}{k^2} - \frac{1}{(k+m)^2}\Big) = H_m^{(2)} - \sum_{j}$$

that is $16A \equiv H_m^{(2)} I - C_m C_m^\top$. As in the proof of Lemma 5.3 (substitute $j = \ell - b$, so $b - a \equiv -(a+j)$), $16\hat{B}\hat{B}^\top \equiv \sum_{j=s+1}^{m} w_j w_j^\top = C_m C_m^\top -$

$C_sC_s^\top$. Therefore $16(A+\hat{B}\hat{B}^\top) \equiv H_m^{(2)}I - C_sC_s^\top$, and $\det(XI + \frac{1}{16}(H_m^{(2)}I - C_sC_s^\top)) = 16^{-K}(-1)^K \det(YI + C_sC_s^\top)$. Combine with Lemma 9.2: the sign is $(-1)^{h-K}(-1)^K = (-1)^h$. □

Theorem 9.4 (choice of h, mirror)**.** *Let $\ell \equiv 1 \pmod 4$ be prime, $\ell \geq 13$, $K = (\ell+3)/4$, $r \in \mathbb{Q}$ with $\ell \nmid \operatorname{den} r$, and H an integer with $K+1 \leq H \leq \ell - K - 1$ (equivalently $\frac{4H+7}{3} \leq \ell \leq 4H-7$). Then one of $h \in \{H, H+1\}$ satisfies $v_\ell(\bar{P}_h(r)) = -2(h-K)$; in particular $\bar{P}_h(r) \neq 0$. For $H = \ell - K - 1$ this is the pair $\{\frac{3\ell-7}{4}, \frac{3\ell-3}{4}\}$ of versions 1–7.*

Proof. Put $s = \ell - H - 1 \in [K, \ell - K - 2]$, so that $\ell - h \in \{s, s+1\}$. Both values satisfy $K+1 \leq h \leq \ell-1$ and $1 \leq \ell - h \leq \ell - K - 1$, so Lemmas 9.2, 9.3 apply: $\ell^{2(h-K)}\bar{P}_h(r)$ is an ℓ-adic unit iff $D_{\ell-h}(Y) = \det(YI_K + N_{\ell-h})$ does not vanish at $Y = y := -(16r + H_m^{(2)}) \in \mathbb{F}_\ell$. Suppose $D_s(y) = D_{s+1}(y) = 0$. Then $-y$ (possibly 0) is a common eigenvalue of N_s and N_{s+1}. Lemma CC holds for any K, any $s \geq K$ and any prime $\ell > K + s$: its proof (Krylov reduction, valid for every eigenvalue including 0, and Theorem A at $\theta = 0$) uses only that $u(K, s)$ is a rational number all of whose prime factors are integers in $[1, K+s]$, and never mentions the residue class of ℓ (Theorem A is an identity in (n, s, θ)). Here $K + s \leq \ell - 2 < \ell$. This contradicts Lemma CC. □

(The same argument gives the three-h version, $h \in \{\ell-1, \ell-2, \ell-K\}$, with $\Xi_{11} = H_K^{(2)}$ and $\Xi_{12} = K/(K+1) \neq 0$; this is the route to 77.13.)

Lemma 9.5 (primes dividing the denominator, mirror)**.** *Let $\ell \equiv 1 \pmod 4$ be prime, $\ell \geq 13$, $K+1 \leq h \leq \ell-1$, $r = p/q$ with $e = v_\ell(q) \geq 1$ and $u = \ell^e r$. If $e = 1$, or $e \geq 3$, or $e = 2$ and $u \not\equiv +1 \pmod \ell$, then $\bar{P}_h(r) \neq 0$ for every admissible h.*

Proof. As for Lemma 5.5, with high block $\ell^2 rI + J$ and now $J \equiv -I$ (Lemma 9.1): the reduced determinants are u^K for $e = 1$, $u^K(u-1)^{h-K}$ for $e = 2$, and u^h for $e \geq 3$. □

Thus the exceptional case is $e = 2$, $u \equiv +1$ (instead of $u \equiv -1$). It is not needed for the bound 52.69, since the proof of Theorem 8.1 chooses $\ell \nmid q$; it would be needed only if $\ell \mid q$ were allowed. (Checked on 574 exact cases for $\ell = 13, 17, 29$, all admissible h and $e \leq 4$; the exception $e = 2$, $u \equiv 1$ does occur. Lemma 9.3 was checked by exact characteristic polynomials for $h = 12, 20$ and all $\ell \equiv 1 \pmod 4$ in $(h, 4h]$.)

9.4. **Decay and node removal.** *The certificate is the same.* The certificates for D_{cert} (§4.4) and for $g_{\min}$ (§7.4) concern one fixed function g on the scaled plane; they do not depend on F, on h, or on where the atoms lie modulo 1. What changes is only the contour identity and the $o(1)$ bookkeeping, as follows.

Reflection. From $\psi(1+z) = \psi(-z) - \pi\cot\pi z$ and $\psi(\frac14+z) = \psi(\frac34 - z) - \pi\cot(\pi z + \frac{\pi}{4})$,

$$\bar F = \tilde F + \bar E, \qquad \tilde F(z) = \psi(-z) - \psi(\tfrac34 - z), \quad \bar E(z) = \frac{-\pi\sin(\pi/4)}{\sin\pi z\,\sin(\pi z + \pi/4)},$$

with $\tilde F$ analytic and $O(1/|z|)$ in $\operatorname{Re} z < 0$, and $\bar E$ 1-periodic with $|\bar E(x+iy)| \le Ce^{-2\pi|y|}$ for $|y| \ge 1$.

The contour. Take $N_0 = \lfloor r_0 h\rfloor$ as before and $x^* = -N_0 + \frac38$, the midpoint of the gap $(-N_0, -N_0 + \frac34)$ between consecutive poles of $\bar F$, so $x^* \equiv \frac38 \pmod 1$. On the part of $\Gamma_h = x^* + u(-1 \pm 3i)$ with $|\operatorname{Im} z| < 1$ one has $\operatorname{Re} z \in (-N_0 + \frac{1}{24}, -N_0 + \frac38]$, inside that gap. The poles to the right of x^* are $-(j+\frac14)$, $0 \le j < N_0$ (residue $+1$), and $-(j+1)$, $0 \le j < N_0 - 1$ (residue -1); so the set of atoms is

$$\bar{\mathcal A} = \{j + \tfrac14 :\ 0 \le j < N_0\} \ \cup\ \{j + 1 :\ 0 \le j < N_0 - 1\}, \qquad |\bar{\mathcal A}| = 2N_0 - 1,$$

with $c_\mu = +1$ on the first part and -1 on the second, and Proposition 4.1 holds verbatim: $\bar L = -\sum_{\mu\in\bar{\mathcal A}} c_\mu u_\mu u_\mu^\top + \frac{1}{2\pi i}\int_{\Gamma_h} \bar E\, vv^\top dz$ (checked entrywise to $2.4\cdot 10^{-17}$ at 25 digits for $N_0 = 1, 2, 7$, and independently for $(N_0, h) = (4,3), (3,4)$).

Changes in Andréief's bound, Lemmas 4.8, 4.9 and Theorem 7.1. All are in $o(1)$ terms:

- the support of the measure has $\operatorname{Re} z \le -\frac14$ (the nearest atom is $-\frac14$), so $|v_a| \le 4$ (instead of $\frac83$), and the corresponding loss is $\log(1+4h)/h \to 0$;
- C_E and c_κ are independent of h, by periodicity and $x^* \equiv \frac38 \pmod 1$ (the contour crosses the real axis at distance $\frac38$ from the poles at 0 and $\frac34$ modulo 1);
- in Lemma 4.9, $x^*/h = -r_0 + \delta_h$ with $\delta_h = (r_0 h - N_0 + \frac38)/h \in [\frac{3}{8h}, \frac{11}{8h})$, so $|\delta_h| \le \omega = 2^{-30}$ as soon as $h \ge \frac{11}{8}\cdot 2^{30}$; the atoms lie in $-(\mathbb Z_{\ge 0} + \frac14)/h \cup -\mathbb Z_{\ge 1}/h \subset [-r_0, 0]$ (the largest is $N_0 - \frac34 \le r_0 h$), two progressions of spacing $1/h$, so $N_\ell \le 2h(b_\ell - a_\ell) + 2$ exactly as before.

The function g, the boxes (R), the segment cells (S), the line cells (L) and $I_\rho(\sigma)$ are identical. Hence Theorem 4.10 holds for $\bar P_h$ with $D^+ = D_{\text{cert}} \ge 3.1113790$, Theorem 7.1 with $c \le 6.8717994$, and Corollary 7.2: $|\bar P_h(\lambda' + t)| \le e^{-(D_{\text{cert}} - o(1))h^2}$ for $|t| \le e^{-ch}$.

Remark 9.6. Exact determinants (Arb, rational entries) give $-\log|\det \bar T_h|/h^2 = 3.030, 3.060, 3.075, 3.084, 3.090, 3.094$ at $h = 32, 48, 64, 80, 96, 112$, against $3.068, 3.084, 3.093, 3.098, 3.101, 3.104$ for T_h. The difference of the logarithms is $\approx 1.06\,h + 4.7$, linear in h (the near atom $-\frac14$ costs $O(h)$, not $O(h^2)$), so the h^2-constants coincide; a margin "$\eta \approx 0.10$" read off from the finite-h slopes is a finite-h artefact. These computations are consistency checks and are not used in the proof.

9.5. **The measure.** The proof of Theorem 8.1 is verbatim with: usable primes $\ell \equiv 1 \pmod 4$ in $[\frac{4H+7}{3}, 4H-7]$ (using $\vartheta(x;4,1) = x/2 + o(x)$, so their product is again $q^{\frac{4}{3}k(1+o(1))} > q$ and some usable ℓ has $\ell \nmid q$; $\ell \geq 13$ for large q); Theorem 9.4 in place of Theorem 6.2; and $h \in \{H, H+1\}$, $H = \lceil kL \rceil$, so that $kL \leq h \leq kL + 2$. Lemma 9.5 is not needed (it would allow $\ell \mid q$ with $v_\ell(q) \neq 2$). With η, c_{cert} as above this yields: for every $\nu > c_{\mathrm{cert}}/\eta$ there are only finitely many p/q with $|\lambda' - p/q| < q^{-\nu}$, and $c_{\mathrm{cert}}/\eta \leq 52.6872 < 52.69$ (52.68713 unrounded). With the three values of h one gets $c_{\mathrm{cert}}(4/(3\eta) + 1) \leq 77.1217$. This proves Theorem 1.4: λ' is irrational and $\mu(\lambda') \leq 52.69$; since $\mu(a\xi + b) = \mu(\xi)$ for rational $a \neq 0$ and b, also $\mu(G + 5\pi^2/48) \leq 52.69$ and $G + 5\pi^2/48 \notin \mathbb{Q}$. No hybrid frame has been built for λ', so 52.69 is the best bound for λ' proved here (versions 1–7: 57.85).

Remark 9.7 (the two measures do not combine). Since $\lambda + \lambda' = G$ one might hope to deduce something about G from Theorems 1.2 and 1.4. This is impossible by any black-box argument. Fix a Liouville number θ. For Lebesgue-almost every real x one has $\mu(x) = 2$ and $\mu(\theta - x) = 2$ (the exceptional sets are null and invariant under $x \mapsto \theta - x$), yet $x + (\theta - x) = \theta$ has $\mu = \infty$. So bounds on $\mu(\xi_1)$ and $\mu(\xi_2)$ say nothing about $\mu(\xi_1 + \xi_2)$, and the pair of results gives no measure for G (nor a new proof that G is irrational, which is OpenAI's theorem [11]). Theorems 1.2 and 1.4 concern two different linear forms, $G - 5\pi^2/48$ and $G + 5\pi^2/48$, in $1, G, \pi^2$.

Part 1. The rectangular hybrid frame and $\mu(\lambda) \leq 46.45$

10. The rectangular hybrid frame

The square frame of §efsec:matrix–§efsec:mirror uses h nodes on both sides of the Löwner matrix. In this part we use a *rectangular* frame: the rows and the columns share a long common block of nodes C, and carry in addition two short, interleaved blocks X' (rows) and Y' (columns) of the same size m. The new feature is the following. The determinant of the square matrix of size $|C|+m$ is divisible, over $\mathbb{Q}(x)$, by $\det$ of the $m \times m$ block $R_{X'Y'}$, and the quotient is a monic polynomial P_r of degree $r = |C|$ in the variable x; but the *decay* of the determinant is that of a square frame with $(1+\alpha)r$ nodes ($\alpha = m/r$), while the ℓ-adic window available for nonvanishing is shorter by the factor $1 + \alpha$. Taking $\alpha = \frac{1}{8}$ gives a much better balance between decay, content and the length of the interval of primes ℓ.

Throughout this part $\alpha = \frac{1}{8}$ unless said otherwise. All statements are labelled as in the rest of the paper; computer-assisted steps are marked as such, and the points at which the argument rests on external citations, on computation, or on material that is new in this text (and so was not covered by the audits of §18) are listed in §17.3, together with the limitations of this part.

10.1. **Notation.**

The kernel. For integers $a, b \geq 0$ put

$$\mathsf{R}(a,b) = \frac{1}{b-a} \sum_{a<k\leq b} \frac{1}{16k(4k-1)} \quad (a<b), \qquad \mathsf{R}(b,a) = \mathsf{R}(a,b), \qquad \mathsf{R}(a,a) = \Psi(a) := \sum_{k=1}^{a} \Big(\frac{1}{16k^2} - \frac{1}{(4}$$

Thus $M_0 = -\mathsf{R}|_{[0,h)^2}$ (the matrix of the Definition in §2), and the entries are exactly (minus) the entries of $\frac{1}{16}L_F$: for $a \neq b$, $\frac{1}{16}\frac{F(b)-F(a)}{b-a} = -\mathsf{R}(a,b)$, and for $a = b$, $\frac{1}{16}F'(a) = \lambda - \Psi(a)$ (by (2.3)).

Geometries. Let $r, m \geq 0$ and $e \in \{0,1\}$. The *geometry* $\mathcal{G}(r,m,e)$ consists of the pairwise disjoint sets of nodes

$$C = \{0,1,\dots,r-1\} \cup \{r+2m \text{ if } e=1\}, \qquad X' = \{r+2i : 0 \leq i < m\}, \qquad Y' = \{r+2i+1 : 0 \leq i < m\}.$$

Its *scale* is $r = |C| - e$ (the length of the interval part of C), and $|C| = r+e$. We write $\mathcal{G}_1 = \mathcal{G}(r,m,0)$ ("member 1") and $\mathcal{G}_2 = \mathcal{G}(r,m,1)$ ("member 2"). In member 2 one extra common node $b = r+2m$ sits directly above the interleaved block $X' \cup Y'$, which occupies $[r, r+2m)$.

The matrices. For a geometry and a variable X let $\mathrm{Full}(X) = \mathrm{Full}_{\mathcal{G}}(X)$ be the square matrix of size $|C|+m$ with rows indexed by $C \sqcup X'$, columns by $C \sqcup Y'$ (in this order), and entries

$$\mathrm{Full}(X)[a][b] = \mathsf{R}(a,b) + X \cdot \mathbf{1}[a=b \in C].$$

(A diagonal entry occurs only at a common node $a = b \in C$; the nodes of X' and Y' are distinct from each other and from C.) For a real t put

$$\mathcal{T}_t = -\mathrm{Full}(-\lambda - t), \qquad \text{i.e.} \quad \mathcal{T}_t[a][b] = -\mathsf{R}(a,b) + (\lambda+t)\mathbf{1}[a=b\in C] = \tfrac{1}{16}L_F[a][b] + t\,\mathbf{1}[a=b\in C].$$

When $m = 0$ and $e = 0$, $\mathcal{T}_t = T_h + tI$ with $h = r$, so the hybrid frame contains the frame of §2–§9. In general

$$\det \mathrm{Full}(-\lambda - t) = (-1)^{|C|+m} \det \mathcal{T}_t. \tag{10.1}$$

The two sign conventions are both used: Full is convenient for p-adic analysis (§12, §14, §16), $\mathcal{T}_t$ for analysis (§13).

Lemma 10.1 (Schur complement; integrality)**.** *Let $m \geq 1$ and suppose $\det \mathsf{R}[X', Y'] \neq 0$ (the $m \times m$ block). Then $\det \mathrm{Full}(X) = \det \mathsf{R}[X', Y'] \cdot P(X)$ where $P(X) = \det(XI_C + M)$, $M = \mathsf{R}[C,C] - \mathsf{R}[C,Y']\mathsf{R}[X',Y']^{-1}\mathsf{R}[X',C]$, is monic of degree $|C|$. In particular $\det \mathrm{Full}(X)$ has degree $|C|$ in X with leading coefficient $\det \mathsf{R}[X',Y'] \neq 0$, and if $\mathrm{den}_F \in \mathbb{Z}_{>0}$ is the least integer with $\mathrm{den}_F \det \mathrm{Full}(X) \in \mathbb{Z}[X]$, then $Q(X) := \mathrm{den}_F \det \mathrm{Full}(X) \in \mathbb{Z}[X]$ has degree $|C|$, and for every rational $x_0 = u/v$ in lowest terms with $\det \mathrm{Full}(x_0) \neq 0$,*

$$|Q(x_0)| \geq v^{-|C|}.$$

Proof. Schur complement on the block of rows X' and columns Y' (which is X-free). The rest is Lemma 2.4: $v^{|C|}Q(u/v) \in \mathbb{Z} \setminus \{0\}$. □

That $\det \mathsf{R}[X', Y'] \neq 0$ for the geometries used below is proved in §16 (Corollary 16.3: it is an ℓ-adic unit for the primes ℓ that we use, hence nonzero).

10.2. **The theorem.** The goal is Theorem 1.3: $|\lambda - a/b| > b^{-46.45}$ for all large b, i.e. $\mu(\lambda) \le 46.45$ (computed value 46.447). This is the audited hybrid theorem of the title. It improves Theorem 1.2 (52.69) for $\lambda = G/2 - 5\pi^2/96$, equivalently $G - 5\pi^2/48 = 2\lambda$ (since $\mu(2\lambda) = \mu(\lambda)$). The hybrid for the mirror constant λ' ($\ell \equiv 1 \pmod 4$) has *not* been done; for λ' Theorem 1.4 (52.69) is the best bound. **The Lean formalization does not cover Theorem 1.3** (§18).

The theorem rests on five propositions about the geometries with $m = r/8 + O(1)$, stated in §10.3, and the deduction in §11. The constants are:

quantity	value	where
content Λ_F	≤ 3.7211401 $(= \frac{313}{105} + \frac{554}{840} + \frac{118/21 + 2\log 2}{64} - \frac{129}{4480})$	§12
decay D_F	≥ 3.9062695	§13 (computer-assisted
window c_{hyb}	≤ 7.6540647	§13 (computer-assisted
margin $\eta_{\mathrm{hyb}} = D_F - \Lambda_F$	≥ 0.1851294	
window of ℓ: $r/\ell \to \kappa$	$\kappa = \frac{3/4}{1+\alpha} = \frac{2}{3}$	§14
exponent $c_{\mathrm{hyb}}(1/\eta_{\mathrm{hyb}} + \kappa)$	$\le 46.4471 < 46.45$	§17

10.3. **The five inputs.** Call a geometry $\mathcal{G}(r, m, e)$ *admissible* if $|m - r/8| \le 1$. All $o(1)$ below are as $r \to \infty$, uniform over admissible geometries. For a geometry let den_F be as in Lemma 10.1, and $N_{\mathrm{tot}} = |C| + m = r + e + m$.

Proposition 10.2 (content)**.** $\log \mathrm{den}_F \le (\Lambda_F + o(1))\, r^2$ *with* $\Lambda_F = 3.7211401$.

Proposition 10.3 (decay and window; computer-assisted)**.** *Let* $D_F = 3.9062695$ *and* $c_{\mathrm{hyb}} = 7.6540647$. *For every real* t *with* $|t| \le e^{-c_{\mathrm{hyb}} r}$,

$$|\det \mathcal{T}_t| \le \exp(-(D_F - o(1))\, r^2).$$

Proposition 10.4 (nonvanishing)**.** *Let* $\ell \equiv 3 \pmod 4$ *be a prime,* $\ell \ge 7$, $K = (\ell + 1)/4$, $0 \le m \le K - 1$ *and* $r_1 = \ell - K - m$. *For every rational* x_0 *with* $\ell \nmid \mathrm{den}\, x_0$, $\det \mathrm{Full}_{\mathcal{G}}(x_0) \ne 0$ *for* $\mathcal{G} = \mathcal{G}(r_1, m, 0)$ *or* $\mathcal{G} = \mathcal{G}(r_1, m, 1)$; *moreover* $v_\ell(\det \mathrm{Full}_{\mathcal{G}}(x_0)) = -2(|C_{\mathrm{high}}| + m)$ *for such a* $\mathcal{G}$.

Proposition 10.5 ($\ell \mid \mathrm{den}$)**.** *In the setting of Proposition 10.4, let* $x_0 = u/v$ *with* $v_\ell(v) = e_0 \ge 1$, $e_0 \ne 2$. *Then* $\det \mathrm{Full}_{\mathcal{G}}(x_0) \ne 0$ *for* both $\mathcal{G} = \mathcal{G}(r_1, m, 0)$ *and* $\mathcal{G}(r_1, m, 1)$ *(for* $m \ge 1$*).*

11. The deduction

Theorem 11.1. *Assume Propositions 10.2–10.5. Let* $\eta_{\mathrm{hyb}} = D_F - \Lambda_F$, $c = c_{\mathrm{hyb}}$ *and* $\kappa = \frac{2}{3}$. *For every* $\nu > c\,(1/\eta_{\mathrm{hyb}} + \kappa)$ *there are only finitely many rationals* p/q *with* $|\lambda - p/q| < q^{-\nu}$.

Proof. Let $\nu > c(1/\eta_{\mathrm{hyb}} + \kappa)$ and suppose $|\lambda - p/q| =: |t| < q^{-\nu}$ with q large; $L = \log q$. Choose $\delta > 0$ with $\nu > c(1+\delta)(1/\eta_{\mathrm{hyb}} + \kappa)$ and put

$$k = \frac{1+\delta}{\eta_{\mathrm{hyb}}}, \qquad X = \tfrac{3}{2}kL, \qquad Y = X + (1+\delta)L.$$

Choice of ℓ. Exactly as in §8: by the prime number theorem in progressions modulo 4, the primes $\ell \equiv 3 \pmod 4$ in $(X, Y]$ have product $> q^{1/2}$, while those with $v_\ell(q) = 2$ have product $\le q^{1/2}$. So there is a prime $\ell \equiv 3 \pmod 4$, $X < \ell \le Y$ (so $\ell \ge 7$), with $v_\ell(q) \ne 2$.

Choice of the geometry. Put $K = (\ell+1)/4$, $m = \mathrm{round}((\ell - K)/9)$ and $r_1 = \ell - K - m$. Then $\ell - K = 9m + \delta''$ with an integer δ'', $|\delta''| \le \frac{9}{2}$ (as $|(\ell-K)/9 - m| \le \frac{1}{2}$), so $r_1 = 8m + \delta''$, whence $|m - r_1/8| \le \frac{9}{16} < 1$: both geometries $\mathcal{G}(r_1, m, e)$, $e \in \{0,1\}$, which have the same scale r_1, are admissible. Also $0 \le m \le K - 1$ (for $\ell \ge 7$: $m = 1 = K - 1$ if $\ell = 7$, and $m \le K - 2$ for all $\ell \ge 11$, since $(\ell - K)/9 + \frac{1}{2} \le K - 2$ for $K \ge 3$; $K = 3$ gives $m = 1$). Moreover $|r_1 - \frac{2}{3}\ell| \le 1$, since $\ell - K = \frac{3\ell - 1}{4}$ and $r_1 = \frac{8}{9}(\ell - K) + \frac{\delta''}{9} = \frac{2}{3}\ell - \frac{2}{9} + \frac{\delta''}{9}$ with $\frac{|\delta''|}{9} \le \frac{1}{2}$.

Let $x_0 = -p/q$. If $\ell \nmid q$, Proposition 10.4 gives $e \in \{0,1\}$ with $\det \mathrm{Full}_{\mathcal{G}}(x_0) \ne 0$ for $\mathcal{G} = \mathcal{G}(r_1, m, e)$. If $\ell \mid q$ then $v_\ell(q) \in \{1\} \cup [3, \infty)$ and Proposition 10.5 gives the same (here $m \ge 1$ because ℓ is large). In every case let $\mathcal{G}$ be such a geometry, $r = r_1$ its scale, $|C| = r + e$. Then

$$kL - 1 < \tfrac{2}{3}X - 1 \le r_1 = r \le \tfrac{2}{3}\ell + 1 \le \tfrac{2}{3}Y + 1 = (1+\delta)\left(\tfrac{1}{\eta_{\mathrm{hyb}}} + \tfrac{2}{3}\right)L + 1, \tag{11.1}$$

(using $r_1 \ge \frac{2}{3}\ell - 1 > \frac{2}{3}X - 1 = kL - 1$), and $r \ge 2^{30}$ for large q.

Lower bound. By Lemma 10.1, with $Q = \mathrm{den}_F \det \mathrm{Full}_{\mathcal{G}}$: $|Q(-p/q)| \ge q^{-(r+e)} = e^{-(r+1)L}$.

Upper bound. Write $-p/q = -\lambda - t$ with $t = p/q - \lambda$. By (10.1), $|\det \mathrm{Full}_{\mathcal{G}}(-p/q)| = |\det \mathcal{T}_t|$. By (11.1) and the choice of δ, $c\,r < \nu L$ for large q, so $|t| < e^{-\nu L} \le e^{-cr}$ and Proposition 10.3 gives $|\det \mathcal{T}_t| \le e^{-(D_F - o(1))r^2}$. With Proposition 10.2,

$$|Q(-p/q)| \le \mathrm{den}_F\, |\det \mathcal{T}_t| \le \exp(-(\eta_{\mathrm{hyb}} - o(1))r^2).$$

Contradiction. Comparing, $(\eta_{\mathrm{hyb}} - o(1))r^2 \le (r+1)L$, i.e. $(\eta_{\mathrm{hyb}} - o(1))r \le L(1 + 1/r)$. But $r > kL - 1$ gives $\eta_{\mathrm{hyb}} r \ge (1+\delta)L - \eta_{\mathrm{hyb}}$. Contradiction for q large. □

Remark 11.2 (where each constant enters). The exponent is $c\,(1/\eta + \kappa)$: $c/\eta = 41.35$ is the cost of a window of exponential radius around λ for a determinant of size ηr^2; $c\kappa = 5.10$ is the cost of the length $(1+\delta)L$ of the interval of ℓ multiplied by $r/\ell \to \kappa = 2/3$. Proposition 10.5 is what allows $\ell \mid q$ (without it the interval has length $(2+\delta)L$ and κ becomes 2κ, giving ≈ 51.5). Compared with the square frame ($\kappa = \frac{3}{4}$, $c = 6.872$, $\eta = 0.1304$) the hybrid has a larger window constant c and a smaller κ, but a much larger η. The derivative of the exponent is $\partial/\partial D_F = -223$, so one more digit of D_F matters.

12. The denominator of the hybrid determinant

This section proves Proposition 10.2. The structure is that of §3: a tropical bound for each prime power with an explicit dual certificate, summed with the prime number theorem in progressions. Two things are new. First, the dual certificate of Lemma 3.3 has to be replaced by asymmetric certificates (D1–D6 below), because rows and columns of the rectangular frame differ. Second, at a positive proportion of primes the tropical bound is not sharp, and a p-adic rank argument (Lemma E, §12.5) removes m units from it; this is what turns the tropical content 3.7499 into $\Lambda_F = 3.7211$.

12.1. **Gauss valuations and the tropical lemma.** Fix a prime p. Let $\mathcal{O}_p \subset \mathbb{Q}(X)$ be the localization of $\mathbb{Z}_{(p)}[X]$ at the prime ideal (p); it is a discrete valuation ring with uniformizer p and residue field $\mathbb{F}_p(X)$. Its valuation v is the *Gauss valuation*: for $f \in \mathbb{Q}[X]$, $v(f) = \min_k v_p(\text{coefficient of } X^k)$, and $v(fg) = v(f) + v(g)$ (Gauss's lemma). For $f \in \mathbb{Q}[X]$ with least common denominator $\operatorname{den} f$ of the coefficients, $v_p(\operatorname{den} f) = \max(0, -v(f))$. In particular, for $f = \det \mathrm{Full}(X)$,

$$v_p(\mathrm{den}_F) = \max\bigl(0, -v(\det \mathrm{Full}(X))\bigr). \tag{12.1}$$

Lemma 12.1 (tropical expansion)**.** *Let B be a square matrix with entries in $\mathbb{Q}[X]$ (rows I, columns J), $W_{ab} = \max(0, -v(B_{ab}))$ and $\tau(W) = \max_\sigma \sum_a W_{a\sigma(a)}$, over bijections $\sigma : I \to J$. Then $v(\det B) \geq -\tau(W)$. If $u_a, v_b \geq 0$ (or real numbers) satisfy $W_{ab} \leq u_a + v_b$ for all a, b, then $\tau(W) \leq \sum_a u_a + \sum_b v_b$.*

Proof. Each term of the Leibniz expansion has $v(\prod_a B_{a\sigma(a)}) = \sum_a v(B_{a\sigma(a)}) \geq -\sum_a W_{a\sigma(a)} \geq -\tau(W)$, and v of a sum is at least the minimum. For the second claim $\sum_a W_{a\sigma(a)} \leq \sum_a (u_a + v_{\sigma(a)})$. □

For $\mathrm{Full}(X)$ the entry $\mathsf{R}(a,b) + X\mathbf{1}[a = b \in C]$ has $v = \min(0, v_p(\mathsf{R}(a,b)))$ (as $v(X) = 0$), so *for every entry*

$$W_{ab} = W_{ab}(p) = \max\bigl(0, -v_p(\mathsf{R}(a,b))\bigr). \tag{12.2}$$

(Compare Lemma 3.2, whose proof by principal minors is replaced here by the multiplicativity of the Gauss valuation; this is the form needed for rectangular matrices.)

12.2. **The level majorant.** Let p be an odd prime and $\nu(k) = v_p(k(4k-1))$, $\mu(a) = \max_{1 \leq k \leq a} \nu(k)$. Since 16 is a p-adic unit, the computation leading to (3.1) gives, for $a < b$ and for the diagonal,

$$W_{ab} \leq v_p(b-a) + \max_{a < k \leq b} \nu(k), \qquad W_{aa} \leq 2\mu(a) \quad (a \in C).$$

For a prime power $q = p^e$ put $S_q = \{k \geq 1 : q \mid k(4k-1)\} = \{k \equiv 0 \text{ or } k_q \ (q)\}$, where $k_q = (j_q q + 1)/4$, $j_q \in \{1, 3\}$, $j_q q \equiv 3 \pmod 4$ (as in §3; $4k_q \equiv 1 \pmod q$), and let $\beta(a) = \#(S_q \cap [1, a])$. The consecutive elements

of S_q are at distance k_q and $q-k_q$ alternately, both $< q$, and the first block $[0,k_q)$ has length $< q$. Define

$$w^{(q)}(a,b) = \mathbf{1}[\beta(a)\neq\beta(b)] + \mathbf{1}[q \mid a-b]\ (a\neq b), \qquad w^{(q)}(c,c) = 2\cdot\mathbf{1}[c\geq k_q]\ (c\in C).$$

Because $v_p(n) = \sum_{e\geq 1}\mathbf{1}[p^e \mid n]$ and $\max_k \nu(k) = \sum_e \mathbf{1}[\exists k \in (a,b] : p^e \mid k(4k-1)]$, we get, exactly as in §3,

$$W(p) \leq \sum_{e\geq 1} w^{(p^e)} \text{ entrywise}, \qquad \tau(W(p)) \leq \sum_{e\geq 1}\tau(w^{(p^e)}), \tag{12.3}$$

and $w^{(q)} \in \{0,1,2\}$: if $q \mid a-b$, $a\neq b$, then $|a-b|\geq q$, so $(a,b]$ contains a multiple of q and $\beta(a)\neq\beta(b)$; hence $w=2$ on congruent pairs and on high diagonal cells, $w=1$ on non-congruent pairs in different blocks, $w=0$ otherwise. Moreover $w^{(q)}\equiv 0$ if $q > 4N'$, where N' bounds all indices (then $k_q > N'$ and no two indices are congruent).

12.3. **Dual certificates.** Let $8m \leq r$, $N = r+2m$, and consider the geometry $\mathcal{G}(r,m,0)$ (indices in $[0,N)$; $Z=[r,N)$ the zone, X' the even and Y' the odd positions of Z relative to r). Fix an odd prime power q, $j=j_q$, $k=k_q$, and write $w = w^{(q)}$ on (rows $C\sqcup X'$) $\times$ (columns $C\sqcup Y'$). As in Lemma 3.3 let

$$\delta_A(q) = \min\bigl(2r,\ 2(r-n) + 2\min(n,(r-q)_+)\bigr), \qquad n = \min(r,k_q),$$

the value of the C-block. Let $\varphi : C\to\{0,1\}$ be the potential of Lemma 3.3 with $h=r$: $\varphi(c)=1$ for $c\geq k_q$ ("high") and $\varphi(c)=\mathbf{1}[c\leq r-1-q]$ for $c<k_q$. By the proof of Lemma 3.3, $w(c,c')\leq\varphi(c)+\varphi(c')$ for $c,c'\in C$ (including the diagonal) and $2\sum_c\varphi(c) = \delta_A(q)$. For $q\leq r$ the set $U=[r-q,k_q)$ (empty if $r-q\geq k_q$) consists of the "low unpaired" indices ($\varphi=0$); a zone node x is *low* if $x-q\in U$, and L denotes the number of low zone nodes; for $q\geq 2m$, $L=\min(|U|,2m)$ and $|U| = (k_q-r+q)_+$.

Proposition 12.2 (dual certificates)**.** *Let q be an odd prime power. Then $\tau(w^{(q)}) \leq B(q) := \delta_A(q)+E(q)$, where*

range	$E(q)$	*certificate*
(R1) $q<2m$	$m+M_q$, $M_q = m-q\lVert m/q\rVert$	*D1*
(R2) $2m\leq q\leq r$	$\min(2m+L,\ m+L+\lvert U\rvert)$	*D2, D3*
(R3) $r<q<N$	$2m+(N-q)$	*D4*
(R4) $q\geq N$, $k_q\leq r$	$2m$	*D5*
(R5) $q\geq N$, $k_q>r$	$(N-k_q)_+$	*D6*

($\lVert\cdot\rVert$ is the distance to the nearest integer.) In (R1) $\delta_A = 2r$.

Proof. In every case we exhibit potentials u_a (rows) and v_b (columns), ≥ 0, with $w(a,b)\leq u_a+v_b$ everywhere, and apply Lemma 12.1. Two remarks used repeatedly: (i) a pair of distinct indices that are not congruent mod q has $w\leq 1$; (ii) pairs of nodes of the zone are not congruent, since $|x-y|\leq 2m-1<q$ when $q\geq 2m$ (and $q\neq 2m$ is odd).

D1 (q arbitrary; used for $q < 2m$). For a residue class ϱ mod q let x_ϱ, y_ϱ be the numbers of nodes of X', Y' in ϱ. Rows in class ϱ get $u = \frac{1}{2}, \frac{3}{2}, 1$ and columns $v = \frac{3}{2}, \frac{1}{2}, 1$ according as $x_\varrho > y_\varrho$, $x_\varrho < y_\varrho$, $x_\varrho = y_\varrho$. Pairs in the same class (congruent pairs and all diagonal cells) have $u + v = 2 \geq w$; pairs in different classes have $w \leq 1 \leq \frac{1}{2} + \frac{1}{2} \leq u + v$. The value is $\sum_\varrho [(\#C_\varrho + x_\varrho)u + (\#C_\varrho + y_\varrho)v] = 2r + 2m - \frac{1}{2}\sum_\varrho |x_\varrho - y_\varrho|$. The nodes of Z are $2m$ consecutive integers alternating between X' and Y', and q is odd, so the members of one class alternate in type: $|x_\varrho - y_\varrho| \in \{0, 1\}$, with 1 iff the class has an odd number of members. Writing $2m = Aq + s'$ $(0 \leq s' < q)$, s' classes have $A + 1$ members and $q - s'$ have A; the number of odd classes is s' if A is even and $q - s'$ if A is odd, which equals $2q\|m/q\|$ in both cases. So the value is $2r + 2m - q\|m/q\| = 2r + m + M_q$. For $q < 2m \leq r/4$ one has $k_q < q \leq r - q$, whence $\delta_A = 2r$.

D2 ($2m \leq q \leq r$). On C take φ (both for rows and columns); a zone node x (row or column) gets $1 + \mathbf{1}[x \text{ low}]$. A zone–$C$ pair with $\varphi(c) = 1$ has $u + v \geq 2 \geq w$. If $\varphi(c) = 0$, i.e. $c \in U$: either $z \not\equiv c$ and $w \leq 1 \leq u_z$, or $z \equiv c$, and then $0 < z - c \leq (N - 1) - (r - q) < 2q$ forces $z = c + q$, so z is low and $u_z = 2$. Zone–zone pairs have $w \leq 1 \leq 1 + 1$. The value is $\delta_A + 2m + L$.

D3 ($2m \leq q \leq r$). Start from φ on C. (1) For each *high* zone node $z \in X'$ put $u = \frac{1}{2}$ on all rows and $v = \frac{3}{2}$ on all columns of its residue class; for each high $y \in Y'$ put $u = \frac{3}{2}$ on rows and $v = \frac{1}{2}$ on columns of its class. These classes contain no other zone node (zone nodes are pairwise incongruent) and no index of U (an index $c \in U$ congruent to a zone node x forces $x = c + q$, i.e. x low); so every C-member of such a class has $\varphi = 1$ and $u + v = 2$ as before. (2) For $c \in U$ put $u_c = v_c = \frac{1}{2}$. (3) A low zone node gets potential $\frac{3}{2}$, a high one $\frac{1}{2}$ (as a row, resp. column; these agree with the class rule in (1)). Check: every potential is $\geq \frac{1}{2}$, which settles all non-congruent pairs ($w \leq 1$). Congruent pairs: inside a class modified in (1), $\frac{1}{2} + \frac{3}{2} = 2$; low x with $x - q \in U$: $\frac{3}{2} + \frac{1}{2} = 2$; low x with $x - 2q, x - 3q, \ldots$ $(< r - q)$, which have $\varphi = 1$ and unmodified class: $\frac{3}{2} + 1$; C–C congruent pairs in unmodified classes: $1+1$ (indices of U have no congruent partner in C since $c+q \geq r$ and $c - q < 0$); diagonal cells: $\frac{1}{2} + \frac{3}{2}$ in a modified class, $1 + 1$ in an unmodified one, and the diagonal cells of U have $w = 0$. The value is $\delta_A + |U| + m + L$ (the potentials of U add $|U|$, the zone adds $\frac{3}{2}L + \frac{1}{2}(2m - L) = m + L$).

D4 ($r < q < N$). $\varphi' = \mathbf{1}[c \geq k_q]$ on C and $1 + \mathbf{1}[x \geq q]$ on the zone. Since $2q > N$ the only congruences inside $[0, N)$ are $x \equiv x - q$ with $x \geq q$, and then $x - q < 2m < r$ lies in C and $u_x = 2$. C–C: no congruences ($q > r$), so $w \leq 1$, and $w = 0$ inside the first block; if one of c, c' is high, φ' gives ≥ 1 (or 2 for both high, and for high diagonal cells). The value is $2(r - k_q)_+ + 2m + (N - q)$, and $\delta_A = 2(r - n) = 2(r - k_q)_+$ here.

D5 ($q \geq N$, $k_q \leq r$). φ' on C, potential 1 on the zone. There are no congruences, so $w \leq 1$ off the diagonal, and the diagonal cells are covered by φ'. Value $2(r - k_q) + 2m = \delta_A + 2m$.

D6 ($q \ge N$, $k_q > r$). Potential 0 on C and $\mathbf{1}[x \ge k_q]$ on zone nodes. The only element of S_q below N is k_q (if $k_q < N$). Then $w(z,c) = \mathbf{1}[z \ge k_q]$ for $c \in C$; for two zone nodes $w = 1$ only if $k_q \in (\min, \max]$, and then the larger node has potential 1; $C \times C$ has $w = 0$ (one block, no congruences, no high diagonal cell). Value $(N - k_q)_+$, and $\delta_A = 0$. □

Remark 12.3. These certificates are also sharp (the corresponding primal assignments exist, and the exact tropical value computed by the Hungarian algorithm equals $B(q)$ for all odd q and all tested (r,m), with τ_p computed from the exact entries equal to $\sum_e B(p^e)$ for every prime $p \ge 17$). Sharpness is not used.

12.4. **Asymptotics of the level bounds.** Put $\theta = q/r$ and $a = m/r$. For $j \in \{1,3\}$ let g_j be the function of §3 (so that $\delta_A(q) = r\,g_j(\theta) + O(1)$ and $\int g_1 = \frac{21}{5}$, $\int g_3 = \frac{37}{21}$), and let $\varepsilon_j(\theta; a)$ be the piecewise function

θ	$\varepsilon_j(\theta; a)$
$(0, 2a)$	$2a - \theta\|a/\theta\|$
$(2a,\ 4/(4+j))$	a
$(\frac{4}{4+j},\ \frac{4(1+2a)}{4+j})$, $\ell := \frac{(4+j)\theta}{4} - 1$	$\min(a + 2\ell,\ 2a + \ell)$
$(\frac{4(1+2a)}{4+j},\ 1)$	$4a$
$(1,\ 1 + 2a)$	$4a - (\theta - 1)$
$(1 + 2a,\ 4/j)$	$2a$
$(4/j,\ 4(1 + 2a)/j)$	$1 + 2a - j\theta/4$
$> 4(1 + 2a)/j$	0

(for $a \le \frac{1}{8}$; for $j = 1$ the fourth interval is empty when $a = \frac{1}{8}$; the formula is the translation of Proposition 12.2: in (R2) $|U| = \ell r + O(1)$, $L = \min(\ell, 2a)r + O(1)$, so that $E = \min(2m + L, m + L + |U|)$ is $\min(a + 2\ell, 2a + \ell)r$ for $\ell \le 2a$ and $4ar$ beyond; in (R3) $E = (2a + (1 + 2a - \theta))r$; in (R4) $E = 2m = 2ar$; in (R5) $N - k_q = (1 + 2a - j\theta/4)r + O(1)$.) One checks directly that ε_j is continuous in θ (and jointly continuous in (θ, a)), bounded by $4a$, supported on $[0, 4(1 + 2a)/j]$, and Lipschitz in a with constant ≤ 8 (each piece has $|\partial_a| \le 4$). For every odd prime power q and all r, m with $8m \le r$,

(12.4)

$$B(q) \le r\, f_j(q/r; a) + C_0, \qquad f_j = g_j + \varepsilon_j, \qquad j = j_q,\ a = m/r, \qquad C_0 = 0.$$

Proof of (12.4). Put $\theta = q/r$ and $n' = \min(r, jq/4)$. The function $\delta(n) = \min\big(2r, 2(r - n) + 2\min(n, (r - q)_+)\big)$ is non-increasing in n (constant for $n \le (r - q)_+$, then of slope -2), and $\delta(n') = r g_j(\theta)$: for $\theta \le \frac{4}{4+j}$ one has $n' \le r - q$ and $\delta = 2r$; for $\frac{4}{4+j} \le \theta \le 1$, $\delta = 2(r - \frac{jq}{4}) + 2(r - q)$; for $1 \le \theta \le \frac{4}{j}$, $\delta = 2(r - \frac{jq}{4})$; beyond, $\delta = 0$. Since $n_q = \min(r, k_q) \ge n'$ ($k_q = \frac{jq}{4} + \frac{1}{4}$), $\delta_A(q) \le r g_j(\theta)$. It remains to compare $E(q)$ with $r\varepsilon_j(\theta; a)$; as $a \le \frac{1}{8}$, $2a < \frac{4}{4+j}$, $\frac{4(1+2a)}{4+j} \le 1$ and $1 + 2a < \frac{4}{j}$, so the intervals of the table are

in the stated order. (R1) $\theta < 2a$: $E = 2m - q\|m/q\| = r(2a - \theta\|a/\theta\|) = r\varepsilon_j$, and $\delta_A = 2r = rg_j$. (R2) $2a \le \theta \le 1$: with $\ell = \frac{(4+j)\theta}{4} - 1$ one has $k_q - r + q = r\ell + \frac{1}{4}$, an integer, so $|U| = (r\ell + \frac{1}{4})_+$ and $L = \min(|U|, 2m)$. The function $\mathcal{E}(x) = \min\big(2m + \min(x, 2m),\, m + \min(x, 2m) + x\big)$ is non-decreasing and 2-Lipschitz, $E = \mathcal{E}(|U|)$, and $\mathcal{E}(r\ell_+) = r\varepsilon_j(\theta; a)$ (it is m, $\min(2m + r\ell, m + 2r\ell)$, $4m$ for $\ell \le 0$, $0 \le r\ell \le 2m$, $r\ell \ge 2m$). If $\ell < 0$, then $r\ell + \frac{1}{4} < \frac{1}{4}$ is a non-positive integer, $|U| = 0$ and $E = m = r\varepsilon_j$. If $\ell \ge 0$, then $|U| = r\ell + \frac{1}{4}$ and $E \le r\varepsilon_j + \frac{1}{2}$; but then $k_q > r - q$, $k_q < r$, and $\delta_A(q) = 2(r - k_q) + 2(r - q) = rg_j(\theta) - \frac{1}{2}$. In both cases $\delta_A + E \le r(g_j + \varepsilon_j)$. (R3) $r < q < N$: $E = 2m + (r + 2m - q) = r(4a - (\theta - 1)) = r\varepsilon_j$. (R4) $q \ge N$, $k_q \le r$: then $1 + 2a \le \theta < \frac{4}{j}$ and $E = 2m = r\varepsilon_j$. (R5) $q \ge N$, $k_q > r$: $E = (r + 2m - k_q)_+$. If $\theta < \frac{4}{j}$ then $E < 2m = r\varepsilon_j$; if $\theta \ge \frac{4}{j}$ then $E \le r(1 + 2a - \frac{j\theta}{4})_+ = r\varepsilon_j$, as $k_q > \frac{jq}{4}$. In every range $B(q) = \delta_A(q) + E(q) \le r f_j(\theta; a)$. □

(This agrees with the exact evaluation: $\max_q[B(q) - rf_j(q/r; a)] = 0$ over all odd q at $r = 10^4, 4 \cdot 10^4$, $m = r/8$, and at 21 further pairs (r, m), $97 \le r \le 10^4$, $m \in \{\lfloor r/8 \rfloor - 5, \lfloor r/8 \rfloor - 1, \lfloor r/8 \rfloor\}$; script `check_hB.py`.)

Lemma 12.4 (summation)**.** *Let $f : [0, \infty) \to [0, \infty)$ be bounded, Riemann integrable and supported in $[0, K_0]$, and $a \in \{1, 3\}$. Then* $\lim_{r\to\infty} \frac{1}{r} \sum_{p \equiv a\,(4)} f(p/r) \log p = \frac{1}{2} \int_0^\infty f.$

Proof. Given $\epsilon > 0$ choose a partition $0 = \theta_0 < \cdots < \theta_n = K_0$ with $\sum_i c_i(\theta_i - \theta_{i-1}) \le \int f + \epsilon$, $c_i = \sup_{[\theta_{i-1}, \theta_i]} f$. Every prime $p \le K_0 r$ lies in some $(\theta_{i-1} r, \theta_i r]$, so $\sum_p f(p/r) \log p \le \sum_i c_i[\vartheta(\theta_i r; 4, a) - \vartheta(\theta_{i-1} r; 4, a)]$. By the prime number theorem in progressions, $\vartheta(x; 4, a) = x/2 + o(x)$, so each bracket is $\frac{1}{2}(\theta_i - \theta_{i-1})r + o(r)$ (n fixed). Hence $\limsup \le \frac{1}{2}(\int f + \epsilon)$. The lower Darboux sum gives the same for $\liminf$. □

(Both an upper and a lower bound for $\vartheta(x; 4, a)$ are needed here, since f_j is not monotone; in §3 monotonicity of g_j made an upper bound enough.)

Proposition 12.5 (tropical content law)**.** *For $\mathcal{G}(r, m, e)$ admissible, the quantity $T(r) := \sum_{p \ge 5} \sum_e \tau(w^{(p^e)}) \log p$ (over all prime powers $p^e \le 4N'$) satisfies*

$$T(r) \le (\Lambda_{\mathrm{trop}} + o(1))r^2, \qquad \Lambda_{\mathrm{trop}} = \frac{313}{105} + \frac{554}{105}\alpha + \Big(\frac{118}{21} + 2\log 2\Big)\alpha^2\Big|_{\alpha = 1/8} = 3.7499347\ldots$$

Proof. Extension to admissible geometries. Let $m_0 = \lfloor r/8 \rfloor$, so $m_0 \le m \le m_0 + 2$ and $8m_0 \le r$. The matrix for $\mathcal{G}(r, m, e)$ is obtained from that of $\mathcal{G}(r, m_0, 0)$ by adding at most $2 + 1$ rows and the same number of columns (the added nodes are $r + 2i, r + 2i + 1$ for $m_0 \le i < m$ and b). Each $w^{(q)} \le 2$, so extending a certificate by the potential 2 on every new row and

column keeps $w \le u + v$ and adds ≤ 4 per new row-column pair. Hence $\tau(w^{(q)}(\mathcal{G}(r,m,e))) \le B_{m_0}(q) + 12$.

Prime powers p^e, $e \ge 2$, and $p \le 3$. For $q = p^e$ with $e \ge 2$ use the trivial potentials $u = v = 1$: $\tau(w^{(q)}) \le 2(N_{\mathrm{tot}})$, and $\sum_{p^e \le 4N', e \ge 2} \log p = \psi(4N') - \vartheta(4N') = O(\sqrt{r})$ (Chebyshev), so these levels contribute $O(r^{3/2})$. For $p = 3$ all levels $3^e \le 4N'$ are bounded by the trivial potentials, total $O(r \log r)$. For $p = 2$: every entry of R has $-v_2 \le 4 + 2\log_2 N'$ (from 16, k, $b - a$ and the ultrametric inequality), so $\tau_2 \le N_{\mathrm{tot}}(4 + 2\log_2 N')$, a contribution $O(r \log r)$.

Level 1. By (12.4) (applied with m_0), for $p \ge 5$, $p \le 4N'$: $\tau(w^{(p)}) \log p \le (r f_{j_p}(p/r; a_0) + C_0 + 12) \log p$, $a_0 = m_0/r \in (\frac{1}{8} - \frac{1}{r}, \frac{1}{8}]$, and f_j is 8-Lipschitz in a, so $r f_j(p/r; a_0) \le r f_j(p/r; \frac{1}{8}) + 8$. The $O(1)$ errors sum to $O(\vartheta(4N')) = O(r)$. By Lemma 12.4, $\sum_{p \equiv a_j (4)} r f_j(p/r; \frac{1}{8}) \log p = \frac{r^2}{2} \int f_j + o(r^2)$. Hence $T(r) \le \frac{r^2}{2}(\int f_1 + \int f_3) + o(r^2)$.

The integral. Put $I = \int_{1/2}^{\infty} \|t\| t^{-3} dt$. On each cell $[n - \frac{1}{2}, n + \frac{1}{2}]$ the antiderivative of $|t - n| t^{-3}$ is elementary; summing, $\sum_n [\frac{1}{n} - \frac{1}{2n-1} - \frac{1}{2n+1}] + \sum_n [\frac{1}{4(n-\frac{1}{2})^2} - \frac{1}{4(n+\frac{1}{2})^2}] = (1 - 2\log 2) + 1$, so $I = 2 - 2\log 2$. The substitution $t = a/\theta$ gives $\int_0^{2a} \theta \|a/\theta\| d\theta = a^2 I$. The seven pieces of $\int \varepsilon_j$ are $4a^2 - a^2 I$; $a(\frac{4}{4+j} - 2a)$; $\frac{22a^2}{4+j}$ (the middle piece: $\frac{4}{4+j} \int_0^{2a} \min(a + 2\ell, 2a + \ell) d\ell = \frac{4}{4+j} \cdot \frac{11a^2}{2}$); $\frac{4a(j-8a)}{4+j}$; $6a^2$; $2a(\frac{4}{j} - 1 - 2a)$; $\frac{8a^2}{j}$. Adding,

$$\int \varepsilon_1 = \tfrac{38}{5} a + (10 - I) a^2, \qquad \int \varepsilon_3 = \tfrac{62}{21} a + (\tfrac{110}{21} - I) a^2.$$

Therefore $\frac{1}{2} \sum_j \int f_j = \frac{1}{2}(\frac{21}{5} + \frac{37}{21}) + \frac{1}{2} \sum_j \int \varepsilon_j = \frac{313}{105} + \frac{554}{105} a + (\frac{160}{21} - I) a^2$, and $\frac{160}{21} - I = \frac{118}{21} + 2\log 2$. At $a = \frac{1}{8}$: $2.9809524 + 0.6595238 + 0.1094585 = 3.7499347$. □

(Negative control: at $\alpha = 0$ every ε_j vanishes and the law is $\frac{313}{105}$, the content constant of Theorem 3.1. The same bound computed from the exact entries, τ_p from exact p-adic valuations, agrees with $\sum_e B(p^e)$ for every prime $p \ge 17$ and $(r, m) \in \{(64, 8), (96, 12), (128, 16), (160, 20)\}$, and the finite-$r$ values of the full tropical bound are $3.7027, 3.6927, 3.7216, 3.7403$, increasing towards 3.7499.)

12.5. **Lemma E: a rank saving at clean primes.** The tropical bound is not sharp: at primes $p \in (\frac{5}{8}r, \frac{4}{5}r)$, $p \equiv 3 \pmod 4$, and $p \in (\frac{5}{7}r, r)$, $p \equiv 1 \pmod 4$, the true denominator is smaller by m units (up to $O(1)$). We prove this in the Gauss valuation of $\mathcal{O}_p$. Note that this is a statement about the polynomial $\det \mathrm{Full}(X)$ with X a *formal variable*: it is false for $\det \mathrm{Full}(x_0)$ at specialisations $x_0 \in \mathbb{Q}$ with $16 x_0 \equiv H^{(2)}_{K'-1} \pmod p$ (below), and it is never applied to a specialised value in this paper.

The case $e = 1$ (member 2, with the extra common node b) is treated below together with $e = 0$. This treatment is new in this text and is not

covered by the audits listed in §18; it was re-derived line by line by a further independent check, and confirmed by two independent exact computations (Remark 12.8).

Set-up. Let $\mathcal{G} = \mathcal{G}(r, m, e)$ be admissible, $p \geq 11$ prime, as specified below. Put $Z_e = [r, r + 2m + e)$ for the zone nodes: $X'' = X' \cup \{b\}$, $Y'' = Y' \cup \{b\}$ if $e = 1$ and $X'' = X', Y'' = Y'$ if $e = 0$. Take the *interval* $C_0 = [0, r)$, $A = \mathsf{R}[C_0, C_0] + XI$, and write

(12.5)

$$\mathrm{Full}(X) = \begin{pmatrix} A & \mathsf{R}[C_0, Y''] \\ \mathsf{R}[X'', C_0] & \mathsf{R}[X'', Y''] + eX\, E_{bb} \end{pmatrix}, \qquad \det \mathrm{Full}(X) = \det A \cdot \det T,$$

$T = \mathsf{R}[X'', Y''] + eX\, E_{bb} - \mathsf{R}[X'', C_0] A^{-1} \mathsf{R}[C_0, Y'']$ (an $m'' \times m''$ matrix, $m'' = m + e$). Here E_{bb} is the matrix unit at (b, b). ($\det A = X^r + \ldots \neq 0$.) By Lemma 12.1,

(12.6)

$$v(\det \mathrm{Full}(X)) \;\geq\; -\tau(W_A) + v(\det T), \qquad v(\det T) \geq \sum_z \min_y v(T_{zy}) \text{ (Leibniz)}.$$

Singular indices and levels. Let $F(n) = -\sum_{k \leq n} \frac{1}{4k(4k-1)}$, so $\mathsf{R}(a, b) = \frac{F(a) - F(b)}{4(b-a)}$. The indices k with $p \mid k(4k-1)$ are $k \equiv 0$ and $k \equiv 4^{-1} \pmod p$. Since $1/(4k-1)$ and $1/k$ have valuation ≥ -1 for $k < p^2/4$, $v(F(n)) \geq -1$ and $v(\Psi(n)) \geq -2$ on the indices considered. For n put $\mathsf{L}(n) = \mathrm{lead}_{-1} F(n) \in \mathbb{F}_p$ and $\pi(n) = \mathrm{lead}_{-2} \Psi(n)$ (residues of pF, $p^2\Psi$). They are step functions, jumping at singular k: if $4k - 1 = j'p$ the jump of L is $-1/j'$ and of π is $-1/j'^2$ (as $4k \equiv 1$); at $k = p$ the jumps are $+\frac{1}{4}$ and $+\frac{1}{16}$. Consequently, for $a \neq b$ among the indices:

(V1) if $p \nmid b - a$: $v(\mathsf{R}(a, b)) \geq -1$ and $\mathrm{lead}_{-1} \mathsf{R}(a, b) = \frac{\mathsf{L}(a) - \mathsf{L}(b)}{4(b-a)}$;

(V2) if $b - a = p$ (and no other multiples of p occur): $v(\mathsf{R}(a, b)) \geq -2$ and $\mathrm{lead}_{-2} \mathsf{R}(a, b) = \frac{\mathsf{L}(a) - \mathsf{L}(b)}{4}$.

Two further elementary facts: the *block-scaling lemma* (if $D = \mathrm{diag}(p^{d_i})$ and $\tilde{A} = DAD \in M(\mathcal{O}_p)$ has $\tilde{A} \bmod p$ invertible over $\mathbb{F}_p(X)$, then $A^{-1} = D\tilde{A}^{-1}D$ with $\tilde{A}^{-1} \in M(\mathcal{O}_p)$, because $\det \tilde{A}$ is a unit) and the *partner decomposition*: for a zone row z and a zone column y, $z \neq y$, whose partners $c_z = z - p$, $c_y = y - p$ lie in C_0 and are distinct, put $\ell_z(c) = \mathsf{R}(z, c) - \mathsf{R}(c_z, c) - X\mathbf{1}[c = c_z]$ ($c \in C_0$), so that $\mathsf{R}[z, C_0] = A[c_z, C_0] + \ell_z$ and likewise $\mathsf{R}[C_0, y] = A[C_0, c_y] + \ell_y^\top$. Since $A[c_z, \cdot]A^{-1} = e_{c_z}^\top$,

(12.7)

$$T_{zy} = \Delta^2\mathsf{R}(z, y) - \ell_z A^{-1} \ell_y^\top, \qquad \Delta^2\mathsf{R}(z, y) := \mathsf{R}(z, y) - \mathsf{R}(z, c_y) - \mathsf{R}(c_z, y) + \mathsf{R}(c_z, c_y).$$

(The variable X cancels in $\Delta^2\mathsf{R}$ because $c_z \neq c_y$.) With $\delta_z := F(z) - F(c_z)$ one computes, for $c \in C_0 \setminus \{c_z\}$ with $p \nmid c - z$,

$$\ell_z(c) = \frac{p\, \mathsf{R}(c_z, c)}{c - z} + \frac{\delta_z}{4(c - z)}, \qquad \ell_z(c_z) = -\frac{\delta_z}{4p} - \Psi(c_z) - X. \tag{12.8}$$

For the diagonal entry (b,b) (when $e=1$), with $c=c_b\in C_0$ and the same decomposition $\mathsf{R}[b,C_0]=A[c,C_0]+\ell_b$, $\mathsf{R}[C_0,b]=A[C_0,c]+\ell_b^\top$ (by symmetry of R): $\mathsf{R}[b,C_0]A^{-1}\mathsf{R}[C_0,b]=A(c,c)+2\ell_b(c)+\ell_bA^{-1}\ell_b^\top$ with $A(c,c)=\Psi(c)+X$ and $\ell_b(c)=\mathsf{R}(b,c)-\Psi(c)-X$, so $T_{bb}=\Psi(b)+X-\mathsf{R}(b,c)\cdot 2+\big(\Psi(c)+X\big)-\ell_bA^{-1}\ell_b^\top$.

Theorem 12.6 (Lemma E)**.** *Let $\mathcal{G}(r,m,e)$ be admissible, r large, $p\ge 11$ a prime, $p\neq 139$, and let v be the Gauss valuation at p.*

(a) If $p\equiv 3 \pmod 4$, $2p>r+2m$ and $5p+1<4r$, then $v(\det \mathrm{Full}(X))\ge -2r-2e$.

(b) If $p\equiv 1 \pmod 4$, $p<r$, $7p+1>4r+8m$ (for $e=0$ it suffices that $7p+1\ge 4r+8m$), and $K_p:=\frac{3p+1}{4}$, then $v(\det \mathrm{Full}(X))\ge -2(2r-p-K_p)-3m-4e$.

(For $m=r/8$, $e=0$ the conditions are exactly $5r<8p$, $5p+1<4r$, resp. $5r\le 7p+1$, $p<r$.)

Proof. Throughout, $V1$, $V2$ are used with all indices $<2p$ (this holds: all indices are $\le r+2m<2p$ in both cases), so that $b-a$ is a p-unit unless $|b-a|=p$.

Case (a), $p\equiv 3 \pmod 4$. Put $k_1=\frac{p+1}{4}$ (so $4k_1-1=p$), $k_2=\frac{5p+1}{4}=p+k_1$ ($4k_2-1=5p$). The singular indices below $r+2m+1$ are k_1,p,k_2: indeed $2p>r+2m$ and $\frac{9p+1}{4}>2p$. Also $k_2<r$ by $5p+1<4r$, and $p<r$. So $[0,r)$ splits into blocks $B_0=[0,k_1)$, $B_1=[k_1,p)$, $B_2=[p,k_2)$, $B_3=[k_2,r)$ and the zone $[r,r+2m]$ lies in the block above k_2, with levels and diagonal leads

block	B_0	B_1	B_2	B_3 (and zone)
L	0	-1	$-\frac{3}{4}$	$-\frac{19}{20}$
π	0	-1	$-\frac{15}{16}$	$-\frac{391}{400}$

(jumps $-\frac{1}{j}$, $+\frac{1}{4}$, $-\frac{1}{5}$ for L and $-\frac{1}{j^2},+\frac{1}{16},-\frac{1}{25}$ for π). The partner map $a\mapsto a+p$ within C_0 sends B_0 onto B_2 and $[k_1,r-p)$ onto B_3 (here $r-p\ge k_1$ by $5p+1\le 4r$); the set $\mathcal{U}=[r-p,p)\subset B_1$ is unpaired (for $a\in\mathcal{U}$: $a+p\ge r$ and $a-p<0$). Each zone node $z\in[r,r+2m]$ has partner $c_z=z-p\in[r-p,r-p+2m]\subset\mathcal{U}$ (as $r+2m-p<p$ by $2p>r+2m$).

Leading form of A. By V1, V2: $p^2A\in M(\mathcal{O}_p)$, and $p^2A \bmod p$ is the matrix M_0 with entries $\pi(a)$ on the diagonal, $\frac{\mathsf{L}(a)-\mathsf{L}(a+p)}{4}$ at $(a,a+p)$ and $(a+p,a)$, and 0 otherwise. After a permutation M_0 is block diagonal: 1×1 blocks (-1) on $\mathcal{U}$; 2×2 blocks $\left(\begin{smallmatrix}0 & 3/16\\ 3/16 & -15/16\end{smallmatrix}\right)$ (determinant $-\frac{9}{256}$) for $a\in B_0$; and $\left(\begin{smallmatrix}-1 & -1/80\\ -1/80 & -391/400\end{smallmatrix}\right)$ (determinant $\frac{6255}{6400}$, $6255=3^2\cdot 5\cdot 139$) for $a\in[k_1,r-p)$. Hence M_0 is invertible for $p\notin\{3,5,139\}$, and $N:=(p^2A)^{-1}\in M(\mathcal{O}_p)$ satisfies $A^{-1}=p^2N$, $N\equiv M_0^{-1} \pmod p$. Let P be the set of paired indices; block inversion of $p^2A=\left(\begin{smallmatrix}\tilde A_{\mathcal{U}\mathcal{U}} & \tilde A_{\mathcal{U}P}\\ \tilde A_{P\mathcal{U}} & \tilde A_{PP}\end{smallmatrix}\right)$ gives (the entries between distinct unpaired indices, and between $\mathcal{U}$ and P, are $p^2A(c,c')$ with $v(A(c,c'))\ge -1$,

i.e. lie in $p\mathcal{O}_p$): $N_{\mathcal{U}\mathcal{U}} = S^{-1}$ with $S = -I + pS_1$, $S_1(c,c') \equiv pA(c,c') = \operatorname{lead}_{-1} A(c,c') = \frac{\mathsf{L}(c)-\mathsf{L}(c')}{4(c'-c)} = 0 \pmod p$ for $c \neq c'$ in $\mathcal{U}$ (same level), and $N_{\mathcal{U}P} \in p\mathcal{O}_p$. Therefore

(12.9)

$$N(c,c) \equiv -1\ (p), \qquad N(c,c') \equiv 0\ (p^2) \text{ for } c \neq c' \in \mathcal{U}, \qquad N(c,c') \in p\mathcal{O}_p\ (c \in \mathcal{U}, c' \neq c).$$

Entries of T. Let $z \in X''$, $y \in Y''$, $z \neq y$, $d = y - z$ (a nonzero integer, $|d| \leq 2m < p$, a unit). Then $c_z \neq c_y$, both in $\mathcal{U}$. The first term of (12.7): z, y lie in one block, hence $v(\mathsf{R}(z,y)) \geq 0$ ($\mathsf{L}(z) = \mathsf{L}(y)$, d a unit); likewise $v(\mathsf{R}(c_z,c_y)) \geq 0$. With $\sigma := \mathsf{L}(z) - \mathsf{L}(c_z)$ (the same for all zone nodes), $\operatorname{lead}_{-1} \mathsf{R}(z,c_y) = \frac{\sigma}{4(d-p)} \equiv \frac{\sigma}{4d}$ and $\operatorname{lead}_{-1} \mathsf{R}(c_z,y) = -\frac{\sigma}{4(d+p)} \equiv -\frac{\sigma}{4d}$; these cancel mod p, so $v(\Delta^2\mathsf{R}) \geq 0$. For the second term, by (12.8): for $c \neq c_z$ we have $c - z \equiv c - c_z \not\equiv 0$, $p\ell_z(c) \equiv \frac{\sigma}{4(c-c_z)} \pmod p$ (since $p^2\mathsf{R}(c_z,c) \in p\mathcal{O}_p$), and $p^2\ell_z(c_z) \equiv -\frac{\sigma}{4} - \pi(c_z) =: -\beta \pmod p$, the same for all z (the signs: $\mathsf{R}(z,c_z) = -\delta_z/(4p)$). Put $\lambda_z = p^2\ell_z \in \mathcal{O}_p^{C_0}$: $\lambda_z(c_z) \equiv -\beta$, $\lambda_z(c) = pu_z(c)$ with $u_z(c) \equiv \frac{\sigma}{4(c-c_z)}$ ($c \neq c_z$). Then $\ell_z A^{-1} \ell_y^\top = p^{-2}\lambda_z N \lambda_y^\top$ and, by (12.9), modulo p^2: the (c_z,c_y) term is $\equiv 0$ because $N(c_z,c_y) \equiv 0\ (p^2)$; the terms with $c \neq c_z$ and $c' \neq c_y$ are $O(p^2)$; the terms (c_z,c'), $c' \neq c_z, c_y$ and (c,c_y) are $O(p^2)$ by $N(c_z,c') \in p\mathcal{O}_p$; the remaining ones are (c_z,c_z) and (c_y,c_y):

$$\begin{aligned}\lambda_z N \lambda_y^\top &\equiv (-\beta)(-1)\, p\, u_y(c_z) + p\, u_z(c_y)(-1)(-\beta) \\ &= p\beta\Big[\frac{\sigma}{4(c_z - c_y)} + \frac{\sigma}{4(c_y - c_z)}\Big] \equiv 0 \pmod{p^2}.\end{aligned}$$

So $v(\ell_z A^{-1} \ell_y^\top) \geq 0$ and $v(T_{zy}) \geq 0$ for all $z \neq y$.

The diagonal entry T_{bb} ($e = 1$). Every term in the formula for T_{bb} has $v \geq -2$: $\Psi(b), \Psi(c)$ by $v(\Psi) \geq -2$, $\mathsf{R}(b,c_b)$ by V2, and $\ell_b A^{-1} \ell_b^\top = p^{-2}\lambda_b N \lambda_b^\top$ with $\lambda_b \in \mathcal{O}_p^{C_0}$, $N \in M(\mathcal{O}_p)$. So $v(T_{bb}) \geq -2$.

Conclusion. For $e = 0$: $v(\det T) \geq 0$ by (12.6); for $e = 1$ every permutation term of $\det T$ either contains T_{bb} (then $v \geq -2$) or $T_{by}T_{zb}$ with all other factors ≥ 0: $v(\det T) \geq -2$. Finally $\tau(W_A) \leq 2r$ since all weights are ≤ 2. This proves (a).

Case (b), $p \equiv 1 \pmod 4$. Now $4^{-1} \equiv K_p$, singular indices $K_p, p, \frac{7p+1}{4} = K_p + p$ ($4k-1 = 7p$) and $2p$. By the hypotheses $K_p < p < r$, $K_p + p > r + 2m$ (since $7p + 1 > 4r + 8m$; in the case $e = 0$, $=$ allowed, because the index $K_p + p = r + 2m$ is then just above the zone $[r, r+2m)$) and $2p > r + 2m$ (as $p > \frac{5}{7}r$). Blocks $B_0 = [0, K_p)$, $B_1 = [K_p, p)$, $B_2 = [p, r)$ and the zone, with $\mathsf{L} = 0, -\frac{1}{3}, -\frac{1}{12}$ and $\pi = 0, -\frac{1}{9}, -\frac{1}{9} + \frac{1}{16} = -\frac{7}{144}$. Partners: $P_0 = [0, r-p) \subset B_0$ ($r - p < K_p$) is paired with B_2; B_1 is unpaired ($K_p + p > r$); $\mathcal{U} = [r-p, K_p)$ is unpaired. Let $P = P_0 \cup B_1 \cup B_2$, $|P| = 2r - p - K_p$. The partner of a zone node z is $c_z = z - p \in [r-p, r-p+2m] \subset \mathcal{U}$, since $r - p + 2m < K_p$ is the hypothesis $4r + 8m < 7p + 1$.

Scaling and the scalar Schur complement. Let $D = \operatorname{diag}(1 \text{ on } \mathcal{U},\ p \text{ on } P)$ and $\tilde{A} = DAD$. By V1, V2: $\tilde{A} \in M(\mathcal{O}_p)$ ($\mathcal{U} \times \mathcal{U}$ entries have $v \geq 0$; $\mathcal{U} \times P$

entries have $v \geq -1$ since no pair in $\mathcal{U} \times P$ differs by p; $P \times P$ entries have $v \geq -2$). Modulo p: $\tilde{A}_{PP} \equiv M_{0P}$, block diagonal with 1×1 blocks $-\frac{1}{9}$ on B_1 and 2×2 blocks $\left(\begin{smallmatrix} 0 & e \\ e & \pi_2 \end{smallmatrix}\right)$, $e = \frac{1}{48}$, $\pi_2 = -\frac{7}{144}$, on the pairs $(a, a+p)$; $\tilde{A}_{\mathcal{U}P} \equiv V$ with $V_{u,c} = \frac{1}{12(c-u)}$ $(c \in B_1)$, $\frac{1}{48(c-u)}$ $(c \in B_2)$, 0 $(c \in P_0)$; and $\tilde{A}_{\mathcal{U}\mathcal{U}} \equiv \bar{A}_{\mathcal{U}\mathcal{U}}$ (the reduction of $\mathsf{R} + XI$). We claim

$$\bar{S} := \bar{A}_{\mathcal{U}\mathcal{U}} - V M_{0P}^{-1} V^{\top} = s_0 I_{\mathcal{U}}, \qquad s_0 = X - \tfrac{1}{16} H^{(2)}_{K_p - 1} \in \mathbb{F}_p(X)^{\times}. \tag{12.10}$$

Proof of (12.10). The inverse of $\left(\begin{smallmatrix} 0 & e \\ e & \pi_2 \end{smallmatrix}\right)$ has zero (B_2, B_2) entry, and V vanishes on P_0; so only the 1×1 blocks contribute, and $\bar{S} = \bar{A}_{\mathcal{U}\mathcal{U}} + 9\, V_{B_1} V_{B_1}^{\top} = \bar{A}_{\mathcal{U}\mathcal{U}} + \frac{1}{16} W W^{\top}$, $W_{u,c} = \frac{1}{c-u}$, $K_p \leq c \leq p-1$. For $n < K_p$, using $4K_p \equiv 1$ so $\frac{4}{4k-1} \equiv \frac{1}{k - K_p}$: $4F(n) \equiv \varphi(n) := H_n + H_{K_p - 1} - H_{K_p - 1 - n}$ and $16\Psi(n) \equiv H^{(2)}_n - H^{(2)}_{K_p-1} + H^{(2)}_{K_p-1-n}$. Hence $16\bar{A}_{uw} = \frac{\varphi(u) - \varphi(w)}{w-u}$ for $u \neq w$ in $\mathcal{U}$. Partial fractions: $\sum_{c=K_p}^{p-1} \frac{1}{(c-u)(c-w)} = \frac{[H_{p-1-u} - H_{K_p-1-u}] - [H_{p-1-w} - H_{K_p-1-w}]}{u-w}$. Adding, $16\bar{S}_{uw} = \frac{[H_{p-1-u} - H_u] - [H_{p-1-w} - H_w]}{u-w} \equiv 0$ since $H_{p-1-n} \equiv H_n \pmod p$. On the diagonal, $16\bar{S}_{uu} = 16X + H^{(2)}_u - H^{(2)}_{K_p-1} + H^{(2)}_{K_p-1-u} + (H^{(2)}_{p-1-u} - H^{(2)}_{K_p-1-u}) \equiv 16X - H^{(2)}_{K_p-1}$, using $H^{(2)}_{p-1-n} \equiv -H^{(2)}_n$ $(p \geq 5)$. $\square$

Hence $\tilde{A}$ mod p is invertible over $\mathbb{F}_p(X)$ (its determinant is $\det M_{0P}\, s_0^{|\mathcal{U}|} \neq 0$: $-e^2$ and $-\frac{1}{9}$ are nonzero for $p \neq 3$), so $A^{-1} = D\tilde{A}^{-1}D$ and $(\tilde{A}^{-1})_{\mathcal{U}\mathcal{U}} \equiv s_0^{-1} I$. Moreover $v(\det A) \geq -2|P|$ (Lemma 12.1 with $u = v = 1$ on P and 0 on $\mathcal{U}$; $W \leq 2$ on $P \times P$, ≤ 1 on $\mathcal{U} \times P$ and 0 on $\mathcal{U} \times \mathcal{U}$).

Entries of T. For $z \neq y$ in the zone: $v(\Delta^2 \mathsf{R}) \geq -1$ (V1; all four terms). For the defect term write $\ell_z A^{-1} \ell_y^{\top} = p^{-4} \mu_z \tilde{A}^{-1} \mu_y^{\top}$ with $\mu_z = p^2 \ell_z D$. By (12.8) and V1, $\ell_z(c)$ has $v \geq -1$ for $c \neq c_z$ (no $c \in C_0$ differs from $c_z \in \mathcal{U}$ by p), so $\mu_z \in \mathcal{O}_p^{C_0}$ and $\mu_z \equiv -\beta e_{c_z} \pmod p$ with $\beta = \sigma/4$ (here $\pi(c_z) = 0$ on B_0). Hence $\mu_z \tilde{A}^{-1} \mu_y^{\top} \equiv \beta^2 (\tilde{A}^{-1})(c_z, c_y) \equiv \beta^2 s_0^{-1} \mathbf{1}[c_z = c_y] = 0 \pmod p$ for $z \neq y$, i.e. $v \geq 1$ and $v(\ell_z A^{-1} \ell_y^{\top}) \geq -3$. So $v(T_{zy}) \geq -3$ for $z \neq y$. For T_{bb} $(e = 1)$: the same expression with $c_z = c_y = c_b$ gives $v(\ell_b A^{-1} \ell_b^{\top}) \geq -4$, and the other terms have $v \geq -2$; so $v(T_{bb}) \geq -4$.

Conclusion. $v(\det T) \geq -3m$ for $e = 0$; for $e = 1$ every term of the Leibniz expansion has $v \geq -3m - 4$ (a term containing T_{bb} has m further factors ≥ -3; others have $m+1$ factors ≥ -3). With $v(\det A) \geq -2|P|$ we obtain (b). $\square$

Remark 12.7 (formal variable). The proof uses that $s_0 = X - \frac{1}{16} H^{(2)}_{K_p - 1}$ is a unit of $\mathcal{O}_p$, which holds because X is an indeterminate: the residue of s_0 in $\mathbb{F}_p(X)$ is nonzero for every p. If X is specialised to $x_0 \in \mathbb{Q}$ with $16x_0 \equiv H^{(2)}_{K_p-1} \pmod p$, then $\bar{S} \equiv 0$ and the bound fails (we have observed $\min v(T_{zy}) = -5, -6$ instead of -3 at $r = 64$, $p = 53, 61$). Likewise (a) uses $p \nmid 6255$. The statement $v_p(\mathrm{den}_F) \leq \ldots$ is about the coefficients of $\det \mathrm{Full}(X) \in \mathbb{Q}[X]$ and therefore about the Gauss valuation; in the proof

of Theorem 11.1 only den_F is used, never the value of a rank-saving bound at a specialisation.

Remark 12.8 (member 2: where the extra node enters). The node $b = r+2m$ enters the proof of Theorem 12.6 only through the following points, each of which uses exactly the hypothesis stated for $e = 1$. (1) b is a zone node with a partner: in (a), $b < 2p$ (the hypothesis $2p > r + 2m$) puts b in the top block and $c_b = b - p \in \mathcal{U}$; in (b), the *strict* inequality $7p + 1 > 4r + 8m$ gives $b < K_p + p$ (so b lies in the block of the zone, with the same level L) and $c_b < K_p$, $c_b \in \mathcal{U}$. Hence the off-diagonal entries of T in row and column b obey the same bounds (≥ 0 in (a), ≥ -3 in (b)) as those between ordinary zone nodes, by the same computation (it uses only that $z \neq y$ are zone nodes with distinct unpaired partners at a unit distance $< p$). (2) b is the only common index of X'' and Y'', so T_{bb} is the only diagonal entry of T; its bound (≥ -2 in (a), ≥ -4 in (b)) uses $\lambda_b \in \mathcal{O}_p^{C_0}$, which holds because no $c \in C_0$ other than c_b differs from b by p and c_b is unpaired. (3) The Leibniz expansion of the $(m+1) \times (m+1)$ matrix T then gives the extra $-2e$ resp. $-4e$. Lemma E is applied (Corollary 12.9) only with the tropical bound $B_{m_0}(p)$ of member 1 as reference, and the extension paragraph in the proof of Proposition 12.5 bounds the tropical value of member 2 by $B_{m_0}(q) + 12$; so the extra node changes each per-prime bound by $O(1)$, and the total by $O(\vartheta(4N')) = O(r) = o(r^2)$, both in the tropical law and in the saving. *Exact checks.* (i) The script `member2_content.py` gives $\tau_p - v_p(\mathrm{den}_F) = m$ (exact Hungarian τ_p for member 2) at every clean prime for $r = 48, 64, 80, 96$ ($m = r/8$). (ii) An independent script, `check_hE_member2.py` (own interpolation nodes $k + \frac{1}{3}$, comparison with the bounds of Theorem 12.6 themselves), gives $v_p(\mathrm{den}_F)$ equal to the bound of Theorem 12.6 (slack 0) in all 27 cases (r, m, e, p) with $r \in \{48, 64, 80\}$, $m \in \{\frac{r}{8} - 1, \frac{r}{8}, \frac{r}{8} + 1\}$, $e \in \{0, 1\}$ and p satisfying the hypotheses of (a) or (b); no violation.

Corollary 12.9 (content saving)**.** *Let $\mathcal{G}(r, m, e)$ be admissible, r large, and let $p \neq 139$ be a prime with*

$$p \equiv 3 \ (4), \quad \tfrac{5}{8}r + 2 < p < \tfrac{4}{5}r - 2, \qquad \text{or} \qquad p \equiv 1 \ (4), \quad \tfrac{5}{7}r + 2 < p < r - 2.$$

Then the hypotheses of Theorem 12.6 hold and $v_p(\mathrm{den}_F) \leq B_{m_0}(p) - m_0 + C_1$ with an absolute constant C_1 ($m_0 = \lfloor r/8 \rfloor$, B_{m_0} as in Proposition 12.2). In these ranges $B_{m_0}(p) = 2r + m_0$ (for $p \equiv 3$) and $B_{m_0}(p) = 2(2r - p - K_p) + 4m_0$ (for $p \equiv 1$).

Proof. The hypotheses: $2p > \frac{5}{4}r + 4 > r + 2m$ since $m \leq \frac{r}{8} + 1$; $5p + 1 < 4r$; $7p + 1 > 5r + 15 > 4r + 8m$ since $8m \leq r + 8$. For the values of B: for $p \equiv 3$ and $\theta = p/r < \frac{4}{5}$ one has $|U| = 0$, $L = 0$, $E = m_0$, $\delta_A = 2r$; for $p \equiv 1$ and $\theta > \frac{5}{7}$, $|U| = \frac{7\theta}{4}r - r + O(1) > 2m_0$, so $L = 2m_0$, $E = \min(2m_0 + L, m_0 + L + |U|) = 4m_0$ and $\delta_A = 2(r - K_p) + 2(r - p)$. Now Theorem 12.6 gives $v_p(\mathrm{den}_F) \leq 2r + 2 \leq B - m_0 + 2$ in the first case and $\leq 2(2r - p - K_p) + 3m + 4 \leq B - m_0 + 3(m - m_0) + 4 \leq B - m_0 + 10$ in the second (by (12.1)). □

12.6. **Proof of Proposition 10.2.** By (12.1), (12.3) and Proposition 12.5, $\log \mathrm{den}_F = \sum_p v_p(\mathrm{den}_F)\log p \le T(r) - \sum_{p\ \mathrm{clean}}(\dots)$, where for every prime p in the ranges of Corollary 12.9 we replace the tropical bound $\sum_e \tau(w^{(p^e)}) \le B_{m_0}(p) + 12$ (note $p^2 > 4N'$ for these p, so only level 1 occurs) by the bound of Corollary 12.9, which is smaller by $m_0 - C_1'$. The primes involved (excluding 139, which lies in the range only for $r < 222$) have

$$\sum_{\substack{p\equiv 3 \\ \frac{5}{8}r<p<\frac{4}{5}r}} \log p + \sum_{\substack{p\equiv 1 \\ \frac{5}{7}r<p<r}} \log p = \frac{r}{2}\Big[\big(\tfrac{4}{5}-\tfrac{5}{8}\big) + \big(1-\tfrac{5}{7}\big)\Big] + o(r) = \frac{129}{560}r + o(r)$$

(PNT in progressions, $\vartheta(x;4,a) = x/2 + o(x)$). Therefore the total saving is $m_0 \cdot \frac{129}{560}r + O(r) = \frac{129}{4480}r^2 + o(r^2)$, and

$$\log \mathrm{den}_F \le \Big(\Lambda_{\mathrm{trop}} - \frac{129}{4480} + o(1)\Big)r^2,$$

$$\Lambda_{\mathrm{trop}} - \frac{129}{4480} = 3.7499347 - 0.0287946 = 3.7211401\ (\le 3.7211401).$$

This proves Proposition 10.2. (More precisely $\Lambda_{\mathrm{trop}} - \frac{129}{4480} = 3.72114002$.) □

Remark 12.10 (range of validity; what is not claimed). The saving $\frac{\alpha}{2}(\frac{51}{70} - \frac{15\alpha}{7})$ for general α is valid only for $\frac{1}{8} \le \alpha \le \frac{1}{6}$; we use it only at $\alpha = \frac{1}{8}$, where it is $\frac{129}{4480}$. Primes $p < \frac{5}{8}r$ (clean primes with additional singular indices) and the non-clean primes in the band (where the saving is m minus the number of zone nodes whose window $(z-p, z]$ meets an extra singular index) probably give a further saving, which is not proved and not used. Any finite-r table of the saving must not count $p = 139$ (for which $\det M_0 \equiv 0$), which is why the exact finite-r values listed in our computations are given only as numerical evidence.

13. Decay and the window for the hybrid frame

This section proves Proposition 10.3. The proof is that of Theorem 4.10 and of Theorem 7.1 (node removal) with two changes: Andréief's identity is used with two different families of functions (rows and columns), and the node set is no longer a segment of integers, so the exact factorial identities are replaced by elementary cell comparisons. Only *common* nodes are removed. Throughout, r is the scale of the geometry, $\alpha_m = m/r$, $L = 2m/r$, $N_{\mathrm{tot}} = |C| + m$, and $\mathcal{T}_t$, $\mathcal{G} = \mathcal{G}(r, m, e)$ are as in §10; the kernel k_ρ, Lemmas 4.3–4.6 and Corollary 4.7 of §4.3 are used unchanged.

13.1. **Contour identity and two-family Andréief.** Fix r and put $N_0 = \lfloor r_0 r \rfloor$, $r_0 = \frac{7}{50}$, $x^* = -N_0 - \frac{3}{8}$, and let Γ, $\mathcal{A}$, m (the complex measure $m = -\sum_{\mu\in\mathcal{A}} c_\mu \delta_{-\mu} + \frac{1}{2\pi i}E(z)dz|_\Gamma$) be as in §4 with h replaced by r. The proof of Proposition 4.1 does not use the nodes except that they lie in $[0, \infty)$,

hence to the right of Γ and away from the poles of F; therefore for *all* integers $a, b \ge 0$

(13.1)
$$\frac{F(b)-F(a)}{b-a} = \int v_a v_b \, dm \ (a \neq b), \qquad F'(a) = \int v_a^2 \, dm, \qquad v_a(z) = \frac{1}{z-a}.$$

For a set $S \subseteq C$ of common nodes put $\mathcal{R}_S = S \sqcup X'$, $\mathcal{Q}_S = S \sqcup Y'$, $n = |S| + m$, and $M_S = \frac{1}{16} L_F[\mathcal{R}_S, \mathcal{Q}_S]$. Since the diagonal of $\mathcal{T}_t$ is $\frac{1}{16}F'(c) + t$ at $c \in C$, multilinearity gives

$$\det \mathcal{T}_t = \sum_{S \subseteq C} t^{|C|-|S|} \det M_S. \tag{13.2}$$

Proposition 13.1.

$$\det L_F[\mathcal{R}_S, \mathcal{Q}_S] = \frac{1}{n!} \int \det[v_a(z_i)]_{i \le n,\, a \in \mathcal{R}_S} \ \det[v_b(z_i)]_{i \le n,\, b \in \mathcal{Q}_S} \prod_i dm(z_i),$$

hence

$$|\det L_F[\mathcal{R}_S, \mathcal{Q}_S]| \le \frac{1}{n!} \int |\mathsf{C}_{\mathcal{R}_S}(z)| \, |\mathsf{C}_{\mathcal{Q}_S}(z)| \prod_i d|m|(z_i),$$

$$\mathsf{C}_{\mathcal{F}}(z) = \frac{\prod_{i<i'}(z_i - z_{i'}) \prod_{a<a' \in \mathcal{F}} (a'-a)}{\prod_{i, a \in \mathcal{F}} (z_i - a)} \ (\textit{up to sign}).$$

Proof. Andréief's identity $\det\big[\int f_a g_b \, d\mu\big]_{a,b} = \frac{1}{n!} \int \det[f_a(z_i)] \det[g_b(z_i)] \prod d\mu(z_i)$ holds for two families of n functions, a finite complex measure μ and bounded f_a, g_b (here $|v_a| \le \frac{8}{3}$ on the support of m, since $\operatorname{Re} z \le -\frac{3}{8}$); apply it with (13.1). The Cauchy determinant evaluates $\det[v_a(z_i)]$. □

13.2. **The potentials and the certified numbers.** Let $\alpha \in [0, \frac{1}{4}]$ and $\nu_\alpha = \mathrm{Leb}|_{[0,1]} + \frac{1}{2}\mathrm{Leb}|_{[1,1+2\alpha]}$ (mass $1+\alpha$). Put $U^{\nu_\alpha}(s) = -\int \log|s-u| \, d\nu_\alpha(u)$ and $I(\nu_\alpha) = \iint -\log|u-u'| \, d\nu_\alpha d\nu_\alpha$. Closed forms: for $s = x + iy$, $x < 0$, $y \ge 0$, with $\Phi(q) = a_q(\frac{1}{2}\log(a_q^2 + y^2) - 1) - y \arctan(y/a_q)$, $a_q = q - x$: $U^{\nu_\alpha}(s) = -[\Phi(1) - \Phi(0)] - \frac{1}{2}[\Phi(1+2\alpha) - \Phi(1)]$; on the real line, with $G(x) = x\log|x| - x$, $U^{\nu_\alpha}(t) = 1 - t\log|t| - (1-t)\log|1-t| - \frac{1}{2}[G(1+2\alpha-t) - G(1-t)]$, so that $U^{\nu_\alpha\prime}(t) = \frac{1}{2}\log|(1-t)(1+2\alpha-t)| - \log|t|$ is decreasing on $(0,1)$ (hence U^{ν_α} is concave there), positive on $[-r_0, 0)$, and negative on $(1, \infty)$; and $I(\nu_\alpha) = \frac{3}{2} + I_{12} + \alpha^2(\frac{3}{2} - \log 2\alpha)$, with $I_{12} = -[\mathsf{F}(1+2\alpha) - \mathsf{F}(1) - \mathsf{F}(2\alpha) + \mathsf{F}(0)]$, $\mathsf{F}(x) = \frac{x^2}{2}\log|x| - \frac{3x^2}{4}$. At $\alpha = \frac{1}{8}$: $I(\nu_{1/8}) = 1.70244575114$. (These were checked against quadrature to 15 digits.)

For a probability measure $\sigma = \sum_k m_k \delta_{c_k}$ and ρ, U^σ_ρ, $I_\rho(\sigma)$ are as in §4.4. Fix $\kappa = 10^{-3}$ and put

$$g_\alpha(s) = (1+\alpha)U^\sigma_\rho(s) - U^{\nu_\alpha}(s) + (1-\kappa)\pi|\operatorname{Im} s|, \qquad \partial_\alpha g_\alpha(s) = U^\sigma_\rho(s) + \log|s-(1+2\alpha)|.$$

The certificate (computer-assisted). Let $\sigma = \sigma_{\mathrm{hyb}}$ be the measure of the file `sig_h500.json` (dyadic data in the format of `sigma300.json`: coordinates $X/2^{24}$, $Y/2^{24}$, masses $n_k/2^{32}$, $\sum m_k = 1$ exactly, conjugation-symmetric, $\rho = 1/2048$; the file is named after its design parameter $K = 500$), $\alpha = \frac{1}{8}$, $L_0 = \frac{1}{4}$, $u_{\max} = \frac{3}{2}$, $\omega = 2^{-30}$. The script `hcert.py` (Arb, 96 bits) certifies, with $g = g_{1/8}$:

(R) $g \geq m'$ on all boxes $[-r_0 - u_b - \omega, -r_0 - u_a + \omega] \times [3u_a, 3u_b]$ covering $u \in [0, u_{\max}]$ (rays $-r_0 + u(-1 \pm 3i)$), and the tail bound $T(u_{\max}) \geq m'$, where for $u \geq u_{\max}$: $|z| \leq r_0 + \omega + u\sqrt{10}$, $U^\sigma_\rho(z) \geq -\frac{1}{2}\log((|z|+R_\sigma)^2+\rho^2)$, $U^\nu(z) \leq -(1+\alpha)\log(|z| - 1 - 2\alpha)$ (all of ν lies in $[0, 1+2\alpha]$ and has mass $1+\alpha$) and the field term is $\geq (1-\kappa)3\pi u$; T is increasing in u (derivative $\geq 9 - 1$).

(S) a partition of $[-r_0, 0]$ into cells $[a_\ell, b_\ell]$ with $g \geq \gamma_\ell$ on them, and $S_{\mathrm{cert}} := 2\sum_\ell (b_\ell - a_\ell)\min(\gamma_\ell - m', 0)$;

(L) a cover of $[0, 1+2\alpha]$ by cells on which $g \geq g_{\min}$.

On a box or cell the lower bound for U^σ_ρ is $-\frac{1}{2}\sum_k m_k \log(d_{x,k}^2 + d_{y,k}^2 + \rho^2)$ ($d_{x,k}, d_{y,k}$ the largest horizontal and vertical distances to c_k), U^ν is bounded above by ball arithmetic on the closed form on ray boxes, by its right endpoint value on segment cells (U^ν is increasing on $(-\infty, 0]$), by its left endpoint value on cells in $[1, \infty)$ (decreasing) and on cells of $[0, 1]$ by the endpoint values or the two tangent bounds (concavity). Put

$$D_F := 2\big((1+\alpha)m' + S_{\mathrm{cert}}\big) - (1+\alpha)^2 I_\rho(\sigma) + I(\nu_\alpha), \qquad c_{\mathrm{hyb}} := 2(m' - g_{\min}).$$

quantity	certified value	cells
$I_\rho(\sigma)$ (exact finite sum)	2.00548597469 ($\pm 6 \cdot 10^{-26}$)	finite sum
$I(\nu_{1/8})$	1.70244575114	–
m': inf g on the rays, $u \in [0, \frac{3}{2}]$	$\geq$ 2.2185001686	11,681 boxes
tail $T(\frac{3}{2})$	$\geq$ 13.595	–
S_{cert}	≥ -0.1248042185	14,000 cells
$g_{\min}$ on $[0, 1.25]$ (attained near $t = 0.8284$)	≥ -1.6085322035	4,512 cells
D_F	$\geq$ 3.9062695066	
c_{hyb}	$\leq$ 7.6540647441	

Negative control: with the plain certificate σ_{300} and $\alpha = 0$ the same script returns $D \geq 3.111379042$, $c \leq 6.871799888$, reproducing the numbers of §4.4 (3.1113790, 6.8717994). The copy of `hcert.py` in `scripts-hybrid/` has been re-run (Python 3.13 with `python-flint` and `numpy`, the libraries supplied through `PYLIBS`): it reproduces $D_F \geq 3.906269506610271$ and $c_{\mathrm{hyb}} \leq 7.654064744147046$ exactly, in 8 seconds; the original was also re-run by an audit agent, with identical output. The script, the data `sig_h500.json`,

the exact run instructions and the expected output are in `scripts-hybrid/` (§13.4). The certificate is valid for $r \geq 2^{30}$ and does not depend on r.

Remark 13.2 (c_{hyb} is below the true first-order slope). The true profile $-r^{-2}\log|\sum_{|S^c|=j}\det M_S|$ has initial slope ≈ 7.687 in $\beta = j/r$ (exact computations at $r \leq 128$ extrapolate to $7.687, 7.682, 7.677, 7.673, 7.654$ for $j = 1, 2, 3, 4, 8$), larger than the certified $c_{\mathrm{hyb}} = 7.654$. There is no contradiction: Theorem 13.4 certifies the *line* $D_F - \beta c_{\mathrm{hyb}}$, which lies below the true profile at every β (the margins at $\beta = 0, \frac{1}{16}, \frac{1}{8}, \frac{1}{4}, \frac{1}{2}, 1$ are $0.008, 0.026, 0.081, 0.29, 1.06, 3.87$; the true profile is strongly convex). The window argument uses only that line (Corollary 13.5).

13.3. **The pointwise inequality.** Let $\mathcal{G} = \mathcal{G}(r, m, e)$ be admissible, $S \subseteq C$, $S^c = C \setminus S$, $j = |S^c|$, $\beta = j/r$. Let $\mathcal{R}, \mathcal{Q}$ be the full row and column node sets ($|\mathcal{R}| = |\mathcal{Q}| = N := r + m + e$), $n = |S| + m = N - j$ the number of free points, $t_a = a/r$. Free points $z_1, \ldots, z_n$ in the support of $|m|$; $s_i = z_i/r$. Split $d|m| = w_\kappa d\Omega_\kappa$ as before ($w_\kappa = |E|^{1-\kappa}$ on Γ, 1 at atoms), Ω_κ of total mass $\leq 2r + c_\kappa$.

Scaling. Counting powers of r in the two Cauchy determinants: free pairs $\frac{n(n-1)}{2}$ and node pairs $\frac{n(n-1)}{2}$ (since $|\mathcal{R}_S| = |\mathcal{Q}_S| = n$) against n^2 cross factors, so $|\mathsf{C}_{\mathcal{R}_S}\mathsf{C}_{\mathcal{Q}_S}| = r^{-2n}\,|\tilde{\mathsf{C}}_{\mathcal{R}_S}\tilde{\mathsf{C}}_{\mathcal{Q}_S}|$ with the same expression in the scaled variables; $r^{-2n} \leq 1$ is dropped. Hence $-\log(|\mathsf{C}_{\mathcal{R}_S}||\mathsf{C}_{\mathcal{Q}_S}|\prod w_\kappa(z_i)) \geq r^2 H_S$ with

$$r^2 H_S = \sum_{i \neq i'} -\log|s_i - s_{i'}| + \sum_{\mathcal{F} \in \{\mathcal{R}, \mathcal{Q}\}} \Big[\tfrac{1}{2} \sum_{a \neq a' \in \mathcal{F}_S} -\log|t_a - t_{a'}| + \sum_i \sum_{a \in \mathcal{F}_S} \log|s_i - t_a|\Big] + \sum_i -\log w_\kappa(z_i).$$

(Ordered pairs; each family has weight $\frac{1}{2}$ in the node–node and 1 in the cross term, and the two together give weight 1 to the free–free term.)

Completing the node set. For each family $\mathcal{F}$ the two identities of the proof of Theorem 7.1 (adding and removing the pairs meeting S^c) hold verbatim, because $S^c \subseteq C \subseteq \mathcal{F}$. Adding the two families with weights $\frac{1}{2}$, the removed–removed pairs enter once with a plus sign (from the identity) and once with a minus sign (from the node energy), so they cancel *exactly* against the removed–removed part of $[B]$ below — this is why only common nodes can be removed (a node removed from one family only would leave a term of weight $\frac{1}{2}$). Let X be the set of the n free points s_i together with the j removed nodes t_a ($a \in S^c$); $|X| = N$, and the points are distinct ($\mathrm{Re}\, s_i < 0 \leq t_a$). Then exactly

$$r^2 H_S = [B] + [R1] + [R2] + [R3] + [R4],$$

$$[B] = \sum_{x\neq y\in X} -\log|x-y|, \qquad [R1] = \sum_{\mathcal{F}}\sum_{i}\sum_{a\in\mathcal{F}} \log|s_i - t_a|,$$
$$[R2] = \tfrac12 \sum_{\mathcal{F}}\sum_{a\neq a'\in\mathcal{F}} -\log|t_a - t_{a'}|, \quad [R3] = \sum_{\mathcal{F}}\sum_{a\in S^c}\sum_{b\in\mathcal{F},b\neq a} \log|t_a - t_b|,$$
$$[R4] = \sum_i -\log w_\kappa(z_i).$$

Error budget. We use the following elementary bounds; $\ell_1 := \log(1 + \frac{(1+L)8r}{3})$ and $N = r + m + e \le \frac98 r + 3$.

- $[B]$. Lemma 4.3 and Corollary 4.7 for X with the measure $N\sigma$ (weights 1 on the N points, $-Nm_k$ on the c_k; the proof of the Corollary is unchanged): $[B] \ge 2N\sum_{x\in X} U^\sigma_\rho(x) - N^2 I_\rho(\sigma) - N|\log\rho|$.
- $[R1]$. For $\operatorname{Re} s \le -\frac38\cdot\frac1r$ the function $u \mapsto \log|s-u|$ is increasing on $[0,\infty)$. For a family $\mathcal{F}$ consisting of the nodes $t_a = a/r$ ($a \in C_0$, step $\frac1r$, weight 1) and of the nodes of X' (step $\frac2r$), a left Riemann sum of an increasing function f loses at most $(f(\text{top}) - f(\text{bottom}))$: $\sum_{a\in C_0} f(t_a) \ge r\int_0^1 f - (f(1) - f(0))$ and $\sum_{i<m} f(1 + \frac{2i}{r}) \ge \frac{r}{2}\int_1^{1+L} f - (f(1+L) - f(1))$. For Y' (shifted by $\frac1r$) the sum is larger, and the extra node b (at $t_b = 1+L$, where $|s - t_b| \ge 1$) contributes ≥ 0. Hence $\sum_{a\in\mathcal{F}} \log|s-t_a| \ge -rU^{\nu_L}(s) - \ell_1$ and $[R1] \ge -2r\sum_i U^{\nu_L}(s_i) - 2n\ell_1$.
- $[R2]$. Give each node a of a family the cell $J_a = [t_a, t_a + d_a]$ of ν-mass $\frac1r$ ($d_a = \frac1r$ in C_0, $\frac2r$ in X'/Y'; for b, $d_b = \frac2r$ in $\mathcal{R}$ and $\frac1r$ in $\mathcal{Q}$); for the families without b the cells tile $\operatorname{supp}\nu_L$ exactly. For $a < a'$, $u \in J_a$, $u' \in J_{a'}$: $u' - u \ge t_{a'} - t_a - d_a$ and hence, if a' is not the successor of a in $\mathcal{F}$, $\log(t_{a'} - t_a) \le \log(u' - u) + \frac{d_a}{u' - u} \le \log(u' - u) + \frac{1}{\#\{\text{nodes between}\}+1}$; summing over pairs, the total error is $\le 2N(1 + \log N)$ per family; adjacent pairs cost $\le \frac32$ each, $\le \frac32 N$ per family; the diagonal cells contribute $N(\log r + \frac32)$ to $r^2 I(\nu_{\mathcal{F}})$ but nothing to the discrete sum. Altogether $[R2] \ge r^2\cdot\frac12\sum_{\mathcal{F}} I(\nu_{\mathcal{F}}) - N(\log r + \frac32) - 4N(1 + \log N) - 3N$, and $|I(\nu_{\mathcal{F}}) - I(\nu_L)| \le \frac4r\cdot\sup|U| \le \frac8r$.
- $[R3]$. For $a \in S^c$ ($t = t_a \in [0, 1+L]$) and a family $\mathcal{F}$: for nodes $b < a$ the cell J_b lies to the left of t_a and $\log(t_a - t_b) \ge \frac{r}{d_b}\int_{J_b}\log(t_a - u)du$ exactly, because $\log(t_a - \cdot)$ is decreasing, so $\log(t_a - t_b) \ge \log(t_a - u)$ for $u \ge t_b$; for $b > a$ the error is $\frac{d_b}{t_b - t_a}$, summing to $\le 2(1 + \log N)$; the self cell contributes ≥ 0. Hence $\sum_{b\in\mathcal{F},b\neq a}\log|t_a - t_b| \ge -rU^{\nu_L}(t_a) - 2(1 + \log N) - \frac12(\log r + 1) - 2$, and $[R3] \ge -2r\sum_{a\in S^c} U^{\nu_L}(t_a) - j\,(4(1 + \log N) + \log r + 5)$. (If $t_a = t_b$ is the extra node, U^{ν_L} is evaluated at $1+L$, where it is continuous.)
- $[R4]$. E is 1-periodic, $x^* \equiv \frac58 \pmod 1$, so $C_E = \sup_\Gamma |E| e^{2\pi|\operatorname{Im} z|}$ is independent of r, and $[R4] \ge (1 - \kappa)(2\pi r\sum_i |\operatorname{Im} s_i| - n\log^+ C_E)$.

Define $g^{(m)} := g_{\alpha_m}$ for the actual $\alpha_m = m/r$ (note $\nu_L = \nu_{\alpha_m}$), put $\bar N = N/r$ and

$$G^{(m)}(x) := g^{(m)}(x) + \tfrac{e}{r}\, U^\sigma_\rho(x).$$

Dividing by r^2 and collecting (the coefficient of $\sum_{x\in X} U_\rho^\sigma(x)$ is $2\bar N = 2(1+\alpha_m)+\frac{2e}{r}$, and the part $\frac{2e}{r}$ is put into $G^{(m)}$; the extra point b is a free or removed point like the others):

$$H_S \ge \frac{2}{r}\sum_{i\le n} G^{(m)}(s_i) + \frac{2}{r}\sum_{a\in S^c} G^{(m)}(t_a) - \bar N^2 I_\rho(\sigma) + I(\nu_{\alpha_m}) - \eta_1(r), \tag{13.3}$$

$$\eta_1(r) = \frac{|\log\rho|\bar N}{r} + \frac{2\bar N\ell_1}{r} + \frac{\bar N(\log r + \frac{9}{2}) + 4\bar N(1+\log N)}{r} + \beta\frac{4(1+\log N)+\log r+5}{r} + \frac{\bar N\log^+ C_E}{r} + \frac{16}{r} + \frac{1}{r}.$$

The term $\frac{1}{r}$ covers the finitely many pair terms involving b that are not of the types treated above (in $\mathcal{Q}$ the cell of b overlaps the top cell of Y') and the self-energy of the mismatch $\nu_{\mathcal F}-\nu_L$ (at most three pieces of mass $\le \frac{1}{r}$ and length $\ge \frac{1}{r}$); together they are $O(\log r)$ in r^2H_S, hence $\le r$ for $r\ge 2^{30}$. *Numerical value.* For $r\ge 2^{30}$: $\bar N\le 1.1251$, $\log N\le \log r+0.118$, $\ell_1\le \log r+1.205$, $\beta\le 1.0001$, $|\log\rho| = \log 2048 \le 7.625$, and $\log^+ C_E\le 3.6$. For the last: Γ consists of the rays $x^*+u(-1\pm 3i)$, $u\ge 0$, with $x^*\equiv\frac{5}{8}$ (mod 1), and $|\operatorname{Im} z| = 3u$; for $u\le\frac18$ the real parts of πz and of $\frac{\pi}{4}-\pi z$ lie in $[\frac{\pi}{2},\frac{5\pi}{8}]$ and $[-\frac{3\pi}{8},-\frac{\pi}{4}]$ modulo π, so $|E|\le \pi\sin\frac{\pi}{4}/(\sin\frac{3\pi}{8}\sin\frac{\pi}{4})\le 3.41$ and $|E|e^{6\pi u}\le 3.41\,e^{3\pi/4}\le 36$; for $u\ge\frac18$, $|\sin w|\ge\sinh|\operatorname{Im} w|$ gives $|E|e^{6\pi u}\le 4\pi\sin\frac{\pi}{4}\,(1-e^{-3\pi/4})^{-2}\le 10.9$. Altogether $\eta_1(r)\le(12.9\log r+52)/r$.

Lemma 13.3 (certificates for the actual geometry)**.** *Let $r\ge 2^{30}$, $|m-\frac{r}{8}|\le 1$, $e\in\{0,1\}$. (i) For every free point s on the rays, $G^{(m)}(s)\ge m'-\frac{12}{r}$. (ii) For the atoms, $\frac1r\sum_{atoms}\min(G^{(m)}(s)-m',0)\ge S_{\rm cert}-\frac{2}{r}\sum_\ell(m'-\gamma_\ell)^+-\frac{3}{r}$. (iii) For $a\in S^c\cap C_0$, $G^{(m)}(t_a)\ge g_{\min}-\frac{11}{r}$; and $G^{(m)}(t_b)\ge g_{\min}-\frac{2400}{r}$.*

Proof. We use $\partial_\alpha g_\alpha(s) = U_\rho^\sigma(s)+\log|s-(1+2\alpha)|$, $|\alpha_m-\frac18|\le\frac1r$, $R_\sigma=\max_k|c_k|\le 0.4767$ (from the data of σ), and $-\log(|x|+R_\sigma+\rho)\le U_\rho^\sigma(x)\le\log\frac1\rho\le 7.63$, since $|x-c_k|^2+\rho^2\le(|x|+R_\sigma+\rho)^2$. (i) Compact part $u\le\frac32$: $|s|\le r_0+\omega+\frac32\sqrt{10}\le 4.9$, so $U_\rho^\sigma(s)\ge -1.7$, and $1.39\le|s-(1+2\alpha)|\le 6.2$; hence $|\partial_\alpha g|\le 7.63+1.83\le 10$, $|g^{(m)}(s)-g(s)|\le\frac{10}{r}$, and $\frac{e}{r}U_\rho^\sigma(s)\ge-\frac{1.7}{r}$. Tail $u\ge\frac32$: by (R), $g(s)\ge T(u)\ge T(\frac32)+8(u-\frac32)\ge m'+11+8(u-\frac32)$, while $|\partial_\alpha g(s)|+|U_\rho^\sigma(s)|\le 15.3+3\log(|s|+1.5)$ with $|s|\le 0.15+3.17u$; for $r\ge 2^{30}$ the loss $(15.3+3\log(3.2u+2))/r$ is far below $11+8(u-\frac32)$, so $G^{(m)}(s)\ge m'$. (ii) The atoms lie in $\{-(\mathbb{Z}_{\ge0}+\frac34)/r\}\cup\{-\mathbb{Z}_{\ge1}/r\}\subset[-r_0,0]$, are distinct, at most $2r_0r+1$ in all, and at most $2r(b_\ell-a_\ell)+2$ of them lie in the closed cell C_ℓ of the segment cover (assign each atom to one cell). As in Lemma 4.9, $\frac1r\sum\min(g-m',0)\ge\sum_\ell\frac{2r(b_\ell-a_\ell)+2}{r}\min(\gamma_\ell-m',0) = S_{\rm cert}-\frac{2}{r}\sum_\ell(m'-\gamma_\ell)^+$ for $g=g_{1/8}$. On $[-r_0,0]$, $|s-(1+2\alpha)|\in[1.24,1.40]$, so $|\partial_\alpha g|\le 10$, and $U_\rho^\sigma\ge-\log 0.62>0$; so passing to $G^{(m)}$ costs at most $\frac1r(2r_0r+1)\frac{10}{r}\le\frac3r$. (iii) For $t\in[0,1)$: $|t-(1+2\alpha)|\in[0.24,1.26]$ and

$U^\sigma_\rho(t) \ge -0.4$, so $|\partial_\alpha g| \le 7.63 + 1.43 \le 10$ and $G^{(m)}(t) \ge g(t) - \frac{10.4}{r} \ge g_{\min} - \frac{11}{r}$ by (L). For $t_b = 1 + 2\alpha_m$ ($|t_b - \frac{5}{4}| \le \frac{2}{r}$, the right end of $\operatorname{supp}\nu_{\alpha_m}$): put $\varphi(\alpha) := U^{\nu_\alpha}(1+2\alpha) = -\int_{2\alpha}^{1+2\alpha} \log v\, dv - \frac{1}{2}\int_0^{2\alpha} \log v\, dv$; then $\varphi'(\alpha) = \log 2\alpha - 2\log(1+2\alpha)$, $|\varphi'| \le 1.9$ for $|\alpha - \frac{1}{8}| \le \frac{1}{r}$, and U^σ_ρ is $\frac{1}{2\rho} = 1024$-Lipschitz. So $|g^{(m)}(t_b) - g(\frac{5}{4})| \le 1.126 \cdot 1024 \cdot \frac{2}{r} + \frac{7.63}{r} + \frac{1.9}{r} \le \frac{2316}{r}$, $\frac{e}{r}U^\sigma_\rho(t_b) \ge -\frac{0.6}{r}$, and $g(\frac{5}{4}) \ge g_{\min}$ by (L). □

Combining (13.3) with Lemma 13.3 ($n = N - j = r(1 + \alpha_m - \beta) + e$ free points, some on the rays and the others at distinct atoms; j removed nodes):

$$H_S \ge 2(1 + \alpha_m - \beta)m' + 2S_{\text{cert}} + 2\beta g_{\min} - \bar{N}^2 I_\rho + I(\nu_{\alpha_m}) - \eta_2(r),$$

$$\eta_2(r) = \eta_1(r) + \frac{24\bar{N} + 6 + 22\beta}{r} + \frac{4}{r}\sum_\ell (m' - \gamma_\ell)^+ + \frac{4800}{r^2}$$

(the term $\frac{24\bar{N}}{r}$: at most $n \le N$ ray points, each losing $\frac{2}{r} \cdot \frac{12}{r}$; $\frac{6}{r}$: twice the last term of (ii); $\frac{22\beta}{r}$: the removed nodes of C_0 in (iii); $\frac{4800}{r^2}$: the node b if removed; we used $\frac{e}{r}m' \ge 0$). Finally we replace α_m by $\frac{1}{8}$: $2(1+\alpha_m)m' \ge \frac{9}{4}m' - \frac{4.44}{r}$; $|\bar{N} - \frac{9}{8}| \le \frac{2}{r}$ gives $|\bar{N}^2 - \frac{81}{64}| \le \frac{4.51}{r}$, and $I_\rho(\sigma) \le 2.006$, a loss $\le \frac{9.05}{r}$; and $\frac{d}{d\alpha}I(\nu_\alpha) = 2\varphi(\alpha)$ (as α grows, ν_α gains mass at rate 1 at the point $1 + 2\alpha$), with $0 < \varphi \le 0.7$ for $|\alpha - \frac{1}{8}| \le \frac{1}{r}$ ($\varphi(\frac{1}{8}) = 0.6728$), so $|I(\nu_{\alpha_m}) - I(\nu_{1/8})| \le \frac{1.4}{r}$. Hence

$$H_S \ge D_F - \beta c_{\text{hyb}} - \eta(r),$$

(13.4)

$$\eta(r) \le \frac{15\log r + K_0}{r}, \qquad K_0 = 52 + 70 + 4\sum_\ell (m' - \gamma_\ell)^+ \le 25{,}100 \le 3 \cdot 10^4,$$

where 52 comes from η_1, $70 \ge 24\bar{N} + 6 + 22\beta + 4.44 + 9.05 + 1.4 + 4800/r$, and the dominant constant is $4\sum_\ell (m' - \gamma_\ell)^+ = 24{,}960.84$ for the 14,000 segment cells (computed from the cell file `hcells_h500.json` written by `hcert.py`). We do not try to optimise $\eta(r)$ or to give an explicit r_0 beyond 2^{30}. All these constants affect only the $o(1)$ in Proposition 10.3, not D_F or c_{hyb}.

Theorem 13.4 (node removal, hybrid; computer-assisted)**.** *For admissible $\mathcal{G}(r, m, e)$ with $r \ge 2^{30}$, every $S \subseteq C$ (with $j = |C \setminus S|$, $\beta = j/r$, $n = |S| + m$),*

$$|\det M_S| \le 16^{-n}(2r + c_\kappa)^n \exp(-r^2[D_F - \beta c_{\text{hyb}} - \eta(r)]),$$

with $\eta(r)$ as in (13.4), independent of S.

Proof. Proposition 13.1, the splitting $d|m| = w_\kappa d\Omega_\kappa$, $\frac{1}{n!} \le 1$, the pointwise bound just proved (it holds for all configurations of n distinct free points on the support of m; coincident points give a vanishing Cauchy determinant), and $M_S = \frac{1}{16}L_F[\mathcal{R}_S, \mathcal{Q}_S]$. □

Corollary 13.5 (window)**.** *For admissible $\mathcal{G}$, $r \ge 2^{30}$, and real t with $|t| \le e^{-c_{\text{hyb}}r}$,*

$$|\det \mathcal{T}_t| \le \exp(-D_F r^2 + \eta(r)r^2 + (r+1)\log(2(2r + c_\kappa))) = e^{-(D_F - o(1))r^2}.$$

This is Proposition 10.3.

Proof. By (13.2) and Theorem 13.4, with $e^{r^2\beta c} = e^{jcr}$ and $(2r+c_\kappa)^n \le (2r+c_\kappa)^{r+1+m}$,

$$|\det \mathcal{T}_t| \le (2r+c_\kappa)^N e^{r^2\eta} e^{-D_F r^2} \sum_{j=0}^{|C|} \binom{|C|}{j} 16^{-(N-j)} (|t|e^{c_{\mathrm{hyb}} r})^j,$$

and for $|t|e^{c_{\mathrm{hyb}} r} \le 1$ the sum is $\le 2^{|C|}$. (Here $N = r+m+e$, and $(2r+c_\kappa)^N = e^{O(r\log r)}$.) □

13.4. **Reproducing the certificate.** The directory `scripts-hybrid/` contains `hcert.py`, `sig_h500.json`, `sigma300.json` (the plain control), the expected logs and a `README`. The run is

```
python hcert.py sig_h500.json 1/8 2.2185 -1.6090 h500 14000 512     # D_F, c_hyb
python hcert.py sigma300.json 0 1.97180 -1.46410 ctl0 14000          # negative contr
```

(arguments: measure file, α, target for m', target for $g_{\min}$, tag, number of segment cells, number of line cells beyond 1; targets must lie below the float plateau minimum). Requirements: Python with `python-flint` (Arb) and `numpy`; set the environment variable `PYLIBS` if these live in a non-standard directory. Run time about 30 s. The output was reproduced independently by an audit agent (identical numbers; 8 s), and its closed forms were compared with quadrature.

14. ℓ-adic nonvanishing for the hybrid frame

This section proves Proposition 10.4, modulo the nonvanishing of two Krylov determinants (Theorem 15.1) which is proved in §15. The structure is that of §5–§6: reduce $\det \mathrm{Full}$ modulo ℓ to a *bordered Cauchy–Gram* determinant (Proposition 14.3); the boundary member is controlled at $Y = 0$ by a product of two Cauchy determinants; the other roots are controlled by a rank-one bordering argument that ends in a Krylov determinant.

Throughout $\ell \equiv 3 \pmod 4$ is prime, $\ell \ge 7$, $K = (\ell+1)/4 \ge 2$, $0 \le m \le K-1$, $s = K-m \ge 1$, $r_1 = \ell - K - m$. Indices $0, \dots, K-1$ are *low*, indices $K \le b \le \ell-1$ *high*; for a high node b let $j_b = \ell - b \in [1, \ell-K]$ and, for $j \ge 1$, $w_j = (\frac{1}{a+j})_{0 \le a < K} \in \mathbb{F}_\ell^K$ (all denominators are units because $a+j \le K-1+\ell-K < \ell$). For a set J of positive integers $W_J = [w_j]_{j\in J}$, a $K \times |J|$ matrix, written C_J when J has K elements. A geometry is *admissible for* ℓ if

(H) $\{0, \dots, K-1\} \subseteq C$ and all nodes are $\le \ell - 1$.

Both $\mathcal{G}(r_1, m, 0)$ and $\mathcal{G}(r_1, m, 1)$ satisfy (H): $r_1 \ge K \iff m \le 2K-1$, and the top node of member 2 is $r_1 + 2m = \ell - K + m \le \ell - 1 \iff m \le K-1$. In j-coordinates, member 1 has $\{j_b : b \notin C \cup X' \cup Y'\} = G_1 := \{1, \dots, s\}$ (G = "absent" nodes), $J_{Y'} = s + \{1, 3, \dots, 2m-1\}$, $J_{X'} = s + \{2, 4, \dots, 2m\}$ (the node $b = r_1 + 2i + 1$ has $j_b = s + (2m - 2i - 1)$ and $r_1 + 2i$ has $s + 2(m-i)$);

member 2 has $G_2 = \{1, \dots, s-1\}$ (its extra node $b = r_1 + 2m$ is $j = s$) and the same $J_{X'}, J_{Y'}$.

14.1. **The ℓ-singular structure.**

Lemma 14.1. *Assume (H). Then the only ℓ-singular quantity among k, $4k-1$, 4, $b-a$ occurring in the entries $\mathsf{R}(a,b)$ ($a, b \le \ell-1$) is $4K-1=\ell$ at $k=K$ (note $3\ell \equiv 1 \pmod 4$ and 2ℓ is even, so $4k-1 \in \{\ell, 2\ell, 3\ell\}$ forces $k=K$). Consequently:*

(a) $\mathsf{R}(a,b) \in \mathbb{Z}_{(\ell)}$ for a, b low, and for $a \neq b$ both high;
(b) for a low and b high: $\ell\,\mathsf{R}(a,b) \equiv -\frac{1}{4}\, w_{j_b}(a) \pmod \ell$;
(c) for b high: $\ell^2 \mathsf{R}(b,b) \equiv -1 \pmod \ell$.

Proof. The term $k=K$ occurs in $\sum_{a<k\le b}$ iff $a < K \le b$, and in $\Psi(b)$ iff $b \ge K$. For (b): $\ell\mathsf{R}(a,b) \equiv \frac{1}{16K(b-a)} \equiv \frac{1}{4(b-a)}$ since $16K = 4\ell+4$, and $b-a = \ell - j_b - a \equiv -(a+j_b)$. For (c): $\Psi(b) \ni -\frac{1}{(4K-1)^2} = -\ell^{-2}$. □

Lemma 14.2 (low block)**.** *Let $A = -\mathsf{R}[\mathit{low}, \mathit{low}]$ (the low block of M_0) and $H_n^{(2)} = \sum_{i \le n} i^{-2}$. Then in $\mathbb{F}_\ell$: $16A = W_{[1,\ell-K]} W_{[1,\ell-K]}^\top + H_{K-1}^{(2)} I_K$.*

Proof. This is the computation of the proof of Lemma 5.3 with $m = \ell - K$ (it only used $a < b < K$): $16A_{ab} \equiv \sum_{j=1}^{\ell-K} \frac{1}{(a+j)(b+j)}$ for $a < b$, and $16A_{aa} \equiv \sum_{j=1}^{\ell-K} \frac{1}{(a+j)^2} - H_{\ell-K}^{(2)}$; and $H_{\ell-K}^{(2)} \equiv -H_{K-1}^{(2)}$ because $H_{\ell-1}^{(2)} \equiv 0$ and $1/(\ell-i)^2 \equiv 1/i^2$. □

Proposition 14.3 (bordered reduction)**.** *Assume (H). Let G be the set of j_b, $b \in [K, \ell-1] \setminus (C \cup X' \cup Y')$, W_G the corresponding matrix ($K \times |G|$, $GG^\top := W_G W_G^\top$ below), $\#C_{\mathrm{high}} = |C \setminus [0,K)|$ and, for an indeterminate Y,*

$$B(Y) = \det \begin{pmatrix} Y\, I_K + W_G W_G^\top & W_{Y'} \\ W_{X'}^\top & 0 \end{pmatrix} \in \mathbb{F}_\ell[Y] \quad (\textit{size } K+m).$$

Then $D\,\mathrm{Full}(X)\,D \in M(\mathbb{Z}_{(\ell)}[X])$ for $D = \mathrm{diag}$ (1 on low, ℓ on high rows $C_{\mathrm{high}} \cup X'$ and high columns $C_{\mathrm{high}} \cup Y'$), and

$$\ell^{2(\#C_{\mathrm{high}}+m)} \det \mathrm{Full}(X) \equiv (-1)^{\#C_{\mathrm{high}}+m} (-16)^{-K}\, B(H_{K-1}^{(2)} - 16X) \pmod \ell \quad \textit{in } \mathbb{F}_\ell[X].$$

Proof. By Lemma 14.1, modulo ℓ the matrix $D\mathrm{Full}(X)D$ (rows L, C_{high}, X'; columns L, C_{high}, Y'; L=low) is

$$\begin{pmatrix} XI - A & -\frac{1}{4}W_C & -\frac{1}{4}W_{Y'} \\ -\frac{1}{4}W_C^\top & -I & 0 \\ -\frac{1}{4}W_{X'}^\top & 0 & 0 \end{pmatrix}$$

(the zero blocks: $\ell^2\mathsf{R}$ between distinct high nodes and $\ell^2 X$ are $\equiv 0$; a diagonal entry occurs only at $C \times C$). Taking the Schur complement of the block $-I$ (factor $(-1)^{\#C_{\mathrm{high}}}$; the blocks $C_{\mathrm{high}} \times Y'$ and $X' \times C_{\mathrm{high}}$ vanish so no other block changes) the upper left block becomes $XI - A + \frac{1}{16} W_C W_C^\top$, which by Lemma 14.2 and the partition $[1, \ell-K] = J_C \sqcup G \sqcup J_{X'} \sqcup J_{Y'}$

equals $-\frac{1}{16}(YI + W_G W_G^\top + W_{X'} W_{X'}^\top + W_{Y'} W_{Y'}^\top)$, $Y = H_{K-1}^{(2)} - 16X$. Adding $-\frac{1}{4} W_{X'} \times$ (bottom rows) to the top rows and (right columns) $\times (-\frac{1}{4} W_{Y'}^\top)$ to the left columns removes $W_{X'} W_{X'}^\top$ and $W_{Y'} W_{Y'}^\top$ (the bottom right block is 0, so the off-diagonal blocks do not change). Finally multiply the top K rows by -16, the m bottom rows by -4 and the m right columns by $\frac{1}{4}$; the determinant changes by $(-16)^K (-1)^m$ and we obtain $B(Y)$. Also $\det D = \ell^{\#C_{\text{high}}+m}$ for rows and the same for columns. □

Hence $\deg B \le K - m$, and, since $X \mapsto Y$ is a bijection of $\mathbb{F}_\ell$ *independent of the geometry*, for a rational x_0 with $\ell \nmid \operatorname{den} x_0$ and $Y_0 = H_{K-1}^{(2)} - 16 x_0 \bmod \ell$: if $B(Y_0) \neq 0$ then $v_\ell(\det \mathrm{Full}(x_0)) = -2(\#C_{\text{high}} + m)$ exactly; in particular $\det \mathrm{Full}(x_0) \neq 0$. (Reduction commutes with evaluation because $x_0 \in \mathbb{Z}_{(\ell)}$.)

Corollary 14.4 (boundary value)**.** *For member 1 ($|G_1| + m = K$): $B_1(0) = \pm \det C_{G_1 \cup Y'} \cdot \det C_{G_1 \cup X'} \neq 0$ in $\mathbb{F}_\ell$.*

Proof. At $Y = 0$ the matrix is $\begin{pmatrix} C_G C_G^\top & W_{Y'} \\ W_{X'}^\top & 0 \end{pmatrix} = PQ$ with $P = \begin{pmatrix} C_G & W_{Y'} & 0 \\ 0 & 0 & I_m \end{pmatrix}$, $Q = \begin{pmatrix} C_G^\top & 0 \\ 0 & I_m \\ W_{X'}^\top & 0 \end{pmatrix}$, both square of size $K + m = s + 2m$; $\det P = \pm \det[C_G, W_{Y'}]$, $\det Q = \pm \det[C_G, W_{X'}]$. Each is a $K \times K$ Cauchy determinant $\prod_{a<a'}(a' - a) \prod_{j<j'}(j' - j) / \prod (a + j)$ whose factors are integers in $[1, 2K + m - 1] \subseteq [1, 3K - 2]$, and $3K - 2 < 4K - 1 = \ell$. So the determinant is an ℓ-unit. □

14.2. **The dual formulation and bordering.** For a member put $\Gamma_{ij} = \sum_{a<K} \frac{1}{(a+i)(a+j)}$, i.e. $\Gamma = W^\top W$, and let

$$\hat{\Gamma} = C_{G \cup X'}^\top C_{G \cup Y'} \ (= \Gamma[G \sqcup J_{X'}, G \sqcup J_{Y'}]), \qquad E = \operatorname{diag}(I_G, 0_m), \qquad D(z) = zE + \hat{\Gamma}.$$

Lemma 14.5. *For $z \neq 0$, $\ker M(z) \cong \ker D(z)$ and the same for left kernels, where $M(z) = \begin{pmatrix} zI + C_G C_G^\top & W_{Y'} \\ W_{X'}^\top & 0 \end{pmatrix}$ is the matrix of $B(z)$. Hence the nonzero roots of B and of $\det D$ coincide.*

Proof. Let $(x, c) \in \ker M(z)$: $(z + C_G C_G^\top)x + W_{Y'} c = 0$ and $W_{X'}^\top x = 0$. Put $\tilde{u} = -(C_G^\top x,\, c)/z$. Then $x = C_{G \cup Y'} \tilde{u}$ and the rows of $D(z)\tilde{u}$ are $C_{G \cup X'}^\top$ applied to x (rows in X' give $W_{X'}^\top x = 0$; rows in G give $C_G^\top x + z\tilde{u}_G = 0$). Conversely, for $\tilde{u} \in \ker D(z)$ put $x = C_{G \cup Y'} \tilde{u}$, $c = -z\tilde{u}_{Y'}$; then $(z + C_G C_G^\top)x + W_{Y'} c = z(x - C_{G \cup Y'} \tilde{u}) = 0$ using $C_G^\top x = -z\tilde{u}_G$, and $W_{X'}^\top x = 0$. Both maps are injective as $z \neq 0$. Left kernels: transpose (exchange $X' \leftrightarrow Y'$). □

Let member 1 have absent set $G_1 = \{1..s\}$ and member 2 $G_2 = G_1 \setminus \{g\}$, $g = s$, with the same X', Y'. Then $\hat{\Gamma}_1$ is a $K \times K$ matrix $= C_{G_1 \cup X'}^\top C_{G_1 \cup Y'}$, invertible over $\mathbb{F}_\ell$ by the proof of Corollary 14.4. Put

$$T = (\hat{\Gamma}_1^{-1})_{G_1 G_1} \in M_s(\mathbb{F}_\ell), \qquad k_R = \det\left[e_g^\top T^k\right]_{0 \le k < s}, \qquad k_L = \det\left[T^k e_g\right]_{0 \le k < s}.$$

Lemma 14.6 (no common root)**.** *If B_1 and B_2 have a common root $z_0 \neq 0$ in $\overline{\mathbb{F}}_\ell$, then $k_R = 0$ or $k_L = 0$.*

Proof. By Lemma 14.5, $\det D_1(z_0) = \det D_2(z_0) = 0$. Now D_2 is D_1 with row g and column g deleted. Write $D_1 = \left(\begin{smallmatrix} D_2 & \gamma \\ \varrho^\top & z+\Gamma_{gg} \end{smallmatrix}\right)$; then $\det D_1 = (z + \Gamma_{gg}) \det D_2 - \varrho^\top \operatorname{adj}(D_2)\gamma$, so $\varrho^\top \operatorname{adj}(D_2)\gamma = 0$ at z_0. If $\operatorname{rank} D_2 \leq n_2 - 2$ (n_2 its size) then $\ker D_2$ has dimension ≥ 2 and contains $x \neq 0$ with $\varrho^\top x = 0$. If $\operatorname{rank} D_2 = n_2 - 1$, $\operatorname{adj}(D_2) = c\, xy^\top$ with $c \neq 0$, $D_2 x = 0$, $y^\top D_2 = 0$, and $(\varrho^\top x)(y^\top \gamma) = 0$. *Right case* ($\varrho^\top x = 0$, $x \neq 0$): $\tilde{u} = [x; 0] \in \ker D_1(z_0)$. From $z_0 E\tilde{u} + \hat{\Gamma}_1 \tilde{u} = 0$: $\tilde{u} = -z_0 \hat{\Gamma}_1^{-1} E\tilde{u}$; restricting to G_1: $T\tilde{u}_G = -z_0^{-1}\tilde{u}_G$, and $\tilde{u}_G \neq 0$ (otherwise $\tilde{u} = 0$), with $\tilde{u}_g = 0$. Thus $e_g^\top T^k \tilde{u}_G = (-z_0)^{-k}\tilde{u}_g = 0$ for all k, i.e. the matrix $[e_g^\top T^k]_{k<s}$ is singular: $k_R = 0$. *Left case* ($y^\top \gamma = 0$): $[y^\top, 0]$ is a left kernel vector of $D_1(z_0)$; likewise $y_G^\top T = -z_0^{-1} y_G^\top$ with $y_g = 0$, so $[T^k e_g]_{k<s}$ is singular and $k_L = 0$. □

(If $s = 1$, i.e. $m = K - 1$, then $G_2 = \emptyset$, T is 1×1 and $k_R = k_L = 1$.)

Theorem 14.7 (ℓ-adic nonvanishing, assuming Theorem 15.1)**.** *Let $x_0 \in \mathbb{Q}$, $\ell \nmid \operatorname{den} x_0$, $Y_0 = H_{K-1}^{(2)} - 16x_0 \bmod \ell$. If $k_R \neq 0 \neq k_L$ in $\mathbb{F}_\ell$, then $B_1(Y_0) \neq 0$ or $B_2(Y_0) \neq 0$; hence one of $\mathcal{G}(r_1, m, 0)$, $\mathcal{G}(r_1, m, 1)$ has $v_\ell(\det \operatorname{Full}(x_0)) = -2(\#C_{\text{high}} + m)$ and in particular $\det \operatorname{Full}(x_0) \neq 0$.*

Proof. If $Y_0 = 0$: $B_1(0) \neq 0$ (Corollary 14.4). If $Y_0 \neq 0$ and $B_1(Y_0) = B_2(Y_0) = 0$, Lemma 14.6 contradicts $k_R k_L \neq 0$. Then apply the consequence of Proposition 14.3 (note $\#C_{\text{high}}$ differs by one between the two members). □

The selection of m, and κ. Theorem 14.7 holds for every $m \in [0, K-1]$ with $r_1 = \ell - K - m$, so we are free to take $m = \operatorname{round}((\ell - K)/9)$ (so that $m/r_1 \to \frac{1}{8}$). Then $r_1/\ell \to \frac{8}{9} \cdot \frac{3}{4} = \frac{2}{3} = \kappa = \frac{3/4}{1+\alpha}$ and the two members have scales equal to r_1. The condition $r + m \leq \ell - K$ is exactly the existence of a forced common root: $\deg B_i$ drops (the top K rows of $M(0)$ have rank $\leq s+m$), so $Y^{\max(0,K-s-m)} \mid B$, and members with $r+m > \ell-K$ would have $B(0) = 0$; this is why κ cannot be raised. (Members with $r+m < \ell-K$ have no forced root but $B(0)$ is then a Gram-type minor which is not a product, so $Y_0 = 0$ is not covered by the argument.) The condition $\ell \nmid \operatorname{den} x_0$ is removed by Proposition 10.5.

The remaining input is the nonvanishing of k_R, k_L modulo ℓ for *every* prime $\ell \equiv 3 \pmod 4$, $\ell \geq 7$ and every $0 \leq m \leq K - 1$. For $m = 0$ this is the nonvanishing of the Krylov determinant of Theorem A; the general case is a closed form of the same kind, proved next.

15. The bordered Krylov determinants

In this section $K \geq 2$, $0 \leq m \leq K-2$ (the case $m = K-1$ is trivial: $k_R = k_L = 1$), $s = K - m \geq 2$, and

$$G = \{1, \dots, s\}, \quad Y = \{s+1, s+3, \dots, s+2m-1\}, \quad X = \{s+2, s+4, \dots, s+2m\},$$
$$Z = X \sqcup Y = \{s+1, \dots, s+2m\}, \quad g = s.$$

For a real parameter θ let $y_a = a + \theta$ ($0 \leq a < K$), $c_j = (1/(y_a + j))_a \in \mathbb{R}^K$, $C_J = [c_j]_{j \in J}$ and

$$\hat{\Gamma}(\theta) = C_{G\cup X}^{\top} C_{G \cup Y}, \qquad T(\theta) = (\hat{\Gamma}(\theta)^{-1})_{GG},$$
$$k_R(\theta) = \det[e_g^{\top} T^k]_{k<s}, \quad k_L(\theta) = \det[T^k e_g]_{k<s}.$$

(For $\theta \geq 0$ all entries $y_a + j$ are positive and the nodes are distinct, so the two $K \times K$ Cauchy matrices are invertible.) For $\theta = 0$ these are the matrices of §14 computed over $\mathbb{Q}$; their reductions modulo $\ell = 4K-1$ are the objects there, see Lemma 15.16. Write $T_2(n) = n(n-1)$ for $n \geq 1$, $T_2(n) = 0$ for $n \leq 0$, $D = \frac{s(s-1)(4s+6m+1)}{3}$ and define, for $1 \leq c \leq 2K-1$,

(15.1)
$$e_c = \begin{cases} c\,\mathbf{1}[c < s] + T_2(\min(c, s)) + T_2(c-m) & 1 \leq c \leq K, \\ T_2(n) + T_2(o) + d'\,\mathbf{1}[d' < s] & c = K + d',\ 1 \leq d' \leq K-1, \end{cases}$$

with $o = (s-d')_+$, $|F| = (d'-m)_+$, $n = s - |F|$.

Theorem 15.1. *$k_R(\theta)$ and $k_L(\theta)$ are polynomials in θ with rational coefficients, and*

$$k(\theta) = \varkappa \prod_{c=1}^{2K-1} (\theta + c)^{e_c}, \qquad k \in \{k_R, k_L\},$$

where $\varkappa \in \mathbb{Q}^{\times}$ has only prime factors $\leq 2K-1$. Consequently $k_R(0), k_L(0)$ are nonzero rationals with numerator and denominator prime to $\ell = 4K-1$, and for ℓ prime $k_R, k_L \not\equiv 0 \pmod{\ell}$ in the sense of §14.

The proof occupies this section: reduction to a symmetric matrix of Hahn type (§15.1), the order of vanishing at each $\theta = -c$ (§15.3, §15.4), the degree and leading coefficient (§15.5), and $\sum_c e_c = D$ (§15.6).

15.1. Polynomiality and the symmetric form.

Lemma 15.2. $k_R, k_L \in \mathbb{Q}[\theta]$.

Proof. The inverse of a Cauchy matrix $[1/(y_a + j_b)]$ has entries

$$\pm \frac{\prod_k (y_a + j_k) \prod_{a'} (y_{a'} + j_b)}{(y_a + j_b) \prod_{k \neq b} (j_k - j_b) \prod_{a' \neq a} (y_{a'} - y_a)};$$

the factor $y_a + j_b$ occurs in both products, so the entry is a polynomial in θ (the differences $y_{a'} - y_a = a' - a$ are constants). Hence $\hat{\Gamma}^{-1} = C_{G\cup Y}^{-1} C_{G\cup X}^{-\top}$, T, k_R, k_L have polynomial entries. □

Put $\omega(x) = \prod_{i=1}^{s}(x-i)$, $\pi_a = \prod_{b\neq a}(b-a) = \pm a!(K-1-a)!$, $\Pi_X(z) = \prod_{x\in X}(z-x)$, Π_Y likewise, $\Pi_Z = \Pi_X\Pi_Y$, and

$$\Omega_a = \frac{(y_a+1)_s(y_a+1)_{s+2m}}{\pi_a^2}, \quad A = C_G^{\top}\operatorname{diag}(\Omega_a)C_G \in M_s(\mathbb{Q}[\theta]), \quad \Psi_j = \frac{(j+\theta)_K}{|\omega'(j)|\sqrt{\Pi_Z(j)}}.$$

($\Pi_Z(j) > 0$ for $j \le s$, as it is a product of $2m$ negative numbers.) Let $\Lambda = \operatorname{diag}\big((\Pi_X(j)/\Pi_Y(j))^{1/2}\big)_{j\in G}$, a *constant* positive matrix (both products have sign $(-1)^m$), and $\mathsf{H} = \Psi A \Psi$, real symmetric, with entries polynomial in θ up to the constant square roots.

Proposition 15.3. *$T = \Lambda\,\mathsf{H}\,\Lambda^{-1}$. Consequently, with $\kappa_{\mathsf{H}} = \det[e_g^{\top}\mathsf{H}^k]_{k<s} = \det[\mathsf{H}^k e_g]_{k<s}$,*

$$k_R = \Lambda_g^{s}(\det\Lambda)^{-1}\kappa_{\mathsf{H}}, \qquad k_L = \Lambda_g^{-s}(\det\Lambda)\kappa_{\mathsf{H}}, \qquad k_R k_L = \kappa_{\mathsf{H}}^2,$$

so k_R/k_L is a nonzero constant independent of θ.

Proof. Step 1. Block inversion of $\hat\Gamma = \begin{pmatrix} C_G^{\top}C_G & C_G^{\top}C_Y \\ C_X^{\top}C_G & C_X^{\top}C_Y \end{pmatrix}$ gives $T = S^{-1}$, $S = C_G^{\top}\Pi C_G$, $\Pi = I - C_Y(C_X^{\top}C_Y)^{-1}C_X^{\top}$, the projector with range $\ker C_X^{\top}$ and kernel $\operatorname{range} C_Y$. If the columns of U span $\ker C_X^{\top}$ and those of V span $\ker C_Y^{\top}$, then $\Pi = U(V^{\top}U)^{-1}V^{\top}$. *Step 2.* For $u \in \mathbb{R}^K$, $\sum_a \frac{u_a}{z+y_a} = \frac{p(z)}{\prod_b (z+y_b)}$ with $\deg p < K$ and $u_a = p(-y_a)/\pi_a$. Hence $u \in \ker C_X^{\top}$ iff $p = \Pi_X q$, $\deg q < s$; similarly $\ker C_Y^{\top}$ with Π_Y. Choose q, r in the Lagrange basis ℓ_i of the nodes $1,\dots,s$: then $C_G^{\top}U = D_X := \operatorname{diag}(\Pi_X(j)/(j+\theta)_K)$, $V^{\top}C_G = D_Y$ (same with Π_Y), and $M := V^{\top}U$ has entries $\sum_a \Pi_X(-y_a)\Pi_Y(-y_a)\ell_i(-y_a)\ell_{i'}(-y_a)/\pi_a^2$. Therefore $S = D_X M^{-1} D_Y$, $T = D_Y^{-1} M D_X^{-1}$. *Step 3.* $\ell_i(-y) = -\omega(-y)/((y+i)\omega'(i))$, $\Pi_Z(-y) = (y+s+1)_{2m}$, $\omega(-y)^2 = (y+1)_s^2$; so $M = D_{\omega'}^{-1} A D_{\omega'}^{-1}$ and $T = \Lambda_1 A \Lambda_2$ with $\Lambda_1 = \operatorname{diag}\frac{(j+\theta)_K}{\Pi_Y(j)\omega'(j)}$, $\Lambda_2 = \operatorname{diag}\frac{(j+\theta)_K}{\Pi_X(j)\omega'(j)}$. Since $\Lambda_1\Lambda_2 = \Psi^2$ and $\Lambda_1/\Lambda_2 = \Lambda^2$, $\Lambda_1 = \Psi\Lambda$, $\Lambda_2 = \Psi\Lambda^{-1}$, i.e. $T = \Lambda\Psi A\Psi\Lambda^{-1}$. The formulas for k_R, k_L: $e_g^{\top}T^k = \Lambda_g e_g^{\top}\mathsf{H}^k\Lambda^{-1}$ and $T^k e_g = \Lambda_g^{-1}\Lambda\mathsf{H}^k e_g$. $\square$

Lemma 15.4 (moment form)**.** *Let $\varrho_a = (y_a+s+1)_{2m}/\pi_a^2$. There is a constant invertible matrix $\mathcal{E} \in M_s(\mathbb{Q})$ with $A = \mathcal{E}\,\mathbb{M}\,\mathcal{E}^{\top}$, $\mathbb{M} = \big[\sum_{a<K}\varrho_a\, y_a^{p+q}\big]_{p,q<s}$. Equivalently, for $f \in \mathbb{R}^G$ and $P_f(x) = \sum_j f_j\,\omega(x)/(x-j)$:*

$$f^{\top}Af = \sum_{a<K}\varrho_a\,P_f(x_a)^2, \qquad f^{\top}\Psi^{-2}f = \sum_{j=1}^{s}\nu_j P_f(j)^2,$$

$$x_a = -y_a, \qquad \nu_j = \frac{\Pi_Z(j)}{\prod_b (j-x_b)^2}, \qquad \varrho_a = \frac{\Pi_Z(x_a)}{\prod_{b\neq a}(x_a-x_b)^2}.$$

Proof. The polynomial $P_y(x) = \frac{1-\omega(x)/\omega(-y)}{x+y} = -\frac{Q(x,y)}{\omega(-y)}$, $Q = \frac{\omega(x)-\omega(-y)}{x+y} = \sum_{r<s} x^r q_r(y)$, interpolates $x \mapsto \frac{1}{y+x}$ at $x = 1,\dots,s$; q_r has degree exactly

$s-1-r$ with leading coefficient ±1, so $(q_r(y))_r = \mathcal{N}(y^p)_p$ with $\mathcal{N}$ constant and anti-triangular. Hence $C_G = -D_{1/\omega(-y)}\hat{Y}\mathcal{N}^\top V_s^\top$ ($\hat{Y} = [y_a^p]$, $V_s = [j^r]$), and $A = \mathcal{E}\hat{Y}^\top D_\varrho \hat{Y}\mathcal{E}^\top$, $\mathcal{E} = V_s\mathcal{N}$, since $\Omega_a/\omega(-y_a)^2 = \varrho_a$. The quadratic-form identities follow from $\sum_j f_j/(y+j) = -P_f(-y)/\omega(-y)$ and $P_f(j) = f_j\omega'(j)$, $\Psi_j^{-2} = \omega'(j)^2\Pi_Z(j)/(j+\theta)_K^2$. □

So the eigenvalues of H are those of the pencil $\langle\cdot,\cdot\rangle_\varrho - \lambda\langle\cdot,\cdot\rangle_\nu$ on $\mathrm{Poly}_{<s}$, and H has entries polynomial in θ (up to constant roots).

15.2. **Analytic tools.** Fix $c \in \{1,\dots,2K-1\}$, $\theta = -c+\epsilon$ (ϵ real), and write $\mathsf{H}(\epsilon)$.

Lemma 15.5 (Rellich)**.** *Let $H(\epsilon)$ be a real symmetric $s\times s$ matrix, real-analytic in ϵ near 0. There are analytic eigenvalue branches $\lambda_i(\epsilon)$ and an analytic orthogonal $U(\epsilon)$ of eigenvectors, and $\kappa(\epsilon) := \det[e_g, He_g, \dots, H^{s-1}e_g] = \pm\prod_i u_{ig}\prod_{i<i'}(\lambda_{i'}-\lambda_i)$ ($u_{ig} = U_{gi}$). Hence $\operatorname{ord}_0\kappa = \sum_i \operatorname{ord} u_{ig} + \sum_{i<i'}\operatorname{ord}(\lambda_i-\lambda_{i'})$. If for all small $\epsilon\neq0$ at least d eigenvalues of $H(\epsilon)$ lie in a window of width $C\epsilon^p$, then some d branches satisfy $\operatorname{ord}(\lambda_i-\lambda_{i'})\ge p$ pairwise.*

Proof. Rellich's theorem; $H^k e_g = \sum_i \lambda_i^k u_{ig}\varphi_i$ so the Krylov matrix is $U\operatorname{diag}(u_{\cdot g})[\lambda_i^k]$ (Vandermonde), $\det U = \pm1$ constant. For the last claim, along a sequence $\epsilon_n\to0$ a fixed d-subset of the finitely many branches lies in the window; a nonzero analytic function of order $q<p$ is $\gg|\epsilon|^q$ near 0. (Identically coinciding branches give $\kappa\equiv0$ and $\operatorname{ord}=\infty$; the inequality remains true.) □

Lemma 15.6 (cluster lemma)**.** *Let $\mathcal{M}(\epsilon) = \mathcal{M}_0 + \epsilon\mathcal{M}_1 + O(\epsilon^2)$ be real symmetric and analytic, suppose $\mathcal{M}_0$ has an eigenvalue 1 with eigenspace $\mathcal{E}_1$ (dimension d) isolated from the rest of its spectrum, and $P_{\mathcal{E}_1}\mathcal{M}_1P_{\mathcal{E}_1}=0$. Then $\mathcal{M}(\epsilon)$ has d eigenvalues $1+O(\epsilon^2)$.*

Proof. Kato's first-order perturbation theory: the splittings of order ϵ are the eigenvalues of $P_{\mathcal{E}_1}\mathcal{M}_1P_{\mathcal{E}_1}$ (or: conjugate by $\exp(\epsilon S)$ with S antisymmetric cancelling the $\mathcal{E}_1$–$\mathcal{E}_1^\perp$ blocks of $\mathcal{M}_1$, then Weyl's inequality). □

Lemma 15.7 (scaled Schur lemma)**.** *Let $H(\epsilon) = \Gamma\hat{H}(\epsilon)\Gamma$ with $\hat{H}$ real symmetric analytic, $\Gamma = \operatorname{diag}(\epsilon$ on J_1, 1 on $J_2)$, and $\hat{H}_{22}(0)$ invertible. Then exactly $|J_1|$ eigenvalues of H are $O(\epsilon^2)$, the other $|J_2|$ tend to the eigenvalues of $\hat{H}_{22}(0)$ (nonzero), and the unit eigenvectors of the latter have J_1-components $O(\epsilon)$; the eigenvectors of the former have J_2-components $O(\epsilon)$.*

Proof. For $|\lambda|\le\delta$, $\det(H-\lambda) = \det(\hat{H}_{22}-\lambda+O(\epsilon^2))\det(\epsilon^2X(\lambda,\epsilon)-\lambda I)$ by the Schur complement with X bounded; the second factor has exactly $|J_1|$ roots in $|\lambda|\le\delta$, all $O(\epsilon^2)$ (Rouché). From $(\hat{H}_{22}-\lambda)u_2 = -\epsilon\hat{H}_{21}u_1+O(\epsilon^2)u_2$ we get $u_2 = O(\epsilon)u_1$ for the small eigenvalues, and symmetrically for the large ones. □

15.3. **Zeros at $\theta = -c$ for $1 \le c \le K$.**

Proposition 15.8. *For $1 \le c \le K$, $\operatorname{ord}_{\theta=-c} \kappa_{\mathsf{H}} \ge e_c$ (with e_c as in (15.1)).*

Proof. With $\theta = -c+\epsilon$: $x_a = c-a-\epsilon$, $x_a^0 = c-a$, $W_0 = \{c-K+1, \dots, c\}$; Ψ_j has a simple zero at $\epsilon = 0$ iff $(j+\theta)_K$ does iff $c-K+1 \le j \le c$, i.e. (as $c \le K$) iff $j \in O := \{1, \dots, \min(c,s)\}$. So $\mathsf{H} = \Lambda_\epsilon \tilde{\mathsf{H}} \Lambda_\epsilon$ with $\Lambda_\epsilon = \operatorname{diag}(\epsilon \text{ on } O, 1)$ and $\tilde{\mathsf{H}} = \tilde{\Psi} A \tilde{\Psi}$ analytic. For $j \in O$, $a = c - j$ collides: $j - x_a = \epsilon$, and $\epsilon^2 \nu_j \to \varrho_a^0$ exactly. The *far* nodes are $F = \{c-K+1, \dots, 0\}$, $|F| = K - c$; nodes of W_0 above s lie in Z (since $c \le K \le s + 2m$) and have $\varrho_a = O(\epsilon)$; $\varrho^0 > 0$ on $O \cup F$ (there $\Pi_Z > 0$).

Case $c < s$. Here $J_1 = O = \{1..c\}$, $J_2 = \{c+1..s\} \ni g$. At $\theta = -c$, Ω_a has a double zero for $a < c$ (both Pochhammers contain the factor $a - c + i = 0$ for $i = c - a \in [1, s]$), cancelling the double pole of the Cauchy factors, and $\Omega_a > 0$ for $a \ge c$; the columns $j > c$ of C_G have $y_a + j = a - c + j > 0$ for all $a \ge 0$. Hence $\tilde{\mathsf{H}}_{22}(0) = \tilde{\Psi}\big(\sum_{a \ge c} \Omega_a c_a c_a^\top\big)_{>c,>c} \tilde{\Psi}$ is positive definite (Cauchy columns $1/(a-c+j)$, $j \in (c, s]$, $a \in [c, K)$, have full rank $s - c \le K - c$). Lemma 15.7: exactly c eigenvalues are $O(\epsilon^2)$ and their eigenvectors have $u_{ig} = O(\epsilon)$ ($g \in J_2$): $\sum \operatorname{ord} u_{ig} \ge c$. *Case $c \ge s$.* Then $O = G$, $\mathsf{H} = \epsilon^2 \tilde{\mathsf{H}}$: all s eigenvalues are $O(\epsilon^2)$, and $\sum \operatorname{ord} u_{ig} \ge 0$. *The cluster.* Put $\mathcal{N}(\epsilon) = \epsilon^2 \langle \cdot, \cdot \rangle_\nu$, $\mathcal{R}(\epsilon) = \langle \cdot, \cdot \rangle_\varrho$ on $\mathrm{Poly}_{<s}$, both analytic at 0. $\mathcal{R}_0(P) \ge \sum_{O \cup F} \varrho^0 P^2$ is positive definite ($|O \cup F| = \min(c,s) + K - c \ge s$), and $\mathcal{N}_0(P) = \sum_{j \in O} \varrho^0_{c-j} P(j)^2 \le \mathcal{R}_0(P)$ with equality iff $P|_F = 0$. With the analytic Cholesky factorisation $\mathcal{R} = L^\top L$, $\mathcal{M} = L^{-\top} \mathcal{N} L^{-1}$ is symmetric analytic, $\mathcal{M}_0$ has maximal eigenvalue 1 with eigenspace $\mathcal{U}_0 = \{P : P|_F = 0\}$, of dimension $s - (K - c) = c - m$ (if $c \ge m$). The eigenvalues μ of $\mathcal{M}$ and λ of H satisfy $\lambda = \epsilon^2/\mu$.

Residue identity. For $P, Q \in \mathcal{U}_0$, $P^\top(\mathcal{N}_1 - \mathcal{R}_1)Q = 0$. Indeed with $F_0 = PQ$ (double zeros on F), $W_0(x) = \prod_b (x - x_b^0)$ and $\mathcal{Y} = \Pi_Z F_0 / W_0^2$: at a collision node $x_a^0 = j$, $\frac{d}{d\epsilon}[\epsilon^2 \nu_j F_0(j) - \varrho_a F_0(x_a)]_0 = (\Pi_Z F_0 / \prod_{b \ne a}(x - x_b^0)^2)'(j) = \operatorname{Res}_{x=j} \mathcal{Y}$ (double pole); at a Z-node $z = x_a^0$: $-\frac{d}{d\epsilon} \varrho_a F_0(x_a) = \Pi_Z'(z) F_0(z) / \prod_{b \ne a}(z - x_b^0)^2 = \operatorname{Res}_z \mathcal{Y}$ (simple pole); far nodes and non-colliding ν-nodes contribute $O(\epsilon^2)$ and $\mathcal{Y}$ has no pole there. So the sum is the sum of all finite residues of $\mathcal{Y}$, which vanishes because $\deg \mathcal{Y} \le 2m + 2(s-1) - 2K = -2$. Since $\mathcal{N}_0(P, \cdot) = \mathcal{R}_0(P, \cdot)$ for $P \in \mathcal{U}_0$, this is $P_{\mathcal{U}_0} \mathcal{M}_1 P_{\mathcal{U}_0} = 0$. Lemma 15.6 gives $c - m$ eigenvalues $\mu = 1 + O(\epsilon^2)$, i.e. $\lambda = \epsilon^2(1 + O(\epsilon^2))$, in a window of width $O(\epsilon^4)$ (these are among the $\min(c,s)$ small eigenvalues).

Counting. By Lemma 15.5: for $c < s$, $\operatorname{ord} \kappa_{\mathsf{H}} \ge c + 2\big[\binom{c}{2} - \binom{c-m}{2}\big] + 4\binom{c-m}{2} = c + c(c-1) + (c-m)(c-m-1)$; for $s \le c \le K$, $\ge s(s-1) + (c-m)(c-m-1)$ (pairs among the s small eigenvalues give 2, pairs inside the cluster 4). This is (15.1). □

15.4. **Zeros at $\theta = -c$ for $K < c \le 2K-1$.** Write $c = K + d'$, $1 \le d' \le K - 1$, $\epsilon = \theta + c$. Now $W_0 = \{d'+1, \dots, c\}$ splits into $O = W_0 \cap V = \{d'+1, \dots, s\}$

($|O| = o$), $Z_c = W_0 \cap Z$ and $F = (s+2m, c]$, $|F| = (d'-m)_+$; the free ν-nodes are $Q = V \setminus O = \{1, \dots, \min(d', s)\}$, and Ψ_j vanishes at $\epsilon = 0$ exactly for $j \in O$. Put $\mathcal{K}_0 = \{P \in \mathrm{Poly}_{<s} : P|_{O \cup F} = 0\}$, $\dim \mathcal{K}_0 = r_0 := s - o - |F|$ ($o + |F| \le s$), $n := o + r_0 = s - |F|$. Since $x_a - x_b$ does not depend on ϵ,

$$A(\epsilon)(P,Q) = \sum_a c_a \, \mathcal{G}(x_a^0 - \epsilon), \quad \mathcal{G} = \Pi_Z P Q, \; c_a = \prod_{b \neq a} (x_a^0 - x_b^0)^{-2}, \tag{15.2}$$

with $A = A_0 + \epsilon A_1 + \epsilon^2 A_2 + \dots$ (Taylor coefficients of this exact form).

Lemma 15.9 (residue facts)**.** *For $P_0 \in \mathcal{K}_0$ and arbitrary $R \in \mathrm{Poly}_{<s}$: $A_0(P_0, R) = 0$ and $A_1(P_0, R) = 0$.*

Proof. A_0: P_0 vanishes on $O \cup F$ and $\varrho^0 = 0$ on Z_c. $A_1 = -\sum_a c_a \mathcal{G}'(x_a^0)$ with $\mathcal{G} = \Pi_Z P_0 R$. The function $\Phi = \Pi_Z P_0 R / W_0^2$ has at most a simple pole at each node of $W_0 = O \cup Z_c \cup F$ (P_0 cancels one factor at $O \cup F$, Π_Z at Z_c) with residue $c_a \mathcal{G}'(x_a^0)$, and $\deg \Phi \le 2m + 2(s-1) - 2K = -2$; the residues sum to 0. □

Proposition 15.10 (L1)**.** *Exactly n eigenvalues of $\mathsf{H}(\epsilon)$ are $O(\epsilon^2)$; the others tend to nonzero limits, and their eigenvectors have components $O(\epsilon)$ on $O \oplus \mathcal{Q}_0$ (defined in the proof). If $d' \ge s$ then $o = 0$ and $\operatorname{ord} \kappa_{\mathsf{H}} \ge n(n-1)$.*

Proof. $\mathsf{H} = \Lambda_\epsilon \tilde{\mathsf{H}} \Lambda_\epsilon$ ($\Lambda_\epsilon = \epsilon$ on O); $\tilde{\mathsf{H}}(0) \succeq 0$ since $A_0 \succeq 0$. Work in value coordinates $f \leftrightarrow P_f$. The kernel $\mathcal{Q}_0 \subseteq \mathbb{R}^Q$ of $\tilde{\mathsf{H}}_{QQ}(0)$ is $\tilde{\Psi}(0)^{-1}\mathcal{K}_0$: for f supported on Q, P_f vanishes on O, and $f^\top A_0 f = \sum_{O \cup F} \varrho^0 P_f^2 = 0$ iff $P_f|_F = 0$. As $\tilde{\mathsf{H}}(0) \succeq 0$, the rows of $\mathcal{Q}_0$ in $\tilde{\mathsf{H}}(0)$ vanish. For $f, f' \in \mathcal{Q}_0$, expanding $(\tilde{\Psi}(\epsilon)f)^\top A(\epsilon)(\tilde{\Psi}(\epsilon)f')$ to first order and using Lemma 15.9 (with $P_0 = P_{\tilde{\Psi}(0)f} \in \mathcal{K}_0$) gives $\tilde{\mathsf{H}}_{\mathcal{Q}_0\mathcal{Q}_0} = O(\epsilon^2)$. So $\mathsf{H} = \Gamma \hat{\mathsf{H}} \Gamma$ with $\Gamma = \epsilon$ on $O \oplus \mathcal{Q}_0$, 1 on $\mathcal{Q}_1 = \mathcal{Q}_0^\perp \cap \mathbb{R}^Q$, $\hat{\mathsf{H}}$ analytic and $\hat{\mathsf{H}}_{\mathcal{Q}_1\mathcal{Q}_1}(0) = \tilde{\mathsf{H}}_{\mathcal{Q}_1\mathcal{Q}_1}(0)$ positive definite. Apply Lemma 15.7: $|O| + \dim \mathcal{Q}_0 = o + r_0 = n$ small eigenvalues. If $d' \ge s$, $O = \emptyset$ and all pairs of small eigenvalues have ord ≥ 2: $2\binom{n}{2} = n(n-1)$. □

For $d' < s$ two more facts are needed (here $g = s \in O$). Fix a complement $\mathcal{K}_0^c = C_1 \oplus C_2$ of $\mathcal{K}_0$ with $C_1 = \{P : P|_F = 0\}/\mathcal{K}_0$, $C_2 = \{P : P|_O = 0\}/\mathcal{K}_0$ ($\dim C_1 = o$, $\dim C_2 = |F|$), let $D_\epsilon(P_\perp + P_0) = P_\perp + \epsilon^{-1} P_0$ and $\mathcal{A} = A \circ D_\epsilon$, $\mathcal{N} = \epsilon^2 N \circ D_\epsilon$ (analytic by Lemma 15.9). At $\epsilon = 0$, in the value coordinates on $O \cup F$ for $\mathcal{K}_0^c$,

$$\mathcal{A}_0 = \mathrm{diag}(\varrho_O^0, \varrho_F^0) \oplus A_2|_{\mathcal{K}_0}, \qquad \mathcal{N}_0 = \mathrm{diag}(\varrho_O^0, 0) \oplus N_Q|_{\mathcal{K}_0}, \qquad N_Q(P_0) = \sum_{j \in Q} \nu_j^0 P_0(j)^2,$$

the cross terms vanishing by Lemma 15.9(b). Thus the pencil $(\mathcal{N}, \mathcal{A} + \mathcal{N})$ has at $\epsilon = 0$: eigenvalue $\mu' = \frac{1}{2}$ on C_1, 0 on C_2 and the eigenvalues of $N_Q(A_2 + N_Q)^{-1}$ on $\mathcal{K}_0$.

Gap condition (G′): $A_2 + N_Q$ is positive definite on $\mathcal{K}_0$ and no eigenvalue τ of $A_2 N_Q^{-1}$ equals 1.

Lemma 15.11 (L2, L3 under (G′))**.** *Assume (G′). Then (L2) o of the small eigenvalues of H lie in a window of width $O(\epsilon^4)$; (L3) $\sum_i \operatorname{ord} u_{ig} \ge d'$.*

Proof. (L2) For $\epsilon \neq 0$ the pencils $(\mathcal{N}, \mathcal{A})$ and $(\mathcal{N}, \mathcal{A}+\mathcal{N})$ have the same eigenvectors and $\mu = \mu'/(1-\mu')$. Under (G′), $\mathcal{A}_0 + \mathcal{N}_0 \succ 0$ so $\mathcal{M}' = L'^{-\top}\mathcal{N}L'^{-1}$ is analytic with eigenvalue $\frac{1}{2}$ of eigenspace exactly C_1 (isolated: the $\mathcal{K}_0$-eigenvalues are > 0 and $\neq \frac{1}{2}$ by $\tau \neq 1$). For $P, Q \in C_1$, D_ϵ is the identity and the first-order term is $\frac{1}{2}P^\top(\mathcal{N}_1 - \mathcal{A}_1)Q$, which vanishes by the residue identity of §15.3 (verbatim: PQ has double zeros on F, double poles of $\Pi_Z PQ/W_0^2$ at O, simple poles at Z_c). Lemma 15.6: o eigenvalues $\mu' = \frac{1}{2} + O(\epsilon^2)$, hence $\mu = 1 + O(\epsilon^2)$ and $\lambda = \epsilon^2/\mu$ lies in a window of width $O(\epsilon^4)$. (L3) The eigenvectors of H correspond to ν-orthonormal eigenpolynomials $\hat{P}$ and $u_{ig} = \pm\sqrt{\nu_g}\,\hat{P}(g)\,\omega'(g)/(\text{const})$. For the r_0 branches whose limit pencil eigenvalue lies in $\mathcal{K}_0$'s spectrum (separated from $\{0, \frac{1}{2}\}$ by (G′)), the group projector shows $\hat{P} \in \mathcal{K}_0 + O(\epsilon)$; as $\hat{P}_0(g) = 0$ $(g \in O)$, $\hat{P}_\perp(g) = O(\epsilon)$ and $\mathcal{N}_0 \geq c > 0$ there, so $u_{ig} = \sqrt{\nu_g}\epsilon\hat{P}_\perp(g)/\mathcal{N}(\hat{P})^{1/2} = O(\epsilon)$ $(\nu_g \sim \epsilon^{-2})$. The $|F|$ large-eigenvalue branches have $u_{ig} = O(\epsilon)$ by Proposition 15.10. Together $|F| + r_0 = d'$ branches. (To avoid ambiguity at coinciding eigenvalues, one applies the argument to the spectral projector of each isolated group: if $\Pi(0)e_g = 0$ then $\sum_{\text{group}} u_{ig}^2 = \|\Pi(\epsilon)e_g\|^2 = O(\epsilon^2)$ and every branch of the group has $\operatorname{ord} u_{ig} \geq 1$.) □

Removing (G′): deformation of the free ν-nodes. (G′) is true for $K \leq 18$ with a margin shrinking like 3^{-K} (exact computation), but we do not need it. Keep the ϱ-side and the collision nodes O and replace the free ν-nodes $j \in Q = \{1..d'\}$ by parameters $y_1, \dots, y_{d'}$: $\nu_j := \Pi_Z(y_j)/W_\theta(y_j)^2$, $A_{jj'} := \sum_a \varrho_a L_j(x_a)L_{j'}(x_a)$ (L_j the Lagrange basis on all s nodes $Q \cup O$), $h_n := ((\mathcal{N}^{-1}A)^n)_{gg}$ and $\mathbb{K}(\theta; y) := \det[h_{i+j}]_{i,j<s}$. Then: (i) $\mathbb{K} \in \mathbb{Q}[y][1/\Delta][\theta]$ with $\Delta = \prod_{i<j}(y_i - y_j)\prod_{i,o\in O}(y_i - o)\prod_i \Pi_Z(y_i)$ (the denominators of the Lagrange basis and of $\nu_j^{-1} = W_\theta(y_j)^2/\Pi_Z(y_j)$; the numerators are polynomial in θ since $x_a - x_b$ is constant); (ii) at $y = (1, \dots, d')$, $\Delta \neq 0$ and $\mathbb{K} = \kappa_{\mathsf{H}}^2$ (the Hankel matrix of the Krylov sequence is $\mathcal{K}^\top\mathcal{K}$ with $\mathcal{K}$ the Krylov matrix of the symmetric matrix $\mathcal{N}^{-1/2}A\mathcal{N}^{-1/2}$, whose powers have the same gg-entries as those of $\mathcal{N}^{-1}A$); (iii) the argument above (L1, L2, L3 and Lemma 15.5) uses only: the ϱ-side facts (Lemma 15.9, exact form (15.2)), the collision data ($\epsilon^2\nu_j \to \varrho^0$ for $j \in O$, $\Lambda_\epsilon = \epsilon$ on O), and for the free nodes only that $y_j \notin W_0$, $\Pi_Z(y_j) > 0$, distinctness; all these are open conditions. So for y in a nonempty open set $\mathcal{Y} \subset \mathbb{R}^{d'}$ on which (G′) holds we get $\operatorname{ord}_{\theta=-c}\mathbb{K}(\theta; y) \geq 2e_c$. (G′) holds on a nonempty open set: let $\mathcal{Z}_c = W_0 \cap Z$ ($|\mathcal{Z}_c| = m + \min(d', m) \geq r_0$, $d' \geq r_0$) and take $y_i = z_i \pm \delta$ for r_0 distinct $z_i \in \mathcal{Z}_c$ (sign chosen so that $\Pi_Z(y_i) > 0$) and the other y_j generic below 1. Then $\nu_i^0 = \Pi_Z(y_i)/W_0(y_i)^2 \sim |\Pi_Z'(z_i)|/(W_0'(z_i)^2\delta) \to \infty$, and $P_0 \mapsto (P_0(z_i))_{i<r_0}$ is injective on $\mathcal{K}_0 = \Pi_{O\cup F}\mathrm{Poly}_{<r_0}$ (as $\Pi_{O\cup F}(z_i) \neq 0$), so $N_Q \geq (c'/\delta)I - O(1)$ on $\mathcal{K}_0$ while A_2 does not depend on y; hence $A_2 + N_Q \succ 0$ and $|\tau| = O(\delta) < 1$. Finally, the Taylor coefficients of order $< 2e_c$ of $\mathbb{K}$ at $\theta = -c$ are elements of $\mathbb{Q}[y][1/\Delta]$ which vanish on the open set $\mathcal{Y}$; their numerators are polynomials vanishing on a nonempty open subset of $\mathbb{R}^{d'}$,

hence identically, in particular at $y = (1, \dots, d')$, where $\Delta \neq 0$. Hence $\operatorname{ord}_{\theta=-c} \kappa_{\mathsf{H}}^2 \geq 2e_c$.

Proposition 15.12. *For $K < c \leq 2K-1$, $\operatorname{ord}_{\theta=-c} \kappa_{\mathsf{H}} \geq e_c$.*

Proof. If $d' \geq s$: Proposition 15.10. If $d' < s$: by Lemma 15.5, $\operatorname{ord} \kappa_{\mathsf{H}} \geq 2\binom{n}{2} + 2\binom{o}{2} + d'$ (all pairs of small eigenvalues ≥ 2; the o-cluster ≥ 4; $\sum \operatorname{ord} u_{ig} \geq d'$), which is (15.1); the deformation argument removes the need for (G′). □

(Numerical confirmation: exact evaluation of the Rellich counts for all $2 \leq K \leq 300$, $0 \leq m \leq K-2$, $1 \leq c \leq 2K-1$ reproduces (15.1); and the exact orders $\operatorname{ord}_{\theta=-c} \mathbb{K}(\theta; y) = 2e_c$ at random rational y were verified for 22 triples (K, m, c).)

15.5. **Degree and leading coefficient.** Let $h_d(\varpi)$ denote the squared norm of the monic orthogonal polynomial p_d of degree d for a positive weight ϖ on finitely many nodes (with at least s nodes), and put the two weights

$$\varpi_a = \frac{1}{(a!\,(K-1-a)!)^2} \quad (a \in \{0, \dots, K-1\}),$$

$$v_j = \frac{1}{\omega'(j)^2 \Pi_Z(j)} = \frac{1}{((j-1)!)^2 (s-j)!\,(s+2m-j)!} \quad (j \in \{1, \dots, s\}).$$

Theorem 15.13. *As $\theta \to +\infty$ the eigenvalues of H satisfy $\lambda^{(d)} = h_d(v) h_d(\varpi)\, \theta^{4K-2-4d}(1 + O(1/\theta))$, $d = 0, \dots, s-1$ (distinct orders), and the unit eigenvector of $\lambda^{(d)}$ has $u_g^2 \to v_s\, p_d^{(v)}(s)^2 / h_d(v)$. Consequently $\deg_\theta \kappa_{\mathsf{H}} = D$ exactly and the leading coefficient $\varkappa_{\mathsf{H}}$ satisfies*

$$\varkappa_{\mathsf{H}}^2 = \prod_{d<s} \frac{v_s p_d^{(v)}(s)^2}{h_d(v)} \prod_{d<s} \left(h_d(v) h_d(\varpi)\right)^{2(s-1-d)}.$$

Proof. (1) *Factorisation.* $\Psi_j = \theta^K \tilde{\psi}_j (1 + O(1/\theta))$ with $\tilde{\psi}_j^2 = v_j$ (indeed $\Psi_j^{-2} \sim \theta^{-2K} \omega'(j)^2 \Pi_Z(j)$); $\varrho_a = \theta^{2m} \varpi_a (1 + O(1/\theta))$. Expanding $y_a^p = (\theta + a)^p$, $\mathbb{M} = R\, \mathbb{M}_c R^\top$ with $R = D_0 Z_0 D_0^{-1}$, $D_0 = \operatorname{diag}(\theta^p)$, $Z_0 = [\binom{p}{n}]$, and $\mathbb{M}_c = [\sum_a \varrho_a a^{p+q}] = \theta^{2m}(L \Delta_w L^\top + O(1/\theta))$, where $L \Delta_w L^\top$ is the $LDL^\top$ factorisation of the moment matrix of ϖ, $(\Delta_w)_d = h_d(\varpi) > 0$. Writing $\mathcal{E} = V_s \mathcal{N}$ ($\mathcal{N} J$ upper triangular with diagonal ± 1, J the reversal), one finds $\mathsf{H}/\theta^{2m+2K} = B(I+F)B^\top$, $F = O(1/\theta)$, $B = \tilde{\psi} V_s D' Y D_0^{-1} \Delta_w^{1/2}$, with $D' = \operatorname{diag}(\theta^{s-1-r})$, $Y = U_\theta J Z_0 L_\theta = \operatorname{diag}(\pm 1) J Z_0 + O(1/\theta)$ ($U_\theta = D'^{-1} \mathcal{N} J D'$ and $L_\theta = D_0^{-1} L D_0$ have entries $O(1/\theta)$ off the diagonal). With $I + F = (I + F_1)(I + F_1)^\top$ the eigenvalues of $\mathsf{H}/\theta^{2m+2K}$ are $\sigma_i(B(I+F_1))^2$, and $\sigma_i(B(I+F_1)) = \sigma_i(B)(1 + O(1/\theta))$. (2) *Compounds.* $\Lambda^k(B(I+F_1))$: the scaling $\Lambda^k(D') \otimes \Lambda^k(D_0^{-1})$ is uniquely maximal at the index sets $\{0..k-1\}$, every other pair being smaller by a factor $\geq \theta$; $\det[\binom{p}{n}]_{p=s-k..s-1,\, n<k} = \pm 1$. Hence $\Lambda^k B(I+F_1) = \theta^{e'_k} [\pm(\wedge_{r<k} \tilde{\psi} V_s e_r) \otimes \prod_{d<k} h_d(\varpi)^{1/2} e_{J_0}^\top + O(1/\theta)]$, and $\| \wedge_{r<k} \tilde{\psi} V_s e_r \|^2 = \det(V_k^\top \tilde{\psi}^2 V_k) = \prod_{d<k} h_d(v)$ (Gram determinant of

the moments of v). Taking norms and ratios gives $\lambda^{(d)}$; the exponent of θ is $2(s-1-2d)+2m+2K = 4K-2-4d$. (3) *Eigenvectors.* The compounds are rank one at leading order with scale gaps θ^2, so (Wedin) the top-k eigenspace of H converges to $\operatorname{span}(\tilde{\psi}V_k)$ [apply this to $\Lambda^k(B(I+F_1))$, whose left singular spaces are the eigenspaces of H]. The d-th eigenvector tends to $\tilde{\psi}\,p_d^{(v)}(j)/h_d(v)^{1/2}$ (Gram–Schmidt of $\tilde{\psi}j^r$), hence the limit of u_g^2. No limit vanishes: if $p_d(s)=0$, put $q=p_d/(x-s)$; $q\neq 0$ at some node $j<s$, and $0=\langle p_d,q\rangle_v=\sum_j v_j(j-s)q(j)^2<0$. (4) By Proposition 15.3's symmetric form, $\kappa_{\mathsf{H}} = \pm\prod_i u_{ig}\prod_{i<i'}(\lambda_{i'}-\lambda_i)$ for the (real symmetric) H at each large θ; the orders of $\lambda^{(d)}$ are strictly separated, so $\prod_{i<i'}|\lambda_i-\lambda_{i'}| = \prod_d(\lambda^{(d)})^{s-1-d}(1+O(\theta^{-4}))$, of degree $\sum_d(4K-2-4d)(s-1-d) = \frac{s(s-1)(4s+6m+1)}{3} = D$. □

Proposition 15.14 (prime factors of $\varkappa$)**.** *$\varkappa_{\mathsf{H}}^2$, and hence $\varkappa = \varkappa_R = \Lambda_g^s(\det\Lambda)^{-1}\varkappa_{\mathsf{H}}$, $\varkappa_L$, are rational numbers whose prime factors are all $\le 2K-1$.*

Proof. Both weights are Hahn weights with negative parameters: using $\binom{\alpha+x}{x}\binom{\beta+N-x}{N-x} = (-1)^N\binom{-\alpha-1}{x}\binom{-\beta-1}{N-x}$, $\varpi_a = \pm((K-1)!)^{-2}\binom{K-1}{a}^2$ is the Hahn weight with $\alpha=\beta=-K$, $N=K-1$ (in the variable $x=a$), and $v_j = \pm\frac{1}{(s+2m-1)!(s-1)!}\binom{s+2m-1}{j-1}\binom{s-1}{s-j}$ the Hahn weight with $\alpha=-s-2m$, $\beta=-s$, $N=s-1$ (variable $x=j-1$, the endpoint $j=s$ being $x=N$). In both cases $\alpha+\beta+1=-2K+1$. We use the Hahn formulas of [6, §9.5] at these negative parameters, with the following justification. For fixed N and n, the orthogonality relation (with its norm h_n), the leading coefficient and the endpoint value $Q_n(N)=(-1)^n(\beta+1)_n/(\alpha+1)_n$ are identities between rational functions of (α,β) (the weight $\binom{\alpha+x}{x}\binom{\beta+N-x}{N-x}$ is a polynomial in (α,β)); they hold on the open set $\alpha,\beta>-1$, hence identically, hence at every point at which no denominator vanishes. Here the quantities that must be nonzero for $0\le n\le s-1$ (denominators, and the factors of h_n and of the leading coefficient) are the factors of $(\alpha+1)_k$, $(\beta+1)_k$, $(-N)_k$ $(k\le n)$, $2n+\alpha+\beta+1 = 2n-2K+1$ (odd), $(n+\alpha+\beta+1)_n$ and $(n+\alpha+\beta+1)_{N+1}$ (last factor $n-K\le -1$ resp. $n-2K+s\le -2m-1$), all nonzero integers of absolute value $\le 2K-1$ (the largest is $|n+\alpha+\beta+1|$ at $n=0$). Since both weights are positive, the monic orthogonal polynomials are unique and equal the Hahn polynomials divided by their leading coefficients, which are nonzero. All remaining constants are factorials of integers $\le K+m-1\le 2K-3$ and $\Lambda_g^2=\Pi_X(s)/\Pi_Y(s)$, a ratio of integers of absolute value $<K+m$. This gives the claim. □

(Checked in exact arithmetic: the formula for $\varkappa_R^2,\varkappa_L^2$ agrees with the leading coefficient obtained by interpolation of k_R,k_L at $D+2$ points, for 20 pairs (K,m) up to $(8,6)$, including $(3,1)$: $\varkappa_R=\frac12,\varkappa_L=\frac38$; and all prime factors are $\le 2K-3$ for $K\le 18$.)

15.6. **The sum $\sum_c e_c = D$.**

Lemma 15.15. $\sum_{c=1}^{2K-1} e_c = D$.

Proof. For $c \le K$: $\sum_{c<s} c = \frac{s(s-1)}{2}$; $\sum_{c\le K} T_2(\min(c,s)) = \sum_{c\le s} c(c-1) + (K-s)s(s-1) = \frac{(s-1)s(s+1)}{3} + ms(s-1)$; $\sum_{c=m+1}^{K} T_2(c-m) = \sum_{n=1}^{s} n(n-1) = \frac{(s-1)s(s+1)}{3}$. For $c = K+d'$, $d' = 1..K-1$: $n = s$ for $d' \le m$ and $n = K-d'$ for $d' > m$, so $\sum T_2(n) = ms(s-1) + \sum_{t=1}^{s-1} t(t-1) = ms(s-1) + \frac{(s-2)(s-1)s}{3}$; $\sum T_2(o) = \sum_{o=1}^{s-1} o(o-1) = \frac{(s-2)(s-1)s}{3}$; $\sum_{d'<s} d' = \frac{s(s-1)}{2}$. The total is $s(s-1)\big[1 + 2m + \frac{2(s+1)+2(s-2)}{3}\big] = \frac{s(s-1)(4s+6m+1)}{3} = D$. □

15.7. **Proof of Theorem 15.1.** By Lemma 15.2 and Theorem 15.13, k is a polynomial of degree exactly D with leading coefficient $\varkappa$ (a rational with primes $\le 2K-1$, Proposition 15.14). By Propositions 15.8, 15.12 (applied to κ_{H}, and $k_{R,L} = \text{const}\cdot\kappa_{\mathsf{H}}$ by Proposition 15.3) $\prod_c (\theta+c)^{e_c}$ divides k, and $\sum e_c = D$ (Lemma 15.15) $= \deg k$. Hence $k = \varkappa \prod_c (\theta+c)^{e_c}$.

Lemma 15.16 (from $\mathbb{Q}$ to $\mathbb{F}_\ell$)**.** *Let $\ell = 4K-1$ be prime. Then $\hat\Gamma(0) \in M_K(\mathbb{Z}_{(\ell)})$ with $\det\hat\Gamma(0) \in \mathbb{Z}_{(\ell)}^{\times}$, $T(0) \in M_s(\mathbb{Z}_{(\ell)})$, $k_R(0), k_L(0) \in \mathbb{Z}_{(\ell)}$, and the reductions modulo ℓ are the Krylov determinants k_R, k_L computed over $\mathbb{F}_\ell$ from $\bar T = (\bar{\hat\Gamma}^{-1})_{GG}$ in §14.*

Proof. $\det\hat\Gamma(0) = \det C_{G\cup X} \det C_{G\cup Y}$, with Cauchy factors $a' - a \le K-1$, $j' - j \le K+m-1 \le 2K-3$ and $a + j \le 2K+m-1 \le 3K-3 < \ell$; all are ℓ-units. The entries $1/(a+j)$ are ℓ-integral, so $\hat\Gamma(0)$ and its inverse (adjugate over a unit) are ℓ-integral; reduction is a ring homomorphism and commutes with products, submatrices, inverses of unit-determinant matrices and determinants. □

By Lemma 15.16 and the product formula, $k_R(0) = \varkappa \prod_{c\le 2K-1} c^{e_c}$ is an ℓ-unit because $\varkappa$ and all $c \le 2K-1 < \ell$ are prime to ℓ ($2K \le \ell - 1$). This is Theorem 15.1. □

Proof of Proposition 10.4. Theorem 14.7 together with Theorem 15.1 (for $0 \le m \le K-2$) and the trivial case $m = K-1$ ($k_R = k_L = 1$), for every prime $\ell \equiv 3 \pmod 4$, $\ell \ge 7$ (for $\ell = 7$, $K = 2$ and $m \in \{0,1\}$). □

16. The case $\ell \mid \mathrm{den}$ and the block $R_{X'Y'}$

This section proves Proposition 10.5 and the statement $\det \mathsf{R}[X',Y'] \ne 0$ used in Lemma 10.1. Notation and hypotheses are those of §14: $\ell \equiv 3$ (4), $K = (\ell+1)/4$, $1 \le m \le K-1$, $s = K-m$, a geometry $\mathcal{G} = \mathcal{G}(r_1, m, e)$ with (H); in j-coordinates $J_{X'} = s + \{2,4,\dots,2m\}$, $J_{Y'} = s + \{1,3,\dots,2m-1\}$. The new ingredient is a closed-form determinant.

Lemma 16.1 (high–high block mod ℓ)**.** *For distinct high nodes x, y:* $\mathsf{R}(x,y) \equiv \frac{1}{16}\sum_{a<K} \frac{1}{(a+j_x)(a+j_y)} = \frac{1}{16} W_{j_x}^{\top} W_{j_y} \pmod\ell$. *Hence* $\mathsf{R}[X',Y'] \equiv \frac{1}{16} W_{X'}^{\top} W_{Y'}$.

Proof. Let $K \le a < b \le \ell - 1$. For $k \in (a, b]$, $K < k \le \ell - 1$ and $4k-1 \equiv 4(k-K)$ (as $4K = \ell+1$), so $\frac{1}{16k(4k-1)} \equiv \frac{1}{64k(k-K)} = \frac{1}{64K}\big(\frac{1}{k-K} - \frac{1}{k}\big)$, all of $k, k-K, b-a$ lying in $[1, \ell-1]$. Summing, $64K(b-a)\mathsf{R}(a,b) \equiv (H_{b-K} - H_b) - (H_{a-K} - H_a)$. By $H_n \equiv H_{\ell-1-n}$ ($1/(\ell-i) \equiv -1/i$, $H_{\ell-1} \equiv 0$) with $j = \ell - b$: $H_b \equiv H_{j-1}$, $H_{b-K} \equiv H_{j+K-1}$, so $H_{b-K} - H_b \equiv F(j_b) := \sum_{a'<K} \frac{1}{a'+j_b}$. As $b-a = j_a - j_b$, $\mathsf{R}(a,b) \equiv \frac{F(j_b)-F(j_a)}{64K(j_a-j_b)} = \frac{1}{64K}\sum_{a'} \frac{1}{(a'+j_a)(a'+j_b)}$ and $64K = 16(\ell+1) \equiv 16$. □

For θ in a field of characteristic 0 (or an integer with all entries invertible mod ℓ) put $X_\theta = \theta + \{2, 4, \dots, 2m\}$, $Y_\theta = \theta + \{1, 3, \dots, 2m-1\}$ and $\Delta_{K,m}(\theta) = \det(C_{X_\theta}^\top C_{Y_\theta}) = \det\big[\sum_{a<K} \frac{1}{(a+x)(a+y)}\big]_{x\in X_\theta, y\in Y_\theta}$. Let $V(Z) = \prod_{z<z'}(z'-z)$, $\mathrm{sf}(n) = \prod_{k=1}^n k!$ and $\mathsf{H}_{K,m} = \det\big[\binom{K}{p+q+1}\big]_{0\le p,q<m}$.

Proposition 16.2 (the block determinant)**.**

$$\Delta_{K,m}(\theta) = (-1)^{m(m-1)/2}\, V(2,4,\dots,2m)\, V(1,3,\dots,2m-1) \times \mathrm{sf}(m-1)^2\, \mathsf{H}_{K,m} \Big/ \prod_{i=1}^{m} (\theta+i)_m (\theta+K+i)_m,$$

and $(-1)^{m(m-1)/2}\mathrm{sf}(m-1)^2\mathsf{H}_{K,m} = \sum_{S\subset[0,K),|S|=m} V(S)^2 = \prod_{n=0}^{m-1} h_n$, $h_n = \frac{(n!)^4}{(2n)!(2n+1)!}\prod_{i=-n}^{n}(K+i)$.

Proof. (1) Cauchy–Binet. $\Delta = \sum_S \det C_X[S] \det C_Y[S]$ over m-subsets $S \subset [0, K)$. By Cauchy's formula $\det C_Z[S] = V(S)V(Z)/\prod_{a\in S, z\in Z}(a+z)$, and $\prod_{x\in X}(a+x)\prod_{y\in Y}(a+y) = (a+\theta+1)_{2m}$. So $\Delta = V(X)V(Y)\Phi$, $\Phi = \sum_S V(S)^2 \prod_{a\in S}\varrho(a)$, $\varrho(a) = 1/(a+\theta+1)_{2m}$. *(2) Hankel form.* By Cauchy–Binet again $\Phi = \det[\langle a^p, a^q\rangle]_{p,q<m}$ with $\langle f, g\rangle = \sum_{a=0}^{K-1} fg\varrho$, and a unitriangular change of the monic bases gives $\Phi = \det[\langle \phi_p, \psi_q\rangle]$, $\phi_p(a) = (a+\theta+1)_p$, $\psi_q(a) = (a+\theta+2m-q+1)_q$. *(3) Telescoping.* $\phi_p\psi_q\varrho = 1/(a+\theta+p+1)_L$, $L = 2m-p-q \ge 2$, and $1/(c)_L = \frac{1}{L-1}\big[\frac{1}{(c)_{L-1}} - \frac{1}{(c+1)_{L-1}}\big]$ telescopes the sum over a. With $u_p = \theta+p+1$, $v_q = \theta+2m-q$, $f(z) = (z)_K$: $\langle\phi_p, \psi_q\rangle = \frac{\Gamma(u_p)}{\Gamma(v_q+K)}\frac{f(v_q)-f(u_p)}{v_q-u_p}$. So $\Phi = \prod_p \Gamma(u_p)/\prod_q \Gamma(v_q+K)\cdot\det[f[u_p, v_q]]$ (divided differences). *(4) Differences.* Reverse the order of rows and of columns (two signs cancel): rows $u = \theta+m-p$, columns $v = \theta+m+1+q$. Replace row p by $(-1)^p\nabla_u^p$ and column q by Δ_v^q (unitriangular, leading coefficient 1). For unit spacing $\nabla^p g(x) = p!\, g[x-p, \dots, x]$, $\Delta^q h(y) = q!\, h[y, \dots, y+q]$, and iterated divided differences compose: the new (p,q) entry is $(-1)^p p!q!\, f[\theta+m-p, \dots, \theta+m+1+q] = (-1)^p p!q!\, \Delta^n f(\theta+m-p)/n!$, $n = p+q+1$. Since $\Delta(z)_k = k(z+1)_{k-1}$, $\Delta^n(z)_K = \frac{K!}{(K-n)!}(z+n)_{K-n}$, so the entry is $(-1)^p p!q!\binom{K}{n}\frac{\Gamma(\theta+m+K-p)}{\Gamma(\theta+m+q+1)}$ (also correct for $n > K$, both sides vanishing). Taking out the row and column factors, $\det[f[u,v]] = (-1)^{m(m-1)/2}\mathrm{sf}(m-1)^2\mathsf{H}_{K,m}\prod_p\Gamma(\theta+m+K-p)/\prod_q\Gamma(\theta+m+q+1)$. *(5) Collecting* the Gamma factors: $\prod_p\Gamma(\theta+p+1)/\prod_q\Gamma(\theta+m+q+1) =$

$\prod_{i=1}^{m} 1/(\theta+i)_m$ and $\prod_p \Gamma(\theta+m+K-p)/\prod_q \Gamma(\theta+2m-q+K) = \prod_{i=1}^{m} 1/(\theta+K+i)_m$. This is the identity. For the constant, multiply by θ^{2m^2} and let $\theta \to \infty$: each Cauchy–Binet term is $V(X)V(Y)V(S)^2\theta^{-2m^2}(1+O(1/\theta))$, giving $\sum_S V(S)^2 = \det[\sum_a a^{p+q}]$ (Hankel determinant of the moments of the uniform weight on $\{0..K-1\}$), which equals $\prod_n h_n$ with h_n the squared norm of the monic orthogonal (discrete Chebyshev, Hahn $\alpha=\beta=0$) polynomial of degree n [6, §9.5]. □

(Checked exactly for $K \le 12$, $1 \le m \le K+1$ and several non-integer θ (540 cases); the factorisation of $\Delta(\theta)$ over $\mathbb{Q}[\theta]$ was also computed independently.)

Corollary 16.3. *Let $\theta = s = K-m$, $1 \le m \le K-1$. Then $\Delta_{K,m}(s)$, computed in $\mathbb{F}_\ell$ from the Cauchy entries, is nonzero; hence $\det(W_{X'}^\top W_{Y'}) = \pm\Delta_{K,m}(s) \in \mathbb{F}_\ell^\times$ and $\det \mathsf{R}[X',Y'] \equiv 16^{-m}\det(W_{X'}^\top W_{Y'}) \not\equiv 0$. In particular $\det \mathsf{R}[X',Y'] \neq 0$, $v_\ell(\det \mathsf{R}[X',Y']) = 0$ and $\mathsf{R}[X',Y']^{-1} \in M_m(\mathbb{Z}_{(\ell)})$.*

Proof. The identity of Proposition 16.2 holds in $\mathbb{Z}_{(\ell)}$ at the integer $\theta = s$ (all entries $\frac{1}{(a+x)(a+y)}$ have $a+x \le K-1+s+2m = 2K+m-1 < \ell$ and are ℓ-integral). The numerator is, by the second part, a product of integers $2(i'-i) < 2m$, $k!$ with $k \le 2m-1$, and $K+i$ with $|i| \le m-1$, all in $[1, \ell-1]$ (as $m \le K < \frac{\ell+1}{2}$), divided by factorials $(2n)!(2n+1)!$ with $2n+1 \le 2m-1 < \ell$; the denominator is a product of integers in $[s+1, s+K+2m-1] \subseteq [1, 3K-2]$, $3K-2 < \ell$. So Δ is an ℓ-unit. Finally use Lemma 16.1 (no diagonal entries occur in $X' \times Y'$). □

(Direct exact computation of $\det(C_X^\top C_Y)$ modulo ℓ for every prime $\ell \equiv 3\ (4)$, $\ell < 200$, all m and $s \in \{K-m-1, K-m, K-m+1\}$, and of the factor ranges for $\ell < 3000$ (223,320 cases) gave no failure.)

Proof of Proposition 10.5. Let $\mathcal{G} = \mathcal{G}(r_1, m, e)$, $n_C = |C|$, $x_0 = u'/\ell^{e_0}$ with $u' = \ell^{e_0}x_0 \in \mathbb{Z}_{(\ell)}^\times$ ($e_0 \neq 2$). By Lemma 10.1 and Corollary 16.3 it suffices to show $P(x_0) \neq 0$ for the monic polynomial $P(X) = \det(XI + M)$, $M = \mathsf{R}[C,C] - \mathsf{R}[C,Y']\mathsf{R}[X',Y']^{-1}\mathsf{R}[X',C]$. By Lemma 14.1, $v_\ell(\mathsf{R}[C,C]) \ge -2$ and $v_\ell(\mathsf{R}[C,Y']), v_\ell(\mathsf{R}[X',C]) \ge -1$ (no diagonal entries occur, since C, X', Y' are disjoint; low–high entries have valuation -1, the rest ≥ 0). By Corollary 16.3 $\mathsf{R}[X',Y']^{-1}$ is ℓ-integral, so $\ell^2 M \in M(\mathbb{Z}_{(\ell)})$.

$e_0 \ge 3$. $\ell^{e_0 n_C} P(u'\ell^{-e_0}) = \det(u'I + \ell^{e_0-2}\cdot \ell^2 M) \equiv u'^{n_C} \not\equiv 0$.

$e_0 = 1$ *(asymmetric scaling).* Let $N = \operatorname{diag}(\ell I_{C\sqcup X'})\mathrm{Full}(u'/\ell)\operatorname{diag}(I_L, \ell I_H, I_{Y'})$ (L low, $H = C_{\text{high}}$): scale all rows by ℓ and the H columns by ℓ, leaving the columns of L and Y' unscaled. By Lemmas 14.1 and 16.1, $N \in M(\mathbb{Z}_{(\ell)})$ and modulo ℓ (rows L, H, X'; columns L, H, Y')

$$N \equiv \begin{pmatrix} u'I & 0 & -\frac{1}{4}W_{Y'} \\ * & -I & 0 \\ -\frac{1}{4}W_{X'}^\top & 0 & 0 \end{pmatrix}$$

(the entries $\ell\mathsf{R}[H, Y']$ and $\ell\mathsf{R}[X', Y']$ are $\equiv 0$ because off-diagonal high–high entries are integral; $\ell^2\mathsf{R}[L, H] \equiv \ell^2\mathsf{R}[X', H] \equiv 0$). The H columns vanish outside the H rows; expanding, $\det N \equiv \pm\det\begin{pmatrix} u'I_K & -\frac{1}{4}W_{Y'} \\ -\frac{1}{4}W_{X'}^{\top} & 0 \end{pmatrix} = \pm 16^{-m} u'^{K-m} \det(W_{X'}^{\top} W_{Y'}) \neq 0$ (Schur complement of $u'I_K$; Corollary 16.3). Since $\det N = \ell^{(n_C+m)+\#H} \det \mathrm{Full}(u'/\ell)$, $v_\ell(\det \mathrm{Full}(x_0)) = -(2n_C - K + m)$. □

Remark 16.4 (Newton polygon; not used). Write $P(X) = \sum a_k X^k$ for a member, $k_0 = K - m$, $v = 2(r - K + m)$ (with $r = |C|$) and $b_k = v_\ell(a_k) + v$. Then $b_k \geq 2(k - k_0)$ for all k (the coefficient a_k is $\pm$ a sum of principal $(r-k)$-minors of M and $\ell^2 M$ is integral), $b_k \geq 0$ for all k with $b_{k_0} = 0$ (this is Proposition 14.3: $\ell^{2(\#C_{\mathrm{high}}+m)} \det \mathrm{Full}(X) \in \mathbb{Z}_{(\ell)}[X]$, whose reduction $\pm 16^{-K} B(H_{K-1}^{(2)} - 16X)$ has exact degree $K-m$ because its leading coefficient is $\pm\det(W_{X'}^{\top} W_{Y'}) \neq 0$; this is the place where the bound $b_k \geq 0$ for $k < k_0$ must be taken from Proposition 14.3, not from the $e_0 = 1$ argument), and $b_r = 2(r - k_0)$. Hence the Newton polygon of P has the slope-2 segment from $(k_0, 0)$ to $(r, 2(r-k_0))$ on $[k_0, r]$, which is another way to see Proposition 10.5: the minimum of $b_k - e_0 k$ is unique at k_0 for $e_0 = 1$ and at r for $e_0 \geq 3$, and all points tie for $e_0 = 2$.

Remark 16.5. For $e_0 = 2$ the same computation mixes $u'I$ with the block $-I$ of H and an exceptional residue of u' can occur (as in Lemma 5.5); it is avoided by the choice of ℓ with $v_\ell(q) \neq 2$. The scaling of Lemma 5.5 cannot be used for $m \geq 1$: scaling the Y' columns by ℓ kills the entries $\ell\mathsf{R}[L, Y']$ which carry the rank.

17. Proof of Theorem 1.3, status, and reproducibility

17.1. **The proof.**

Proof of Theorem 1.3. Propositions 10.2 (§12), 10.3 (§13), 10.4 (§14, §15) and 10.5 (§16) give the inputs of Theorem 11.1. With the certified numbers

$$D_F \geq 3.90626950661, \qquad c_{\mathrm{hyb}} \leq 7.65406474415,$$

$$\Lambda_F = \tfrac{313}{105} + \tfrac{554}{840} + \tfrac{118/21 + 2\log 2}{64} - \tfrac{129}{4480} = 3.72114001606 \ (\leq 3.7211401),$$

we get $\eta_{\mathrm{hyb}} = D_F - \Lambda_F \geq 3.9062695 - 3.7211401 = 0.1851294$ (indeed 0.18512949) and, rounding every constant in the direction that increases the bound,

$$c_{\mathrm{hyb}}\Big(\frac{1}{\eta_{\mathrm{hyb}}} + \frac{2}{3}\Big) \leq 7.6540648\Big(\frac{1}{0.1851294} + \frac{2}{3}\Big) = 7.6540648 \times 6.06829\ldots = 46.4471\ldots < 46.45.$$

(With the unrounded values the exponent is 46.44709.) By Theorem 11.1, every $\nu > 46.4471$ admits only finitely many p/q with $|\lambda - p/q| < q^{-\nu}$; in particular $\nu = 46.45$ does. □

Note that the rounding is upward: the stated bound 46.45 is larger than the computed 46.4471. The sensitivity is $\partial\mu/\partial D_F = -223$ and $\partial\mu/\partial c_{\mathrm{hyb}} = 6.07$.

Remark 17.1 (flexibility in the hybrid frame; a conjecture, not a result). In the square frame the freedom to choose $h = H$ first and ℓ afterwards removes the ℓ-interval term and gives $\kappa = 0$ (Theorem 8.1). The hybrid theorem keeps $\kappa = 2/3$. The reason is that the geometry pair $G(r_1, m, e)$, $e = 0, 1$, of Theorem 14.7 has absent set of size $s = K - m$, so $|G_1| + m = K$ exactly, and this one-parameter family is already used up by the admissibility condition $|m - r/8| \le 1$ of the certificates (decay, window and content are certified only at $\alpha = 1/8$). If the hybrid could be used with an arbitrary target $r \approx kL$ and an ℓ chosen afterwards, the exponent would become $c_{\mathrm{hyb}}/\eta_{\mathrm{hyb}} = 7.6540648/0.1851294 = 41.3444$, i.e. $\mu(\lambda) \le 41.35$ (unrounded 41.34438). That requires absent sets $S > K - m$, i.e. $r + m < \ell - K$. There $B(0)$ is not a product of Cauchy determinants, the matrix $\hat{\Gamma}_1 = C^{\top}_{G\cup X'} C_{G\cup Y'}$ is $(S+m)\times(S+m)$ of rank at most $K < S+m$ and hence singular, so k_R, k_L and Lemma 14.6 are undefined, and the product formula of Theorem 15.1 does not apply. An independent audit gave a reduction (matrix determinant lemma applied to the bordered matrix $M_e(z) = \left[\begin{smallmatrix} zI+N_e & W_Y \\ W_X^{\top} & 0 \end{smallmatrix}\right]$ with $N_e = W_{G_e} W_{G_e}^{\top}$, $G_1 = \{1, \dots, S\}$, $G_2 = \{1, \dots, S-1\}$, valid also at $z = 0$) of the nonvanishing of the pair to the nonvanishing modulo ℓ of two generalized bordered Krylov determinants $k_R^{\mathrm{gen}}(K, m, S)$, $k_L^{\mathrm{gen}}(K, m, S)$. We state the remaining input as a conjecture; it is *not* proved here, and nothing in this paper depends on it.

Conjecture. For all K, $0 \le m \le K - 1$ and $S \ge K - m$, the rationals $k_R^{\mathrm{gen}}(K, m, S)$ and $k_L^{\mathrm{gen}}(K, m, S)$ are nonzero and all their prime factors are at most $S + K + m - 1$ (which is $< \ell$ whenever $r \ge K$).
Evidence: exact computation for all $K \le 11$, all $m \in [0, K-1]$ and all $S \in [K - m, K - m + 22]$ (1494 cases, both k_R^{gen} and k_L^{gen}) found no zero and always the stated bound on the prime factors; for $m = 0$ it is a basis constant times $u(K, S-1)$ of Theorem A. If the conjecture holds, the selection of the geometry (target $r = \lceil kL \rceil$, $m = \mathrm{round}(r/8)$, usable primes $\ell \equiv 3 \pmod 4$ with $\ell \in [\frac{3r}{2}, 4r]$ roughly, $\ell \nmid q$ by a prime-product count) would give $\mu(\lambda) \le 41.35$ in the same way. Until it is proved, the best bound for λ is 46.45.

Remark 17.2 (a second certificate). The certificate measure optimised for $D - 0.04c$ (file `sig_h500w.json`) gives $D_F \ge 3.9045139$, $c \le 7.5928685$ and the worse exponent 46.468; the optimal weight on c is about 0.027, so a re-optimisation could gain only marginally.

17.2. **What was proved by computer, and what was checked numerically.**

- **Computer-assisted (interval arithmetic).** The numbers m', S_{cert}, $g_{\min}$, $I_\rho(\sigma)$, $I(\nu_{1/8})$ in §13.2 (script `hcert.py`, Arb); everything else in the deduction is pen-and-paper, with explicit constants where stated.
- **Exact checks (rational arithmetic), not used in the proofs.** Reduction mod ℓ (Proposition 14.3) including the constant, at every residue of X for $\ell = 11, \dots, 43$ and several geometries; the product formula of Theorem 15.1 (exact polynomial interpolation at $D+1$ points for all m at $K = 11, \dots, 14$, up to $D = 3458$; $K \le 10$ via the closed form of the exponents); $k_R(0), k_L(0) \bmod \ell$ for all $\ell \equiv 3$ (4), $11 \le \ell < 800$ and every m, and $\ell < 3000$ in the α-window; end-to-end: $\det \mathrm{Full}(x_0) \neq 0$ for one of the two members for every residue class of x_0 and random rationals, $\ell = 7, 11, 19, 43, 59, 83, 107, 131$, no failure; Lemma E and the content: $v_p(\mathrm{den}_F) = \tau_p - m$ exactly at every clean prime for $r = 24, 48, 56, 72$ (formal X), and for the member with the extra node ($e = 1$) at $r = 48, 64, 80, 96$ (the script `member2_content.py`); independently, $v_p(\mathrm{den}_F)$ equals the bound of Theorem 12.6 in all 27 cases (r, m, e, p) of Remark 12.8 (`check_hE_member2.py`); the bound (12.4) with $C_0 = 0$ for all odd q at 21 pairs (r, m) (`check_hB.py`); Proposition 10.5 in 3086 valuation checks.
- **Cross-checks of c_{hyb} and D_F.** Exact values of $-r^{-2} \log |\det \mathcal{T}_t|$ extrapolate to $D \approx 3.914$ (the certificate loses 0.008), see Remark in §13.2.

17.3. **Status of the individual steps, and limitations.** We list the points at which the written proof rests on something other than the audited arguments of §18.

S1. *Constants of $\eta(r)$.* The $o(r^2)$ term in Theorem 13.4 is explicit, (13.4): $\eta(r) \le (15 \log r + K_0)/r$ with $K_0 \le 25{,}100$, for $r \ge 2^{30}$. The Lipschitz bounds in α (Lemma 13.3), the treatment of the coefficient discrepancy e/r (the function $G^{(m)}$), the constant C_E and the extra node b are proved in §13; the dominant constant $4 \sum_\ell (m' - \gamma_\ell)^+$ is read off the certificate data. These constants were re-derived by hand after the audits; they affect only the $o(1)$.

S2. *Member 2 (the extra common node).* The content saving for the geometry with the extra common node b is proved in Theorem 12.6 (case $e = 1$); this proof is new in this text. It was re-derived line by line by a further independent check (Remark 12.8, which also explains why the extra node changes the content by only $O(r)$), and it is confirmed by two independent exact computations, in which the bound of Theorem 12.6 is attained with equality. For the decay the extra node is handled by the perturbation argument of §13.

S3. *Per-prime bound* (12.4). Proved by hand, with $C_0 = 0$; this agrees with the exact evaluation.

S4. *Classical external inputs not proved here:* the norms h_n of the discrete Chebyshev polynomials and the Hahn polynomial formulas at negative

parameters, cited from [6, §9.5]; the latter are used through the rational-identity (continuation) argument of §15.5: for fixed N, n they are identities between rational functions of (α, β), valid on $\alpha, \beta > -1$ and hence at every point without vanishing denominators, which is the case for our parameters. Also Rellich's theorem and first-order degenerate perturbation theory (Kato), and the two-sided prime number theorem in progressions.

S5. *The zero orders for $K < c \le 2K - 1$.* The gap condition (G′) of §15.4 is not assumed: the deformation of the free ν-nodes removes it. The deformation argument was audited (with a correction of the denominator Δ by the factors $\prod(y_i - o)$, included above); in addition the counts of Rellich orders were verified by computer for $K \le 300$, and the formulas (not the proof) of Proposition 15.14 were verified for $K \le 18$ and in 20 exact cases.

S6. *The certificate.* The copy of `hcert.py` in `scripts-hybrid/` has been re-run and reproduces $D_F \ge 3.906269506610271$ and $c_{\mathrm{hyb}} \le 7.654064744147046$ exactly (§13.2).

Limitations. The following is not done in this paper: the hybrid for the mirror constant $\lambda' = G/2 + 5\pi^2/96$ (primes $\ell \equiv 1 \pmod 4$), for which Theorem 1.4 (52.69) remains the best bound; values $\alpha \ne \frac{1}{8}$ (the saving formula holds only for $\frac{1}{8} \le \alpha \le \frac{1}{6}$, the tropical law as stated only for $\alpha \le \frac{1}{8}$); the savings from non-clean primes and from primes $p < \frac{5}{8}r$, which probably exist but are neither proved nor used; and a Lean formalization of any part of Part 1.

17.4. **Reproducing the hybrid computations.** The directory `scripts-hybrid/` (in the repository next to `paper-lambda/scripts/`) contains: `hcert.py` (the Arb certificate of §13.2); `sig_h500.json` (the certificate measure), `sig_h500w.json` (second measure), `sigma300.json` (negative control); `expected_h500.log`, `expected_ctl0.log` (expected output); `member2_content.py` (exact content check for the extra-node member), `check_hE_member2.py` (independent exact check of Theorem 12.6 for both members), `check_hB.py` (the bound (12.4)); and a `README` with the commands of §13.4. The value $D_F \ge 3.906269506610$, $c \le 7.654064744147$ appears in the last lines of `expected_h500.log`. The directory also contains the exact checks written for this part: `check_hE.py` (Theorem 12.6 with formal X for admissible (r, m, e), $e \in \{0, 1\}$: 27 primes, 0 failures, several with equality), `check_k.py` (the product formula of Theorem 15.1 with the exponents (15.1), exact, for seven pairs (K, m)), `sum_ec.py` ($\sum e_c = D$ for $K \le 200$ and agreement of (15.1) with the min-formula of our earlier derivation for $K < 80$) and `bsum.py` (the sum of the per-prime bounds of Proposition 12.2: $3.6497, 3.7227, 3.7418, 3.7458$ at $r = 10^3, 10^4, 10^5, 4 \cdot 10^5$, converging to $\Lambda_{\mathrm{trop}} = 3.7499$).

18. Formalization status

We state this carefully. **The hybrid theorem (Theorem 1.3, $\mu(\lambda) \le 46.45$) and everything in Part 1 are NOT formalized in Lean.** The Lean results cover only the square-frame results of §2–§9 with the weak nonvanishing route, namely the weaker exponents 77.28 (λ), 78 (λ') and 79 ($G + 5\pi^2/48$) below. The improvement of the square-frame bound from 57.85 to 52.69 (the $\kappa = 0$ selection of §8) does not change the Lean bound: Lean uses Theorem 5.4, with three values of h spread by a factor 4/3, whose ℓ-selection is the old one, and not the free choice of h of Theorem 6.2. The companion paper [7] on Catalanś constant is fully formalized in Lean 4, and so are the irrationality of λ and the bound $\mu(\lambda) \le 77.28$ proved here; Theorem A, Lemma CC and the bound 52.69 are not. Each of the Lean results below compiles without `sorry` and depends only on the axioms `propext`, `Classical.choice` and `Quot.sound`. The library uses the Lean module system; a full build of the λ library takes about 14 minutes, and the kernel-checked numerical certificate about one CPU-hour.

- **Formalized (Lean 4 and Mathlib): nonvanishing.** The ℓ-adic nonvanishing theorem, Theorem 5.4: for a prime $\ell \equiv 3 \pmod 4$, $\ell \ge 7$, and $r \in \mathbb{Q}$ with $\ell \nmid \operatorname{den} r$, one of $P_{\ell-1}(r), P_{\ell-2}(r), P_{(3\ell-1)/4}(r)$ is nonzero. The theorem is `CatalanLambda.lambda_nonvanishing`. The formalized proof is organized slightly differently from §5: it works in the subring $\mathbb{Z}_{(\ell)}$ with the reduction map to $\mathbb{Z}/\ell$, uses the telescoping $\frac{1}{k} - \frac{1}{k+\ell-K} = -\frac{1}{k(4k-1)}$ directly, and absorbs the constant $-H_m^{(2)}$ into a single constant. It does not formalize the valuation statement $v_\ell = -2(h-K)$.
- **Formalized: the prime number theorem in arithmetic progressions.** The theorem for every modulus, used in Theorems 3.1 and 8.1, as `CatalanLambda.pnt_AP`, together with the upper bound `CatalanLambda.theta_mod4_upper`. It is derived from the vendored Wiener–Ikehara theorem and Mathlib's results on Dirichlet L-functions.
- **Formalized: the content bound.** The integrality estimate (Lemma 2.4) as `CatalanLambda.integrality`, and the denominator bound of Theorem 3.1 in the form `CatalanLambda.log_Dden_le_eventually`.
- **Formalized: the Löwner form.** The off-diagonal and diagonal formulas for the matrices T_h as Löwner matrices (Proposition 2.2), as `CatalanLambda.Decay.lowner_offdiag` and `CatalanLambda.Decay.lowner_diag`, and the identity $\lambda = G/2 - 5\pi^2/96$ as `CatalanLambda.Decay.lamS_eq`.
- **Formalized: the contour identity.** Proposition 4.1 as `CatalanLambda.Decay.contour_identity`
- **Formalized: Andréief and Cauchy.** The Andréief-type determinant bound (Proposition 4.2) as `CatalanLambda.Decay.tent_andreief_bound`, and the Cauchy determinant as `CatalanLambda.Decay.cauchy_det`.
- **Formalized: the kernel k_ρ, Theorem 7.1 and the deduction.** The positive-definiteness lemma for the kernel k_ρ (Lemmas 4.4, 4.5, 4.6 and Corollary 4.7); the contour identity for a

general ray angle (including the direction $(-1, 3)$ used here); Theorem 7.1 (node removal N_ρ), as `CatalanLambda.Decay.theoremN`, together with `CatalanLambda.Decay.window_liminf_P`; Lemma 5.5 as `CatalanLambda.Measure.lemma3_pq`; and the deduction of the irrationality and of the bound of Remark 8.3 from the certificate facts, as `CatalanLambda.Measure.lambda_irrational` and `CatalanLambda.Measure.lambda_measure_one`.

- **Formalized: the numerical certificate.** The structure `CatalanLambda.Cert.certFacts` collecting the numerical facts of §4.4 and §7.4 (the certified values of D_{cert}, m', S_{cert}, $g_{\min}$) is inhabited by a kernel-checked proof: about 27.6k ray boxes, 2.8k segment cells and 512 line cells, the bound on I_ρ and the tail estimate. The proof uses a far-field first-order Taylor model with verified rational tables for the logarithm and the arctangent, and takes about one CPU-hour in the kernel. Lean works with the short rational constants $D_{\mathrm{cert}} = 31111/10000$ and $c_{\mathrm{cert}} = 68726/10000$ obtained by rounding the certified values, which is why it proves 77.28 rather than 77.13.
- **Formalized: the final theorems.** `CatalanLambda.lambda_irrational_final` (λ is irrational), `CatalanLambda.catalan_sub_irrational` ($G - 5\pi^2/48$ is irrational), `CatalanLambda.lambda_measure_final` ($\mu(\lambda) \le 31682445459/409965000 \approx 77.28$) and `CatalanLambda.lambda_measure_78_catalan` (the round exponent 78), all unconditional. The statements are in `ChallengeLambda.lean`, which needs only Mathlib and writes G as the series $\sum_{j\ge 0}(-1)^j/(2j+1)^2$; the proofs are in `SolutionLambda.lean`.
- **Formalized: the mirror theorem (weaker exponent).** The irrationality of $\lambda' = G/2 + 5\pi^2/96$ and of $G + 5\pi^2/48$, and the measure with the same exponent $31682445459/409965000 \approx 77.28$ (hence 78; and 79 for $G + 5\pi^2/48$) are formalized in `CatalanLambda/Mirror/`, with the three-h nonvanishing for $\ell \equiv 1 \pmod 4$ (no Theorem A), the mirror contour identity, and the *same* kernel-checked certificate `CatalanLambda.Cert.certFacts`. The final statements are `CatalanLambda.Mirror.lambda_mirror_measure_78_catalan`, `CatalanLambda.Mirror.catalan_` and `CatalanLambda.Mirror.catalan_add_measure_79`, with challenge statements `CatalanLambdaChallenge.lambdaMirror_irrationalityMeasure_78` and `CatalanLambdaChallenge.catalan_add_irrationalityMeasure_79`; axioms `propext`, `Classical.choice`, `Quot.sound`. The bound 52.69 for λ' (via Theorem A) is not formalized.
- **Not formalized.** Theorem A, Lemma CC and Theorem 6.2 (§6), the deduction of the bound 52.69 in Theorem 8.1, and *all of Part 1* (the hybrid theorem $\mu(\lambda) \le 46.45$ with its tropical content bound, Lemma E, the two-family Andréief estimates, the bordered Krylov determinants and the $\ell \mid$ den analogue; the Arb certificate `hcert.py` is not re-proved in Lean either). The Lean statement is the weaker bound 77.28, obtained with the simpler choice of h (Remark 8.3) and the rounded constants.

- **Computer-assisted.** Theorem 4.10 and Theorem 7.1 were established by interval-arithmetic computations in Arb (§4.4 and §7.4), carried out by the Python script `kcert.py` on the certificate measure `sigma300.json` (produced from a float optimum by the dyadic rounding script `roundsig.py`; `mkdata.py` re-derives the data in exact rationals; `scan.py` is the supplementary content scan). The script was re-run by an independent audit agent, which reproduced the numbers of §4.4 and sampled the certified cell bounds against floating-point values at about 5×10^5 ray points, 4×10^4 segment points and 10^4 points of the line; but this sampling is not a proof; the certificate facts themselves are re-proved in Lean (the item on the numerical certificate above), by a separate kernel-checked computation. The float construction of σ need not be correct since the certificate is checked as it stands. The scripts are archived in the directory `paper-lambda/scripts/` of https://github.com/slitsey/catalan-irrationality-measure.

The Lean sources are in the directory `CatalanLambda/` of the repository https://github.com/slitsey/catalan-irrationality-measure, with the files `ChallengeLambda.lean` and `SolutionLambda.lean` at its top level.

All other claims were audited by independent Claude (Anthropic) agents who re-derived the statements by hand and re-ran the numerical checks: the content bound; Lemmas 5.2, 5.3 and Theorem 5.4 (exact characteristic polynomials at $h = 8, 12, 16, 20$ and all admissible ℓ; gcd $= 1$ for $\ell = 7, \dots, 383$; no violation of the tropical bound against the exact $v_p(\operatorname{den} P_h)$ at $h = 32, 64$); Lemmas 4.4, 4.5, 4.6 and Corollary 4.7 (by hand, with a random numerical test of the inequality in 2000 trials); the contour identity for the direction $(-1, 3)$ (by hand and to 10^{-17} numerically); Lemma 4.9 and Theorem 7.1, checked line by line against the circle-smoothing version; the certificate re-run (reproducing $D_{\mathrm{cert}} \geq 3.11137905$, $c \leq 6.87179932$); Lemma 5.5 (1044 exact cases); and the simpler deduction of Remark 8.3 ($4/(3\eta) + 1 = 11.2228582$, $77.12123 < 77.13$). The new material of §6 and Theorem 8.1 was audited by a further independent Claude agent (full pass): Theorem A (including the sign) was checked in exact rational arithmetic at 90 random rational pairs (θ, s), including non-integer and negative s, for $n \leq 10$, and against the direct Cauchy Gram matrices ($\theta = 0$, $s \leq 2n + 1$, $n \leq 10$); $u(K, K)$ modulo ℓ is nonzero and equals the closed form for all 86 primes $\ell \equiv 3 \pmod 4$ with $7 \leq \ell < 1000$; the resultant identity (R) was checked exactly; the characteristic polynomials of N_K, N_{K+1} are coprime for 39 primes $\ell < 400$; Lemma 5.3 was checked for $\ell = 7, 11, 19, 23$ and $h = \ell - K, \ell - K - 1$; and the numbers $6.8717994\,(1/0.1304266 + 0.75) = 57.84095$, 62.9948 (without Lemma 5.5) and 77.1213 were recomputed. The formula of Theorem A was also verified exactly for every $n \leq 12$. Exact computation of $u(K, K)$ modulo ℓ for all $\ell \equiv 3 \pmod 4$, $7 \leq \ell < 4500$, gave no zero. The deduction of Theorem 8.1 is written out in §8 from the audited arithmetic. The audit

listed only cosmetic issues, which are corrected in this text (notably the factor $4/h$ in η).

The mirror section §9 was audited by a separate Claude agent (full pass, no fatal issues; four cosmetic points, corrected here). Lemmas 9.1–9.3 were re-derived by hand (Schur complement with high block $-I$); N_0 was checked against $-(\lambda' I + \bar{T}_h)$ to 10^{-51} at $h = 9$; the reflection $\bar{F} = \tilde{F} + \bar{E}$ was checked by hand and to 10^{-22}, and the contour identity independently to $2 \cdot 10^{-17}$, with the atom set compared to the poles right of x^*; the valuation $2(h-K)$ and degree K were checked for all 131 admissible (ℓ, h) with $\ell = 13, 17, 29, 37$; the characteristic polynomials of N_K, N_{K+1} are coprime over $\mathbb{F}_\ell$ for all 58 primes $\ell \equiv 1 \pmod 4$, $13 \le \ell < 700$; Lemma 9.5 deviates only for $e = 2$, $u \equiv 1$; $v_p(\operatorname{den} \bar{P}_h) \le \sum_e \delta(p^e)$ for all $p \ge 5$ at $h = 16, 32, 64, 128$; and $6.8717994\,(1/0.13042662 + 0.75) = 57.84094$, 77.12126 were recomputed. This is not a substitute for human refereeing.

The $\kappa = 0$ improvement of version 8 (the general-s form of Theorem 6.2, the selection of ℓ in §8, and the mirror analogue) was audited by a further independent Claude agent, who read the proofs line by line and re-ran exact computations with its own code. The only uses of s in the proof of Lemma CC are the ranges of the bases of Theorem A at $\theta = 0$ (all integers in $[1, K+s]$, hence ℓ-units) and the ℓ-integrality of the entries of N_s, w_{s+1}; nothing else depends on s. The audit corrected the range of the pair to $s \in [K, \ell - K - 2]$ and dropped the step "$D_K(0) \ne 0 \Rightarrow y \ne 0$", replaced by Lemma CC at $z = 0$. Checks: $u(K, s) \ne 0$ modulo ℓ for all 640 pairs (ℓ, s) with $\ell \equiv 3 \pmod 4$, $7 \le \ell < 160$, $s \in [K, \ell - K - 1]$, with Theorem A at $\theta = 0$ reduced modulo ℓ equal to the Krylov determinant (up to sign) in each case; Theorem A at $\theta = 0$ exact over $\mathbb{Q}$ (sign $(-1)^{\lfloor n/2 \rfloor}$) for $n \le 9$ and $s \in [n, n+25]$ (234/234); gcd $= 1$ for the 622 pairs of characteristic polynomials of N_s, N_{s+1}; an end-to-end test from the definition of M_0 (no lemma used) for every $\ell \equiv 3 \pmod 4$, $7 \le \ell \le 151$ and every consecutive pair $(H, H+1)$, $K+1 \le H \le \ell - K - 1$, with gcd $= 1$ and $v_\ell(\operatorname{den} P_h) = 2(h-K)$; for the mirror ($\ell \equiv 1 \pmod 4$, $K = (\ell+3)/4$) $u(K, s) \ne 0$ for $s \in [K, \ell - K - 2]$, $\ell < 200$ (960 cases), and the end-to-end test for $13 \le \ell \le 109$. The audit also verified that the content, decay, node removal, window and integrality statements depend on h only, that $c_{\mathrm{cert}}/\eta = 6.8717994/0.1304266 = 52.6871 < 52.69$, and that the prime product in §8 is about q^{10}. It found that the hybrid improvement $\kappa \to 0$ is not covered by the existing results (Remark 17.1).

The material of Part 1 was audited by the following independent Claude agents (each re-derived the statements by hand and re-ran exact computations with its own code): the reduction modulo ℓ, the bordered Lemma CC analogue and the selection of the pair of members (exact constant checked at every residue of X for $\ell = 11, \dots, 43$, 0 mismatches; end-to-end exact nonvanishing for $\ell = 7, \dots, 131$, 0 failures; corrected the member pair to $\{r_1, r_1 + 1\}$, $r_1 = \ell - K - m$); the closed form of the bordered Krylov determinants (the constant $\varkappa$: 20 cases by interpolation, no product assumption; ℓ-integrality bridge; $\sum e_c = D$, which we now prove by direct summation;

the zero orders: Rellich counts for $K \le 300$, 17,954,950 cases, the product formula verified exactly for all m at $K = 11, \dots, 14$, and the deformation argument, with the correction of the denominator Δ by the factors $\prod(y_i - o)$); Lemma E (hand check, exact X-formal checks at $r = 24, 48, 56, 72$, specialised checks at $X \in \{\frac{2}{13}, -7, 5\}$, and the observation that the statement fails at specialised $X \equiv H^{(2)}_{K_p-1}/16$, Remark 12.7); the tropical law (the duals D1–D6 re-derived and checked against certified Hungarian optima for 1300 prime powers, the integral by quadrature to 10^{-15}, the two-sided PNT paragraph of Lemma 12.4); the decay certificate (the Arb output reproduced exactly: $D_F \ge 3.906269506610$, $c \le 7.654064744147$; negative control $\alpha = 0$ reproduces 3.1113790/6.8717994; closed forms compared with quadrature; the window falsification test of the line $D_F - \beta c_{\mathrm{hyb}}$ against exact profiles for $r \le 128$); and the $\ell \mid \mathrm{den}$ analogue (3086 exact valuation checks, 0 failures). The audits found only cosmetic points, all corrected here, with the exceptions listed in §17.3.

Acknowledgments and provenance

The irrationality of Catalan's constant, and the Lean formalization on which the author's earlier work builds, are due to OpenAI [11]. The mathematics of this paper —the matrices T_h, the content, decay, nonvanishing and node-removal arguments, the numerical certificates and the text—were derived by Claude (Anthropic) agents working at the request of the author, who posed the problem, directed the work, and is responsible for this submission; independent Claude agents performed the audits described in §18. The same applies to Part 1: the rectangular hybrid frame, the content, decay, bordered nonvanishing and $\ell \mid \mathrm{den}$ arguments and the numerical certificate $\mu(\lambda) \le 46.45$ were derived by Claude agents at the request of the author, and audited by further Claude agents; the hybrid theorem is not formalized in Lean. The observation that the choice of h can be made before the choice of ℓ (the $\kappa = 0$ square-frame improvement from 57.85 to 52.69, version 8, for λ and λ') was likewise made by a Claude agent at the request of the author, who directed the work, and audited by a further Claude agent; it is not formalized in Lean. The Lean formalization was likewise written by Claude agents working with the author. The repository infrastructure, vendored formalizations and certificate tooling of the companion project, which this work reuses, were developed with OpenAI Codex as described in [7]. We use Lean 4 [9], Mathlib [8], the PNT+ project [12] (whose Wiener–Ikehara theorem we use to formalize the prime number theorem in progressions), and Arb [1] for interval arithmetic. The use of AI in this work is disclosed here in the same spirit as in [7].

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